AN ASYMPTOTIC FORMULA FOR THE THETA-FUNCTION SUM $S(p)$, AND A PROOF OF A CONJECTURE OF BERNDT, BHAT, MEYER, XIE AND ZAHARESCU

SAMSON FRASER
samsonmfraser@gmail.com

Abstract. For a prime p, the sum $S(p)$ adds up $(p-1)^2$ signs: the term indexed by $h,j \in \{1,\dots,p-1\}$ is $(-1)^{j+1+\lfloor hj/p \rfloor}$. Sums of this kind come from the transformation formula of a classical theta function. A sum of that many random signs would have size about p and either sign. Berndt, Bhat, Meyer, Xie and Zaharescu found numerically that $S(p)$ is positive and grows slightly faster than p. They conjectured that $S(p) > 0$ for $p > 5$, that $S(p) > 2p$ for $p > 233$ and $S(p) > 3p$ for $p > 3119$, and that $C_1 p < S(p) < C_2 p^{1+\varepsilon}$ for any fixed $C_1, C_2, \varepsilon > 0$ and all large p. They wrote that already the positivity "appears to be very difficult to prove".

We prove the last of these conjectures, and we prove the three inequalities for all sufficiently large primes. The threshold is effective but far beyond the range of computation; we have verified the inequalities for all $p \le 6 \cdot 10^6$, and for the primes in between they remain open. The main result is that $S(p) = \frac{p}{2}\log p + O\big(p(\log p)^{2/3}(\log\log p)^{17/3}\big)$. Outside a set of primes of density zero the error term is only $O(p(\log\log p)^2)$. We also give an exact decomposition of $S(p)$ into a smooth main term $\frac{p}{2}\log p + c_0 p + O(1)$ and a weighted sum of tangent sums, and we show that $c_0 = 0.2399\dots$ is the mean value of $S(p)/p - \frac{1}{2}\log p$ over the primes.

The tangent sums are controlled by Weil's bound for Kloosterman sums, by the bound of Bourgain and Garaev for short Kloosterman sums, and by a new elementary estimate for the large divisors of the terms of an arithmetic progression. The last conjecture needs only a power-of-log saving for short Kloosterman sums of length $\exp((\log p)^\theta)$, $\theta < 1$; a formalization in the Lean proof assistant checks its proof assuming only this saving. The Bourgain–Garaev bound itself is not formalized, and neither are Theorems E and F. The work was done with substantial AI assistance (see the end of the paper), and it has not yet been checked by a human expert. The Lean code, the data and the verification programs are available at https://github.com/samsonmaximus/theta-sum.

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1. Introduction

A sum of signs. Let p be an odd prime. Each pair (h, j) with $1 \le h, j \le p-1$ contributes a sign $(-1)^{j+1+\lfloor hj/p \rfloor}$, which is $+1$ or -1 according to the parity of $j+1+\lfloor hj/p \rfloor$. This paper is about the sum of these $(p-1)^2$ signs,

$$S(p) = \sum_{h=1}^{p-1} \sum_{j=1}^{p-1} (-1)^{j+1+\lfloor hj/p \rfloor}.$$

For a few small primes and some larger ones it takes the following values.

p	3	5	7	11	13	101	233	1009	3119	999983
$S(p)$	2	4	10	14	20	268	440	4408	9194	6306294

For example, for $p = 3$ the four terms are $+1, -1, +1, +1$, so $S(3) = 2$. If the signs behaved like independent coin tosses, $S(p)$ would have size about p, and it would be negative about half of the time. In fact $S(p)$ has been positive for every prime computed, and it grows slightly faster than p. We shall see that $S(p)$ is about $\frac{p}{2} \log p$. So the signs cancel almost completely, and what is left is a bias only a factor of order $\log p$ above the size of a random sum.

In the computed range the sum fluctuates around $\frac{p}{2} \log p$ by a few multiples of p. For all odd primes $p \le 6 \cdot 10^6$ the quantity $S(p)/p - \frac{1}{2} \log p$ lies between -2.2 and 2.4, and over the primes $10^3 < p \le 6 \cdot 10^6$ it has mean 0.24 and standard deviation 0.63 (Section 10). For example, $p = 999983$, the largest prime below 10^6, has $S(p) \approx 6.31\, p$, while $\frac{1}{2} \log p \approx 6.91$. This prime lies below the average because $2p - 1 = 3 \cdot 5 \cdot 11 \cdot 17 \cdot 23 \cdot 31$ has many small divisors; we explain below why such divisors push $S(p)$ down.

The sign of $S(p)$ has an elementary meaning. Put $H = \frac{p-1}{2}$, reduce the H^2 numbers $2ab$ with $1 \le a, b \le H$ modulo p, and let N_+ of the residues lie in the lower half $(0, p/2)$ and N_- in the upper half $(p/2, p)$. Then $S(p) = 2(N_- - N_+)$ (Lemma 2.6). So $S(p) > 0$ says that the upper half receives more of these products than the lower half, and $S(p) \approx \frac{p}{2} \log p$ says that the surplus is only a fraction of about $(\log p)/p$ of all the products. A proof that $S(p) > 0$ must detect this small surplus for every prime.

1.1. The sum and the conjectures.

For integers h and $k \ge 1$ let

$$S(h, k) = \sum_{j=1}^{k-1} (-1)^{j+1+\lfloor hj/k \rfloor}. \tag{1.1}$$

It is one of the arithmetic sums attached to theta functions that were introduced by Hardy [14] and studied by Berndt [2] and Berndt and Goldberg [3]. It belongs to the classical theta function $\vartheta_3(z) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 z}$: under a substitution $z \mapsto (az+b)/(cz+d)$ with $c > 0$ and c, d coprime of opposite parity, Berndt's transformation formula for $\log \vartheta_3$ contains, besides elementary terms, the term $\frac{\pi i}{4} S(d, c)$ ([2, Theorem 4.1]; see [1, Section 2]). In the same way the Dedekind sum $s(d, c)$ appears in the transformation formula of the Dedekind eta function. For coprime h, k of opposite parity, $S(h, k)$ satisfies the reciprocity law $S(h, k) + S(k, h) = 1$ [2, Theorem 4.2], an analogue of Dedekind's reciprocity law. Remark 2.2 expresses $S(h, k)$ and $S(p)$ through Hardy sums and Dedekind sums. Berndt, Bhat, Meyer, Xie and Zaharescu [1] studied the complete sums

$$S(k) = \sum_{h=1}^{k-1} S(h, k), \tag{1.2}$$

so that $S(p)$ above is the case $k = p$. On the basis of numerical evidence they formulated the following conjectures, in which k denotes a prime.[1]

Conjecture 1.1 (positivity)**.** *$S(k) > 0$ for every odd prime $k > 5$.*

Conjecture 3.1 (the bound $2k$)**.** *$S(k) > 2k$ for every prime $k > 233$.*

Conjecture 3.2 (the bound $3k$)**.** *$S(k) > 3k$ for every prime $k > 3119$.*

Conjecture 3.3 (growth)**.** *For any fixed $C_1, C_2, \varepsilon > 0$ one has $C_1 k < S(k) < C_2 k^{1+\varepsilon}$ for all sufficiently large primes k.*

The authors of [1] write that "Conjecture 1.1 appears to be very difficult to prove". A priori, they note, a sum of $(k-1)^2$ signs should favour neither sign, "But this intuition is apparently misleading." They trace the difficulty to "our lack of knowledge about the distribution of the residues of hj modulo k", that is, to the points on the modular hyperbolas $hj \equiv c \pmod k$ [25]. They made Conjecture 3.3 "with some temerity", and they compare its upper half with conjectures in prime number theory "for which 'square-root cancellation' is surmised". Their data for $k \leq 50000$ did not support a conjecture $S(k) > 4k$: the largest exception they found was $k = 49139$. The last exception below $6 \cdot 10^6$ is $k = 58013$ (Proposition 1.3). The conjectures are also stated in the *formal-conjectures* repository of Google DeepMind [11], where at the time of writing all of them are listed as open.

Why this sum, and where the difficulty lies. Multiplication modulo p scrambles intervals: as h and j run over intervals, the residues of hj modulo p spread over all residues, and how evenly the points of a modular hyperbola $hj \equiv c \pmod p$ fill a box is a subject of its own [25]. The sum $S(p)$ measures a small, systematic departure from randomness in this picture, as the count of the products $2ab$ above shows. A classical question of the same kind is D. H. Lehmer's problem on the parity of an integer and its inverse modulo p (see [30]), and $S(p)$ is a twisted first moment of the error term in a form of that problem (Lemma 2.4).

Our proof starts from an exact formula. A finite Fourier expansion and a reciprocity law for cotangent sums split $S(p)$ into a smooth main term and a weighted sum of the tangent sums $F_k(p) = \sum_{t=1}^{(k-1)/2} \tan(\pi p t/k)$ over odd k. Up to an error $O(p/\log p)$ only the moduli $k < \sqrt{p}(\log p)^4$ matter, and

$$S(p) = \frac{p}{2}\log p + c_0 p + \frac{4p}{\pi} \sum_{\substack{3 \leq k < \sqrt{p}(\log p)^4 \\ k \text{ odd}}} \frac{F_k(p)}{k^2} + O\Big(\frac{p}{\log p}\Big)$$

with an explicit constant $c_0 = 0.2399\ldots$ (Theorem B). The whole problem is therefore to show that the sum over k is $o(\log p)$. The trivial bound $|F_k(p)| \leq F_k(1) \sim \frac{k}{\pi}\log k$ gives only $O((\log p)^2)$.

This bound is needed for each prime, and averaging over p cannot give it, because some terms are large for every prime. The quotient $F_k(p)/k$ depends only on $\theta_k = ((2p)^{-1} \bmod k)/k$ (Lemma 6.1), and it is large when θ_k is very close to a rational number with a small denominator. Iterating the reciprocity law makes this precise: up to a bounded error, $F_k(p)/k$ is a sum of terms $\pm\frac{1}{\pi m}\log\frac{k}{|r|m}$, one for each pair of integers (m, r) with $m \geq 1$ odd, $r \neq 0$, $k \mid rp - m$ and $|r|m$ small compared with k (Theorem 4.10). For instance, $F_k(p) = -F_k(1)$ if $k \mid p+1$ and $F_k(p) = F_k(1)$ if $k \mid p-1$, so that $F_k(p)/k = \mp\frac{1}{\pi}\log k + O(1)$ (Corollary 4.3); the same holds with the minus sign if k divides $2p-1$ and with the plus sign if k divides $2p+1$ (Corollary 4.11 and the remark after it). Which moduli these are depends on the factorizations of $p \pm 1$, $2p \pm 1$ and similar numbers, and such moduli occur for every p.

[1] We keep the numbering of the third arXiv version of [1]; Appendix A compares it with the first version and with the formal statements in [11].

We write $F_k(p)/k$ as a sum of signs along the multiples of θ_k and split it into a head, its first terms, and a tail. For the moduli k between $\exp((\log p)^{2/3+o(1)})$ and $\sqrt{p}(\log p)^4$ we bound the heads on average over k. This leads to sums of modular inverses over short intervals, $\sum_{n\le N} e(bn^*/p)$ with N much smaller than $\sqrt{p}$, where n^* is the inverse of n modulo p and $e(x) = e^{2\pi ix}$. Little is known about these short Kloosterman sums. Nontrivial bounds are known only when $\log N$ is at least about $(\log p)^{2/3}$ [12, Section 1.1], and the bound of Bourgain and Garaev [8] that we use saves only a power of $\log p$. This range is the source of the exponent $\frac{2}{3}$ in our error term, and the small saving, with constants that are not explicit, is the reason why our proof applies only to very large primes. For smaller moduli we bound the heads trivially. The tails are large only when θ_k lies near a fraction with a small denominator, and we show that such moduli are rare with a new elementary estimate for the large divisors of the terms of an arithmetic progression (Theorem D).

1.2. **Main results.** The upper half of Conjecture 3.3 is easy: it follows in a few lines from a formula for $S(p)$ as a sum of products of cotangents (Section 3). Let γ be Euler's constant and

$$K_0 = \gamma + \log\frac{4}{\pi} = 0.8187801401\ldots. \tag{1.3}$$

Theorem A. *For every odd prime p,*

$$|S(p)| \;\le\; \frac{2}{p}\Big(\sum_{\substack{0<a<p\\ a\ odd}} \cot\frac{\pi a}{2p}\Big)^2 \;\le\; \frac{2}{\pi^2}\,p\,(\log p + K_0)^2.$$

In particular, for all $C_2 > 0$ and $\varepsilon > 0$ we have $S(p) < C_2 p^{1+\varepsilon}$ for all sufficiently large primes p.

The exact decomposition of $S(p)$ uses, for odd $k \ge 3$ and $n \in \mathbb{Z}$, the tangent sums

$$F_k(n) = \sum_{t=1}^{(k-1)/2} \tan\frac{\pi n t}{k} \tag{1.4}$$

(no term is singular: $nt/k \equiv \frac{1}{2} \pmod 1$ is impossible for odd k), the function

$$I(\lambda) = \frac{1}{\pi}\int_0^\infty \frac{\cosh v - \cosh \lambda v}{\sinh v \,\sinh \lambda v}\,dv \qquad (\lambda > 0), \tag{1.5}$$

and the constant

$$c_0 = \frac{K_0}{2} + \frac{\log 2}{6} + \frac{3}{\pi^2}\zeta'(2) = 0.2399341036\ldots. \tag{1.6}$$

Theorem B. (i) *For every odd prime p we have the identity $S(p) = \Psi(p) + \Phi(p)$, where*

$$\Psi(p) = \frac{2}{p}\sum_{\substack{0<k<p\\ k\ odd}} \cot\frac{\pi k}{2p}\, I\Big(\frac{k}{p}\Big), \qquad \Phi(p) = 2\sum_{\substack{3\le k<p\\ k\ odd}} \cot\frac{\pi k}{2p}\,\frac{F_k(p)}{k},$$

and $\Psi(p) = \frac{p}{2}\log p + c_0 p + O(1)$.

(ii) *For every odd prime p and every real K with $\sqrt{p}(\log p)^4 \le K \le p^{9/10}$,*

$$S(p) = \frac{p}{2}\log p + c_0 p + \frac{4p}{\pi}\sum_{\substack{3\le k<K\\ k\ odd}} \frac{F_k(p)}{k^2} + O\Big(\frac{p}{\log p}\Big). \tag{1.7}$$

The same holds with the sum extended over all odd k with $3 \le k < p$. All implied constants are absolute.

Part (i) splits $S(p)$ into a smooth part $\Psi(p)$, which depends on p only through the sizes of the numbers k/p, and an arithmetic part $\Phi(p)$, in which $F_k(p)$ depends only on the residue of p modulo k. Numerically, $\Psi(p) - \frac{p}{2}\log p - c_0 p$ tends to a constant $c_1 \approx -0.4076$, which we have not identified (Remark 5.1). In part (ii) the upper limit $p^{9/10}$ is a matter of convenience: for $k < K$ the proof replaces $\frac{2}{k}\cot\frac{\pi k}{2p}$ by $\frac{4p}{\pi k^2}$, at a cost $O(K^2 \log K/p)$. Our main result is that, for every large prime, the sum in (1.7) is of smaller order than the main term.

Theorem C*. *There are absolute constants C and p_0 such that for every prime $p \geq p_0$*

$$\Big|S(p) - \frac{p}{2}\log p - c_0 p\Big| \leq C\, p\, (\log p)^{2/3} (\log\log p)^{17/3}.$$

Since $(\log\log p)^3 \leq (\log p)^{\gamma_0}$ for large p, Theorem C* contains the following form, with the same constant C. Its proof is a little simpler (Section 8).

Theorem C. *There is an absolute constant C such that for every $\gamma_0 \in (0, \frac{2}{3})$ and every prime $p \geq p_0(\gamma_0)$*

$$\Big|S(p) - \frac{p}{2}\log p - c_0 p\Big| \leq C\, p\, (\log p)^{2/3+\gamma_0} (\log\log p)^{8/3}.$$

In particular, for every $\varepsilon > 0$ we have $S(p) = \frac{p}{2}\log p + c_0 p + O_\varepsilon\big(p(\log p)^{2/3+\varepsilon}\big)$ for all primes p.

The term $c_0 p$ in Theorem C* is smaller than the error term. So Theorem C* says no more than $S(p) = \frac{p}{2}\log p + O(p(\log p)^{2/3}(\log\log p)^{17/3})$; it does not show that $c_0 p$ is a second-order term of $S(p)$ for an individual prime. In fact no error term $o(p)$ is possible. In a companion paper [32], which has not yet been refereed, we show that $S(p)/p - \frac{1}{2}\log p$ has a limiting distribution over the primes, symmetric about c_0 and with support unbounded in both directions [32, Theorem 1, Corollary 3.3 and Remark 3.4]. The constant c_0 carries information in two other places: in Theorem B, where the rest of $S(p)$ is given explicitly up to $O(p/\log p)$, and on average over the primes, where c_0 is the mean value of $S(p)/p - \frac{1}{2}\log p$ (Theorem F with $q = 1$) and the deviations from it are bounded in mean square (Theorem E(ii)).

Corollary 1.1. *$S(p)/p \to \infty$ as $p \to \infty$. Hence the lower half of Conjecture 3.3 holds, and together with Theorem A, Conjecture 3.3 holds. Moreover, for every $M > 0$ we have $S(p) > Mp$ for all sufficiently large primes p; in particular, $S(p) > 0$, $S(p) > 2p$ and $S(p) > 3p$ for all sufficiently large primes.*

Corollary 1.1 needs much less than the bound of Bourgain and Garaev.

Theorem C$^\flat$. *Suppose that there are constants $\delta > 0$, $0 < \theta < 1$ and C such that for every sufficiently large prime p, every $b \not\equiv 0 \pmod p$ and every $N < p$ with $\log N \geq (\log p)^\theta$,*

$$\Big|\sum_{n \leq N} e\Big(\frac{bn^*}{p}\Big)\Big| \leq C\, N (\log p)^{-\delta}.$$

Then $S(p) = \frac{p}{2}\log p + O\big(p(\log p)^{\max(\theta, 1-\delta)}(\log\log p)^4\big)$ for all sufficiently large primes p, with an implied constant depending only on δ, θ and C. In particular $S(p)/p \to \infty$, and Conjecture 3.3 holds.

The bound of Bourgain and Garaev gives this hypothesis with $\theta = \frac{9}{10}$ and $\delta = \frac{1}{4}$, and for this it is needed only for $\log N \geq (\log p)^{9/10}$ (the Lean proof assumes it for $\log N \geq \frac{1}{2}(\log p)^{9/10}$), far inside the range that their proof covers (Section 8). So Conjecture 3.3 does not depend on the exact range of their theorem, but only on a fixed power-of-log saving that any future bound for short Kloosterman sums of this length would also give. This is the route of the Lean proof (Appendix A).

Further results. The new ingredient in the proofs of Theorems C* and C is the following elementary estimate, and its variant Theorem D*. We state it for an arbitrary modulus q, since primality plays no role in it. Divisors are positive: for a nonzero integer n, $d \mid n$ means that d is a positive divisor of $|n|$.

Theorem D. *Let $0 < \gamma_0 < 1$. There is $L_0(\gamma_0)$ such that the following holds for every integer q with $\log q \ge L_0(\gamma_0)$. Put $T = \exp((\log q)^{\gamma_0})$. Let a be an integer coprime to q, let $R \ge 1$, and let I be a set of at most R consecutive integers such that $0 < |a + qt| \le q^2$ for all $t \in I$. Then*

$$\sum_{t\in I} \sum_{\substack{d \mid a+qt \\ d > TR}} \frac{1}{d} \;\le\; \exp\Bigl(-\frac{\gamma_0}{4}(\log q)^{\gamma_0}\Bigr) = T^{-\gamma_0/4}.$$

For a random set of R integers of size at most q^2 the left-hand side would be about $1/T$. Theorem D therefore says that the terms of a progression with difference q never carry more than $T^{1-\gamma_0/4}$ times this expected mass of divisors larger than T times the length of the progression, uniformly in that length. The same proof gives the bound $T^{-\gamma_0+\varepsilon}$, and the case $R = 1$, with $a+qt$ a product of the first few primes, shows that this is optimal up to ε (Remark 7.4). We state the weaker form because it suffices and because it is the form checked in Lean. The method also works for much smaller T, with a smaller saving, and this gives Theorem C*.

Theorem D*. *There is an absolute constant q_1 such that the following holds for every integer $q \ge q_1$. Put $L = \log q$, and let T satisfy $\log T \ge (\log L)^3$. Let a be an integer coprime to q, let $R \ge 1$, and let I be a set of at most R consecutive integers such that $0 < |a + qt| \le q^2$ for all $t \in I$. Then*

$$\sum_{t\in I} \sum_{\substack{d \mid a+qt \\ d > TR}} \frac{1}{d} \;\le\; \exp\Bigl(-\frac{\log T}{4\log L}\Bigr).$$

In Section 8, Theorems D and D* are applied with $q = p$ to the progressions $\{pt \mp m : R \le t < 2R\}$, where $1 \le m \le \lceil (\log p)^2 \rceil$ and R is a power of 2. If $\|2m\theta_k\| < 1/T$, then k is a divisor larger than TR of a term of one of them, for a suitable R (Lemma 6.9). Shiu's theorem [24] bounds sums of multiplicative functions over progressions of length at least q^η; Theorem D has no condition on the length, but it bounds only the $1/d$-weighted mass of the divisors above TR. We do not know whether it can be deduced from the results on divisors in short intervals and progressions in Hall and Tenenbaum's book [36] or in Nair and Tenenbaum [37]; we would be glad to learn of such a deduction.

On average over p the error term in Theorem C* can be reduced to $p(\log\log p)^2$.

Theorem E. (i) *Let $A > 0$. For $X \ge X_0(A)$, all primes $p \in (X, 2X]$ except at most $CX(\log X)^{-A}$ of them satisfy*

$$\Bigl|S(p) - \frac{p}{2}\log p\Bigr| \le \frac{(A+15)^2}{\pi^2}\, p\,(\log\log p)^2,$$

where C is an absolute constant.

(ii) *For $X \ge 2$,*

$$\sum_{X < p \le 2X} \Bigl(\frac{S(p)}{p} - \frac{1}{2}\log p - c_0\Bigr)^2 \le C' \frac{X}{\log X}$$

with an absolute but ineffective constant C'.

The constant C' is ineffective because the proof uses the Siegel–Walfisz theorem.

Corollary 1.2. (a) *Let $\mathcal{P}$ be the set of primes p with $|S(p) - \frac{p}{2}\log p| \le 30\,p(\log\log p)^2$. Then $\#\{p \le x : p \notin \mathcal{P}\} \ll x(\log x)^{-2}$.*

(b) *For every function $\xi(x) \to \infty$, $\#\{p \le x :\ |S(p) - \frac{p}{2}\log p - c_0 p| > \xi(p)\,p\} = o(\pi(x))$.*

The heuristics and the numerical evidence (Section 10) suggest that the bound $S(p) = \frac{p}{2}\log p + O(p(\log\log p)^2)$ holds for every prime, and that the extreme values come from primes p for which $p+1$, $p-1$, $2p-1$ or $2p+1$ is divisible by many small odd numbers.

Our last theorem gives the mean value of $S(p)/p - \frac{1}{2}\log p$ in residue classes. For odd $k \geq 3$, a divisor g of k and an integer r coprime to g, let $\langle F_k\rangle_{g,r}$ be the average of $F_k(x)$ over the reduced residues x modulo k with $x \equiv r \pmod g$.

Theorem F. *Let $q \geq 1$ and $(r,q) = 1$. As $x \to \infty$,*

$$\frac{1}{\pi(x;q,r)} \sum_{\substack{p \leq x \\ p \equiv r \pmod q}} \Big(\frac{S(p)}{p} - \frac{1}{2}\log p\Big) \longrightarrow c_0(q;r) := c_0 + \frac{4}{\pi} \sum_{\substack{k \geq 3 \\ k \text{ odd}}} \frac{\langle F_k\rangle_{(k,q),r}}{k^2},$$

and the series converges absolutely. Moreover $c_0(q;r) + c_0(q;-r) = 2c_0$, and $c_0(q;r) = c_0$ if q is a power of 2.

Table 2 compares $c_0(q;r)$ with observed means; for instance $c_0(3;1) = 0.6384$ and $c_0(3;2) = -0.1585$, while the means over the primes $10^3 < p \leq 6\cdot 10^6$ in these classes are 0.6384 and -0.1588.

Finally, we computed $S(p)$ exactly for all odd primes $p \leq 6 \cdot 10^6$ and for 56 selected primes between $3 \cdot 10^6$ and $5.4 \cdot 10^{10}$ (Section 10).

Proposition 1.3. *For all odd primes $p \leq 6 \cdot 10^6$ we have $S(p) > 0$; $S(p) > p$ if $p > 5$; $S(p) > 2p$ if $p > 233$; $S(p) > 3p$ if $p > 3119$; $S(p) > 4p$ if $p > 58013$; and $S(p) > 5p$ if $p > 869399$. None of the thresholds can be lowered: $S(5) = 4$, $S(233) = 440$, $S(3119) = 9194$, $S(58013) < 4 \cdot 58013$ and $S(869399) < 5 \cdot 869399$.*

1.3. What is and what is not proved.

Statement	Status
Growth conjecture (3.3), upper half	proved for all p (Theorem A)
Growth conjecture (3.3), lower half	proved (Corollary 1.1); needs only a power-of-log saving for short Kloosterman sums (Theorem C$^\flat$)
$S(p) = \frac{p}{2}\log p + O(p(\log p)^{2/3}(\log\log p)^{17/3})$	proved for all large p (Theorem C*)
the constant c_0	second-order constant of $\Psi(p)$ (Theorem B); mean of $S(p)/p - \frac{1}{2}\log p$ over the primes (Theorem F)
$S(p) = \frac{p}{2}\log p + O(p(\log\log p)^2)$	proved for all p outside a set of relative density zero (Theorem E); open for all p
$S(p) > 0$, $S(p) > 2p$, $S(p) > 3p$ beyond the thresholds of [1] (Conjectures 1.1, 3.1, 3.2); $S(p) > p$ for $p > 5$	proved for all $p \geq p_1$, an uncomputed threshold (Corollary 1.1); verified for $p \leq 6 \cdot 10^6$ (Proposition 1.3); open in between

We know of no earlier proof of Conjecture 3.3. Our proof gives more than the conjecture: the asymptotic formula of Theorem C*, which rests on the exact decomposition of Theorem B. For the conjecture itself it needs only a fixed power-of-log saving for short Kloosterman sums (Theorem C$^\flat$). Theorems D and D* are estimates for large divisors of the terms of arithmetic progressions of any length; they do not involve $S(p)$, their proofs are elementary, and they may be of independent interest. The values of $S(p)$ for all primes up to $6 \cdot 10^6$ were computed exactly, and checked by independent programs for all $p < 2 \cdot 10^5$ and for 640 values from the whole range (Section 10). All values for $4 \cdot 10^6 < p \leq 6 \cdot 10^6$ were also recomputed by a separately written program based on the same formula (Lemma 2.5), with no discrepancy.

A variant of the proof has also been checked by computer. Lean is a proof assistant: a program that checks every step of a formal proof. In Lean 4 with its library Mathlib, starting from the definitions of S in [11], the file that also states the conjectures formally, we have verified the upper half of Conjecture 3.3 and Theorems D and D* unconditionally. The whole of Conjecture 3.3 is verified assuming only the hypothesis of Theorem C$^\flat$, which the formalization derives from the Bourgain–Garaev bound for $\log N \geq \frac{1}{2}(\log p)^{9/10}$. Theorem C* (and hence Theorem C) is verified assuming the Bourgain–Garaev bound in the range of Lemma 6.5. That bound is the only input of the formal proofs that is not itself formalized. The formal proofs differ from ours in some details; for instance, they use Kloosterman's bound $2p^{3/4}$ in place of Weil's bound. The positivity, $2k$ and $3k$ conjectures are verified for all $p \leq 2 \cdot 10^4$ by a computation in the Lean kernel. Appendix A gives the details.

The main open questions (Section 11) are the positivity, $2k$ and $3k$ conjectures for the primes between $6 \cdot 10^6$ and p_1; the error term $O(p(\log\log p)^2)$ for every prime, which Theorem E gives for almost all primes; and a proof of Conjecture 3.3 that avoids bounds for short Kloosterman sums. The threshold p_1 is effective in principle but has not been computed: it depends on the constants of [8, Theorem 16], which are not given explicitly.

What a reader should check first. Apart from the parts checked in Lean, the proof of Conjecture 3.3 uses one external result: Lemma 6.5, our statement of [8, Theorem 16] together with the range of N covered by its proof. An earlier version of this paper used that bound outside this range. A reader who checks one thing should check Lemma 6.5 against [8], and then Section 8.

The Lean code, the data and the verification programs are available at `https://github.com/samsonmaximus/theta-sum`.

1.4. **Outline of the method.** *The tools.* For a prime p and $b \not\equiv 0 \pmod p$, sums of $e(bn^*/p)$ over n are *Kloosterman sums*. Over a complete residue system, and with an additional factor $e(\ell n/p)$, Weil's bound gives square-root cancellation: such a sum is at most $2\sqrt{p}$ in absolute value [29]. By completion, this bounds the sums over intervals of length N by about $\sqrt{p}\log p$, which is useful only for N larger than $\sqrt{p}\log p$. For the *short* Kloosterman sums $\sum_{n\leq N} e(bn^*/p)$ with smaller N, the bound of Bourgain and Garaev [8] is $C_{BG}N(\log p)(\log\log p)^3/(\log N)^{3/2}$, a saving of the factor $(\log N)^{3/2}/(C_{BG}(\log p)(\log\log p)^3)$ over the trivial bound N. Their Theorem 16 is stated without a restriction on N, but its proof covers only $\log N \geq A_{BG}(\log p)^{2/3}(\log\log p)^{8/3}$, for an absolute constant A_{BG} (Lemma 6.5). We therefore apply it to the moduli $k \geq K_* = \exp(A(\log p)^{2/3}(\log\log p)^{8/3})$ with $A = A_{BG} + 1$. This is the source of the exponent $\frac{2}{3}$ in Theorems C* and C.

An exact formula. The starting point (Section 3) is the finite Fourier expansion of $n \mapsto (-1)^{\lfloor n/p \rfloor}$ on $\mathbb{Z}/2p\mathbb{Z}$. It turns the sum of signs into a sum of products of cotangents,

$$S(p) = \frac{2}{p} \sum_{\substack{0<k<p \\ k \text{ odd}}} \cot\frac{\pi k}{2p}\, C_p(k), \qquad C_p(k) = \sum_{\substack{0<a<p \\ a \text{ odd}}} \cot\frac{\pi k a}{2p}, \tag{1.8}$$

or, equivalently,

$$S(p) = \frac{2}{p} \sum_{\substack{0<k,n<p \\ k,n \text{ odd}}} \cot\frac{\pi k}{2p} \cot\frac{\pi n}{2p}\, \sigma(n\overline{k}), \tag{1.9}$$

where $\overline{k}$ is the inverse of k modulo $2p$ and $\sigma(m) = \pm 1$ according as $m \bmod 2p$ lies in $(0,p)$ or in $(p,2p)$. Since $|C_p(k)| \leq C_p(1) \sim \frac{p}{\pi}\log p$, Theorem A follows at once. Formula (1.9) also shows where the difficulty lies: the signs $\sigma(n\overline{k})$ depend on modular inverses, which is the modular-hyperbola problem of [1] in another form.

Reciprocity and the main term. The half-range cotangent sums satisfy a reciprocity law (Section 4), which we prove by contour integration:

$$C_p(k) = \frac{p}{k}F_k(p) + I\Big(\frac{k}{p}\Big), \qquad \pi\lambda I(\lambda) = \log\frac{1}{\lambda} + K_0 + O(\lambda^2).$$

Inserted into (1.8), the terms $I(k/p)$ produce the main term $\frac{p}{2}\log p + c_0 p$, and the terms $F_k(p)$ produce $\Phi(p)$ (Section 5).

Large moduli: Weil's bound. For $k \geq K_W = \sqrt{p}(\log p)^4$ we do not estimate the $F_k(p)$ one by one. Summed over k in an interval, the sums $C_p(k)$ become sums of $e(c\overline{k}/2p)$ over k, which are incomplete Kloosterman sums, and Weil's bound gives square-root cancellation in them. These moduli contribute only $O(p/\log p)$ (Theorem B(ii)), although the trivial bound for them is of order $p(\log p)^2$.

Heads and tails. For $k < K_W$ we write $F_k(p)/k$, via a sign form of the tangent sums (Lemma 6.1), as

$$\frac{F_k(p)}{k} = -\sum_{\substack{0<n<k\\ n \text{ odd}}} \frac{1}{k}\cot\frac{\pi n}{2k}\,\operatorname{sgn}\Big(\frac{1}{2} - \{n\theta_k\}\Big),$$

a sum of signs along the multiples of θ_k, with weights close to $\frac{2}{\pi n}$. We split it at $n \leq N$, where $\log N \approx \log T$ and $T = \exp((\log\log p)^3)$ (for Theorem C, $T = \exp((\log p)^{\gamma_0})$). The head ($n \leq N$) is at most $\frac{1}{\pi}\log N + O(1)$ in absolute value. Summed with the weights $1/k$ over $k < K_*$, the heads therefore cost $O(\log N \log K_*) = O((\log p)^{2/3}(\log\log p)^{17/3})$, which is the error term in Theorem C*.

Middle moduli: the Bourgain–Garaev bound. For $K_* \leq k < K_W$ we average the heads over k in dyadic blocks. Expanding the sign function into a trigonometric polynomial turns each head into sums of $e(\ell n\overline{k}/2p)$ over k in a block $[K, 2K)$, that is, into Kloosterman sums of length about K. These sums are short, of length at most $\sqrt{p}(\log p)^4$, and for most of them Weil's bound is trivial. The bound of Bourgain and Garaev applies to them because $K \geq K_*$ (Lemmas 6.4 and 6.5). The power $\frac{3}{2}$ of $\log N$ in the saving matters: it makes the contributions of the dyadic blocks above K_* add up to the same order as the heads below K_*.

The tails: Theorems D and D.* The tail ($N < n < k$) is small whenever the points $n\theta_k$ are well distributed. An inequality of Erdős–Turán type, which bounds the discrepancy of a sequence by its exponential sums, shows that this is the case unless $\|2m\theta_k\| < 1/T$ for some $m \leq (\log p)^2$; the tails of the other moduli contribute $O(p)$ to $S(p)$. At this scale no estimate for exponential sums is available, so we count the exceptional moduli instead. Such a modulus k has a *witness*: an integer r with $0 < |r| < k/T$ and $k \mid pr - m$. Theorem D*, applied as described after Theorem D, shows that the total $1/k$-weighted mass of these moduli is at most a power of $\log p$ times $\exp(-\frac{1}{4}(\log\log p)^2)$, so that their tails contribute only $o(p)$. Bounds for divisor sums along progressions, such as Shiu's theorem [24], need progressions of length at least a fixed power of the modulus. Ours have every length from 1 to about $\sqrt{p}$, and this is why we need Theorems D and D*. Their proof (Section 7) is a recursion on the length of the progression, which ends with Rankin's trick, the bound $\mathbf{1}_{d>Y} \leq (d/Y)^\alpha$.

Averages over primes. On average over p the problem is much easier. Since $F_k(p)$ depends only on $p \bmod k$, each truncation $R_K(p)$ of the sum in (1.7) is a periodic function of p, whose frequencies are fractions with odd denominators below K. The large sieve gives Theorem E(i); together with the Siegel–Walfisz theorem it gives Theorem E(ii), and the Siegel–Walfisz theorem gives Theorem F (Section 9).

1.5. **Related work.** Hardy sums and their reciprocity laws go back to Hardy [14] and were developed by Berndt [2], Berndt and Goldberg [3] and Sitaramachandrarao [26], who expressed them in terms of Dedekind sums; Lägeler [18] related $S(d, c)$ to even continued fractions, which

also govern the expansion of Subsection 4.1. Mean values of Dedekind sums $s(h,k)$ over h for fixed k are classical (see e.g. [10]); they are closely related to mean values of $|L(1,\chi)|^2$ over odd characters [31], and Walum [28] gave an exact formula for $\sum_{\chi \text{ odd}} |L(1,\chi)|^2$. Those mean values are quadratic in $L(1,\chi)$, whereas $S(p)$ is cubic: by orthogonality it is a weighted sum of $|A(\chi)|^2 A(\chi)$ over the odd characters χ modulo p, where $A(\chi) = \sum_{0<b<p/2} \chi(b)$ (Lemma 2.7). Averages of Dedekind sums over restricted sets of Farey fractions, which are of order $\log Q$, were evaluated by Minelli, Sourmelidis and Technau [19]; reciprocity formulas, pointwise bounds and limiting distributions for cotangent sums were obtained by Bettin and Conrey [5], Bettin [4] and Bettin and Drappeau [6, 7]; in particular [6] bounds partial sums of the cotangent function in terms of continued fraction data, in the spirit of Subsection 4.1. The distributional results among these average over complete residue systems or over all denominators, whereas $S(p)$ involves essentially one point per modulus. We did not find the twisted first moment of Lemma 2.4 in the literature. For short Kloosterman sums see Korolev [17], who also describes Karatsuba's method, Bourgain and Garaev [8, 9] and Garaev and Shparlinski [12]; for modular hyperbolas see Shparlinski's survey [25]. Theorem D is related in spirit to Shiu's Brun–Titchmarsh theorem for multiplicative functions [24], which bounds divisor sums along progressions whose length is a power of the modulus; Theorem D has no such restriction, but controls only the $1/d$-weighted mass of the large divisors.

How to read this paper. Sections 2 and 3 are elementary; they give several exact formulas for $S(p)$ and the proof of Theorem A. Section 4 proves the reciprocity law; its Subsection 4.1 explains the size and the singularities of $F_k(p)$ but is not used in the proof of Theorem C*. Section 5 proves Theorem B. The proofs of Theorems C*, C and C$^\flat$ occupy Sections 6–8: Section 6 collects the tools, Section 7 proves Theorems D and D* and can be read on its own, and Section 8 puts the pieces together. Section 9 proves the results on average (Theorems E and F), Section 10 describes the computations, Section 11 lists open questions, and Appendix A describes the formalization. A reader interested only in Corollary 1.1 can skip Subsection 4.1 and Sections 9 and 10.

1.6. **Notation.** We write $e(x) = e^{2\pi i x}$, $\|x\|$ for the distance from x to the nearest integer, $\{x\} = x - \lfloor x \rfloor$, and $((x)) = x - \lfloor x \rfloor - \frac{1}{2}$. For $m \in \mathbb{Z}$ with $p \nmid m$ we put

$$\sigma(m) = (-1)^{\lfloor m/p \rfloor};$$

thus $\sigma(m) = 1$ if $m \bmod 2p \in (0,p)$ and $\sigma(m) = -1$ if $m \bmod 2p \in (p,2p)$, and σ is $2p$-periodic and odd. We use two kinds of inverses. For k coprime to $2p$, $\overline{k}$ denotes an integer with $k\overline{k} \equiv 1 \pmod{2p}$. For n coprime to p, n^* denotes the inverse of n modulo p. For coprime odd positive integers m, k let

$$C_m(k) = \sum_{\substack{0<a<m \\ a \text{ odd}}} \cot \frac{\pi k a}{2m}.$$

The sums $F_k(n)$ are defined in (1.4) for odd $k \geq 3$, and $F_1 = 0$. For $K > 3$ we put

$$R_K(n) = \sum_{\substack{3 \leq k < K \\ k \text{ odd}}} \frac{F_k(n)}{k^2}.$$

We use $h(x) = \frac{\pi x}{2} \cot \frac{\pi x}{2}$ for $0 < x \leq 1$ and $h(0) = 1$; h decreases from 1 to 0 on $[0,1]$, is continuously differentiable there, and $0 \leq 1 - h(x) \leq x^2$ (the Taylor coefficients of $1 - y\cot y$ are positive, so $(1-h(x))/x^2$ increases on $(0,1]$, and its value at $x = 1$ is $1 - h(1) = 1$). $P^+(n)$ denotes the largest prime factor of n ($P^+(1) = 1$), and n is *y-smooth* if $P^+(n) \leq y$ and *y-rough* if all its prime factors exceed y; $\Omega(n)$ is the number of prime factors of n counted with multiplicity, and $\sigma_s(n) = \sum_{d \mid n} d^s$. The letter ℓ always denotes a prime. Finally $\tau, \omega, \varphi, \mu$ are the divisor function,

the number of distinct prime factors, Euler's function and the Möbius function, $\pi(x;q,r)$ counts primes $p \le x$ with $p \equiv r \pmod q$, and C denotes an absolute constant, not necessarily the same at each occurrence.

2. Elementary reformulations

This section rewrites $S(p)$ in several equivalent forms, each of which shows a different side of it: a sum of Hardy sums, a sum of sawtooth values, a signed count in Lehmer's problem on an integer and its inverse, a count of the products $2ab \bmod p$ that fall into the lower half of $(0,p)$, and a cubic moment of half-range character sums. The main device is the reflection $b \mapsto p-b$, which makes the terms with even a cancel. The later sections use only Lemma 2.1, the identity (2.1), and Lemma 2.5, which is the basis of our computations.

Lemma 2.1. *For every odd prime p:*
(a) $S(p) = \sum_{a,b=1}^{p-1} (-1)^{\lfloor ab/p \rfloor} = \sum_{a,b=1}^{p-1} \sigma(ab)$;
(b) *the inner sums* $T(a) := \sum_{b=1}^{p-1} (-1)^{\lfloor ab/p \rfloor}$ *vanish for even a, so that* $S(p) = \sum_{0<a<p,\ a\ odd} T(a)$;
(c) $S(p-a,p) = T(a)$ *for* $1 \le a \le p-1$; *in particular* $S(h,p) = 0$ *for odd h (take $a = p-h$, which is even).*

Proof. For $1 \le a, j \le p-1$ we have $p \nmid aj$, hence $\lfloor (p-a)j/p \rfloor = j - 1 - \lfloor aj/p \rfloor$ and $j + 1 + \lfloor (p-a)j/p \rfloor \equiv \lfloor aj/p \rfloor \pmod 2$. This proves (c), and (a) follows by summing over a. If a is even, then $\lfloor a(p-b)/p \rfloor = a - 1 - \lfloor ab/p \rfloor \equiv 1 + \lfloor ab/p \rfloor \pmod 2$, so the terms b and $p-b$ of $T(a)$ cancel; since p is odd, $b \mapsto p-b$ has no fixed point. This proves (b) and the last assertion of (c). □

Remark 2.2 (Hardy and Dedekind sums)**.** In Berndt's notation for Hardy sums, $T(a) = s_4(a,p)$, so that $S(p) = \sum_{a \text{ odd}, 0<a<p} s_4(a,p)$. Sitaramachandrarao [26] expressed the Hardy sums through Dedekind sums $s(h,k)$; in particular $s_4(h,k) = 8s(h,2k) - 4s(h,k)$ for odd h coprime to k, whence $S(p) = \sum_{a \text{ odd}, 0<a<p} \big(8s(a,2p) - 4s(a,p)\big)$, and $S(h,k) = 4s(h,k) - 8s(h+k,2k)$ for coprime h,k with $h+k$ odd. Combined with the identity $s(h,2k)+s(h+k,2k)+s(2h,k) = 3s(h,k)$ for coprime h,k, a case of Knopp's identity [16], the latter becomes $S(h,k) = 8s(h,2k) + 8s(2h,k) - 20s(h,k)$. These identities are not used below.

Lemma 2.3 (Sawtooth form)**.** $S(p) = -4 \sum_{a,b=1}^{p-1} ((\frac{ab}{2p}))$.

Proof. For $y \in \mathbb{R} \setminus \frac{1}{2}\mathbb{Z}$ we have

$$\operatorname{sgn}\Big(\frac{1}{2} - \{y\}\Big) = 2\,((y + \tfrac{1}{2})) - 2\,((y)), \tag{2.1}$$

and $\sigma(m) = \operatorname{sgn}(\frac{1}{2} - \{m/2p\})$ for $p \nmid m$. Hence $\sigma(ab) = 2((\frac{ab+p}{2p})) - 2((\frac{ab}{2p}))$. Replacing b by $p-b$ shows that $\sum_b ((\frac{ab+p}{2p})) = \sum_b ((\frac{a+1}{2} - \frac{ab}{2p}))$; for odd a this equals $-\sum_b ((\frac{ab}{2p}))$, while for even a it equals $-\sum_b ((\frac{ab+p}{2p}))$ and therefore vanishes. In the same way $\sum_b ((\frac{ab}{2p})) = 0$ for even a. Summing over a gives the claim. □

Lemma 2.4 (Lehmer form)**.** *For $1 \le c \le p-1$ let $N(c,p)$ be the number of pairs $(a,b) \in [1,p-1]^2$ with $ab \equiv c \pmod p$ and $a+b$ odd. Then*

$$S(p) = \sum_{c=1}^{p-1} (-1)^c N(c,p) = \sum_{c=1}^{p-1} (-1)^c \Big(N(c,p) - \frac{p-1}{2}\Big).$$

Proof. Write $ab = p\lfloor ab/p \rfloor + c$ with $c \equiv ab \pmod p$, $0 < c < p$. As p is odd, $\lfloor ab/p \rfloor \equiv ab + c \pmod 2$, so $S(p) = \sum_c (-1)^c \sum_{ab \equiv c} (-1)^{ab}$. For fixed c there are $p-1$ pairs with $ab \equiv c$, and $(a,b) \mapsto (p-a, p-b)$ exchanges the pairs with a,b both odd and those with a,b both even. If n_c

denotes the number of the former, then $\sum_{ab\equiv c}(-1)^{ab} = p-1-2n_c = N(c,p)$. The second form follows from $\sum_{c=1}^{p-1}(-1)^c = 0$. □

Lemma 2.5 (A counting formula)**.** *For integers $x \ge 0$, $y \ge 1$ put $F(x,y) = \sum_{c=1}^{\lfloor (y-1)/2\rfloor} \lfloor cx/y \rfloor$. For odd a with $1 \le a < p$,*

$$T(a) = (p-1)a - 4F(2p,a), \qquad \text{hence} \qquad S(p) = \frac{(p-1)^3}{4} - 4 \sum_{\substack{3\le a\le p-2 \\ a \text{ odd}}} F(2p,a).$$

Moreover, with $U = \lfloor (y-1)/2 \rfloor$ we have $F(qy+r,y) = q\binom{U+1}{2} + F(r,y)$ for integers $q, r \ge 0$, and

$$F(r,y) + F(y,r) = \left\lfloor \frac{y-1}{2} \right\rfloor \left\lfloor \frac{r-1}{2} \right\rfloor \qquad \text{for coprime integers } r, y \ge 1.$$

Proof. Let a be odd. For $0 \le c \le a$ let N_c be the number of $b \in [1, p-1]$ with $ab < cp$; thus $N_0 = 0$, $N_a = p-1$, and $N_c = \lfloor cp/a \rfloor$ for $1 \le c \le a-1$ (as $a \nmid cp$). Since $\lfloor ab/p \rfloor = c$ for exactly $N_{c+1} - N_c$ values of b,

$$T(a) = \sum_{c=0}^{a-1} (-1)^c (N_{c+1} - N_c) = -(-1)^a N_a - 2\sum_{c=1}^{a-1} (-1)^c N_c = (p-1) - 2\sum_{c=1}^{a-1} (-1)^c \left\lfloor \frac{cp}{a} \right\rfloor.$$

Now $\sum_{c=1}^{a-1} \lfloor cp/a \rfloor = (p-1)(a-1)/2$, and the terms with even $c = 2c'$ give $\sum_{c'=1}^{(a-1)/2} \lfloor 2c'p/a \rfloor = F(2p,a)$; hence $\sum_{c=1}^{a-1}(-1)^c \lfloor cp/a \rfloor = 2F(2p,a) - \frac{(p-1)(a-1)}{2}$, which gives the formula for $T(a)$. Summing over odd a (note $F(2p,1) = 0$ and $\sum_{a \text{ odd},\, a<p} a = ((p-1)/2)^2$) and using Lemma 2.1(b) gives the formula for $S(p)$. The relation for $F(qy+r,y)$ follows from $\lfloor c(qy+r)/y \rfloor = cq + \lfloor cr/y \rfloor$. For the last identity, let $V = \lfloor (r-1)/2 \rfloor$. For $1 \le c \le U$ we have $cr/y < r/2$, so $F(r,y)$ counts the lattice points (c,d) with $1 \le c \le U$, $1 \le d \le V$ and $dy < cr$; similarly $F(y,r)$ counts those with $cr < dy$. No point of $[1,U] \times [1,V]$ satisfies $dy = cr$, because that would force $y \mid c$. Hence $F(r,y) + F(y,r) = UV$. □

Iterating the last two relations evaluates $F(2p,a)$ in $O(\log a)$ arithmetic operations, as in the Euclidean algorithm; this is how we computed $S(p)$ (Section 10).

The following identities are not used in the proof of Theorem C, but they give useful alternative descriptions of $S(p)$. For a character χ modulo p put

$$A(\chi) = \sum_{0<a<p/2} \chi(a), \qquad B_{1,\chi} = \frac{1}{p} \sum_{a=1}^{p-1} a\,\chi(a),$$

let $s(x,p) = \sum_{c=1}^{p-1} ((\frac{c}{p}))((\frac{xc}{p}))$ be the Dedekind sum, and let $2^* = \frac{p+1}{2}$ be the inverse of 2 modulo p.

Lemma 2.6 (Gauss-lemma form)**.** *Let $H = \frac{p-1}{2}$. Then*

$$S(p) = 2\sum_{a,b=1}^{H} \operatorname{sgn}\left(\left\{ \frac{2ab}{p} \right\} - \frac{1}{2} \right) = 2H^2 - 4N_+,$$

$$N_+ = \#\{(a,b,c) \in [1,H]^3 : \ 2ab \equiv c \pmod p\}.$$

Thus $S(p) > 0$ if and only if fewer than half of the products $2ab \bmod p$, $1 \le a,b \le H$, lie in the lower half $(0, p/2)$.

Lemma 2.7 (Cubic moments)**.** *We have*

$$S(p) = -\frac{4}{p-1}\sum_{\chi\ odd}\chi(2)\,|A(\chi)|^2A(\chi) = \frac{4}{p-1}\sum_{\chi\ odd}\big(2\chi(2)-1\big)\,|2-\chi(2)|^2\,|B_{1,\chi}|^2B_{1,\chi}$$
$$= 4\sum_{a=1}^{p-1}((\tfrac{a}{p}))\Big(-4s(4a,p)+12s(2a,p)+2s(2^*a,p)-9s(a,p)\Big).$$

Proof of Lemmas 2.6 and 2.7. For $p\nmid m$ let $r\in(0,p)$ be the residue of m modulo p. The two integers in $(0,2p)$ that are congruent to m modulo p are r and $r+p$, and they have different parities; hence $m \bmod 2p \in (0,p)$ if and only if $m\equiv r \pmod 2$, that is, $\sigma(m)=(-1)^m\lambda(m)$ with $\lambda(m)=(-1)^r$. By Lemma 2.1(a), $S(p)=\sum_{a,b=1}^{p-1}(-1)^{ab}\lambda(ab)$. Expand $\lambda=\sum_\chi c_\chi\chi$ on $(\mathbb{Z}/p\mathbb{Z})^\times$, $c_\chi=\frac{1}{p-1}\sum_{x=1}^{p-1}(-1)^x\overline{\chi}(x)$. For χ nontrivial, $\sum_x(-1)^x\overline{\chi}(x)=2\sum_{x\text{ even}}\overline{\chi}(x)=2\overline{\chi}(2)\overline{A(\chi)}$, which vanishes for even χ because then $A(\chi)=\frac12\sum_x\chi(x)=0$; for χ trivial the sum is 0. For odd χ we have $\sum_{a=1}^{p-1}\chi(a)=0$ and $\sum_{a\text{ odd}}\chi(a)=-\sum_{a\text{ even}}\chi(a)=-\chi(2)A(\chi)$, so

$$\sum_{a,b=1}^{p-1}(-1)^{ab}\chi(ab)=\Big(\sum_a\chi(a)\Big)^2-2\Big(\sum_{a\text{ odd}}\chi(a)\Big)^2=-2\chi(2)^2A(\chi)^2.$$

Hence $S(p)=\sum_{\chi\text{ odd}}\frac{2\overline{\chi}(2)\overline{A(\chi)}}{p-1}\big(-2\chi(2)^2A(\chi)^2\big)$, the first formula of Lemma 2.7. Next, by orthogonality, $\sum_{\chi\text{ odd}}\chi(x)=\frac{p-1}{2}(\delta_{x\equiv1}-\delta_{x\equiv-1})$, so

$$\sum_{\chi\text{ odd}}\chi(2)A(\chi)^2\overline{A(\chi)}=\sum_{a,b,c=1}^{H}\sum_{\chi\text{ odd}}\chi(2ab\,c^{-1})=\frac{p-1}{2}\,(N_+-N_-),$$

where N_- counts the triples with $2ab\equiv -c$. For each (a,b) exactly one $c\in[1,H]$ satisfies $2ab\equiv\pm c$, so $N_++N_-=H^2$, and N_- counts the pairs with $2ab \bmod p > p/2$. This gives Lemma 2.6. For the second formula of Lemma 2.7, let χ be odd. As b runs over $1,\dots,p-1$, so does $2b \bmod p$; hence

$$pB_{1,\chi}=\sum_{b=1}^{p-1}(2b \bmod p)\chi(2b)=\chi(2)\Big(2pB_{1,\chi}-p\sum_{p/2<b<p}\chi(b)\Big)=\chi(2)\big(2pB_{1,\chi}+pA(\chi)\big),$$

because $\sum_{p/2<b<p}\chi(b)=\sum_{b<p/2}\chi(p-b)=-A(\chi)$. Thus $A(\chi)=-(2-\overline{\chi}(2))B_{1,\chi}$, and $\chi(2)|A(\chi)|^2A(\chi)=-(2\chi(2)-1)|2-\chi(2)|^2|B_{1,\chi}|^2B_{1,\chi}$. For the third formula, $(2z-1)|2-z|^2=-4z^2+12z+2\overline{z}-9$ for $|z|=1$. Expanding $|B_{1,\chi}|^2B_{1,\chi}=p^{-3}\sum_{a,b,c}abc\,\chi(ac\,b^{-1})$ and using orthogonality as above, the term $\chi(\lambda)$, $\lambda\in\{4,2,2^*,1\}$, contributes

$$\frac{4}{p-1}\cdot\frac{p-1}{2p^3}\sum_{a,c=1}^{p-1}ac\sum_{b=1}^{p-1}b\,(\delta_{b\equiv\lambda ac}-\delta_{b\equiv-\lambda ac})=\frac{2}{p^3}\sum_{a,c}ac\,\big(2(\lambda ac \bmod p)-p\big)=\frac{4}{p^2}\sum_{a,c}ac((\frac{\lambda ac}{p}))$$

times its coefficient. Finally $\sum_c c((\frac{xc}{p}))=p\,s(x,p)$ for $p\nmid x$ (because $\sum_c((\frac{xc}{p}))=0$), and $\sum_a a\,s(\lambda a,p)=p\sum_a((\frac{a}{p}))s(\lambda a,p)$ (because $\sum_a s(\lambda a,p)=0$, $s(\cdot,p)$ being odd), so $\sum_{a,c}ac((\frac{\lambda ac}{p}))=p^2\sum_a((\frac{a}{p}))s(\lambda a,p)$. □

3. The dual formula and the proof of Theorem A

Here we expand the sign $(-1)^{\lfloor n/p\rfloor}$ into its finite Fourier series modulo $2p$. This turns $S(p)$ into the cotangent sum (3.1), the starting point of everything that follows, and it gives the upper bound of Theorem A in a few lines. That bound exceeds the truth by a factor of order $\log p$, because it replaces every $C_p(k)$ by its largest possible value $C_p(1)$.

Lemma 3.1 (Dual formula)**.** *For odd a with $0 < a < p$,*

$$T(a) = \frac{2}{p} \sum_{\substack{0<k<p \\ k \text{ odd}}} \cot\frac{\pi k}{2p} \cot\frac{\pi k a}{2p},$$

and consequently

$$S(p) = \frac{2}{p} \sum_{\substack{0<k<p \\ k \text{ odd}}} \cot\frac{\pi k}{2p}\, C_p(k). \tag{3.1}$$

Proof. The function $f(n) = (-1)^{\lfloor n/p \rfloor}$ has period $2p$ and satisfies $f(n+p) = -f(n)$. Hence its Fourier coefficients $\hat f(\ell) = \frac{1}{2p}\sum_{n=0}^{2p-1} f(n) e(-\ell n/2p)$ vanish for even ℓ, and for odd ℓ, with $\zeta = e(-\ell/2p)$ (so that $\zeta^p = -1$),

$$\hat f(\ell) = \frac{1}{2p}(1-\zeta^p)\sum_{n=0}^{p-1}\zeta^n = \frac{2}{p(1-\zeta)} = \frac{1}{p}\Big(1 - i\cot\frac{\pi\ell}{2p}\Big); \tag{3.2}$$

for $\ell = p$ this equals $1/p$. Let a be odd, $0 < a < p$. Then $T(a) = \sum_\ell \hat f(\ell) \sum_{b=1}^{p-1} e(\ell a b/2p)$, where ℓ runs over the odd residues modulo $2p$. For $\ell = p$ the inner sum is $\sum_b (-1)^{ab} = \sum_{b=1}^{p-1}(-1)^b = 0$. For odd $\ell \neq p$ put $\vartheta = \ell a/2p$; then $e(p\vartheta) = -1$ because ℓa is odd, $e(\vartheta) \neq 1$ because $p \nmid \ell a$, and

$$\sum_{b=1}^{p-1} e(b\vartheta) = \frac{1 - e(p\vartheta)}{1 - e(\vartheta)} - 1 = \frac{1 + e(\vartheta)}{1 - e(\vartheta)} = i\cot(\pi\vartheta).$$

Hence

$$T(a) = \frac{1}{p} \sum_{\substack{0<\ell<2p,\ \ell\neq p \\ \ell \text{ odd}}} \Big(\cot\frac{\pi\ell}{2p}\cot\frac{\pi\ell a}{2p} + i\cot\frac{\pi\ell a}{2p}\Big).$$

Under $\ell \mapsto 2p-\ell$ the function $\cot(\pi\ell a/2p)$ changes sign (because a is an integer), so the imaginary part vanishes, while the product of the two cotangents is invariant. This gives the formula for $T(a)$, and (3.1) follows by summing over odd a and using Lemma 2.1(b). □

Lemma 3.2 (Sign form)**.** *Let k be odd with $p \nmid k$. Then*

$$C_p(k) = \sum_{\substack{0<n<p \\ n \text{ odd}}} \sigma(n\overline{k}) \cot\frac{\pi n}{2p}. \tag{3.3}$$

In particular $|C_p(k)| \le C_p(1)$.

Proof. For odd n with $0 < n < p$ let $w_n \in (0, 2p)$ be the residue of $n\overline{k}$ modulo $2p$; it is odd and different from p. Put $a_n = w_n$ if $w_n < p$ and $a_n = 2p - w_n$ otherwise. Then a_n is odd, $0 < a_n < p$, and $k a_n \equiv \sigma(n\overline{k})\, n \pmod{2p}$, so that $\cot(\pi k a_n/2p) = \sigma(n\overline{k})\cot(\pi n/2p)$. The map $n \mapsto a_n$ is a bijection of the set of odd integers in $(0,p)$ onto itself; its inverse sends a to the element of $\{ka \bmod 2p,\ \ 2p - (ka \bmod 2p)\}$ that lies in $(0,p)$. Summing over n gives (3.3), and the bound follows since $\cot(\pi n/2p) > 0$ for $0 < n < p$. □

Combining (3.1) and (3.3) gives the symmetric formula (1.9).

Lemma 3.3. *For odd $m \ge 3$, $C_m(1) \le \frac{m}{\pi}(\log m + 2)$.*

Proof. We have $\cot y \le 1/y$ for $0 < y \le \pi/2$, and $\sum_{a \text{ odd}, 3 \le a \le N} 1/a \le \frac12 \log N$ because $1/a \le \frac12 \int_{a-2}^{a} dx/x$. Hence $C_m(1) \le \frac{2m}{\pi}\big(1 + \frac12 \log m\big)$. □

Proof of Theorem A. By (3.1), Lemma 3.2 and the positivity of $\cot(\pi k/2p)$ for $0<k<p$,

$$|S(p)| \le \frac{2}{p}\sum_{\substack{0<k<p\\ k \text{ odd}}} \cot\frac{\pi k}{2p}\, C_p(1) = \frac{2}{p}\, C_p(1)^2.$$

By Corollary 4.3(a) below, $C_p(1) \le \frac{p}{\pi}(\log p + K_0)$, which gives the second inequality. (Lemma 3.3 gives, in an elementary way, the slightly weaker bound with K_0 replaced by 2.) The last assertion follows since $(2/\pi^2)(\log p + K_0)^2 < C_2 p^{\varepsilon}$ for large p. □

Remark 3.4. For $p = 7$ the ratio of $|S(p)|$ to the second bound of Theorem A is 0.92, its maximum over all $p \le 6\cdot 10^6$. For large p the bound exceeds the true size $\frac{p}{2}\log p$ of $S(p)$ by a factor of order $\log p$.

4. A reciprocity law for half-range cotangent sums

The sums $C_p(k)$ in (3.1) have length about $p/2$, like the original problem. The reciprocity law of this section exchanges the roles of p and k: it writes $C_p(k)$ as $\frac{p}{k}F_k(p)$, where $F_k(p)$ is a tangent sum of length $(k-1)/2$, plus an explicit term $I(k/p)$. The terms $I(k/p)$ will produce the main term of $S(p)$, and the tangent sums its fluctuations. We prove the law by integrating a product of a tangent and a cotangent around a rectangle. We then derive the asymptotics of I and the trivial bound $|F_k(n)| \le F_k(1)$, and Subsection 4.1 iterates the law.

Lemma 4.1 (Reciprocity)**.** *Let m and k be coprime odd positive integers. Then*

$$C_m(k) = \frac{m}{k}\, F_k(m) + I\Big(\frac{k}{m}\Big).$$

Proof. Consider the meromorphic function $f(z) = -\frac{\pi}{2}\tan\frac{\pi z}{2}\cot\frac{\pi k z}{2m}$. In the strip $0<\operatorname{Re} z<m$ its poles are the following. (i) The odd integers $a \in (0,m)$, where $\tan(\pi z/2)$ has a simple pole with residue $-2/\pi$ and $\cot(\pi k a/2m)$ is finite because ka is odd; the residue of f is $\cot(\pi k a/2m)$. (ii) The points $z_n = 2mn/k$, $1 \le n \le (k-1)/2$, where $\cot(\pi k z/2m)$ has a simple pole with residue $2m/(\pi k)$; these points are not integers, since $k \nmid 2mn$, so $\tan(\pi z_n/2) = \tan(\pi mn/k)$ is finite, and the residue of f is $-\frac{m}{k}\tan(\pi mn/k)$. Hence the sum of the residues in the strip is $C_m(k) - \frac{m}{k}F_k(m)$.

On the lines $\operatorname{Re} z = 0$ and $\operatorname{Re} z = m$ the function f is analytic: at $z=0$ a simple zero of $\tan(\pi z/2)$ cancels a simple pole of $\cot(\pi k z/2m)$, at $z=m$ a simple pole of $\tan(\pi z/2)$ (m odd) cancels a simple zero of $\cot(\pi k z/2m)$ (k odd), and there are no other singularities on these lines. For real $y \ne 0$,

$$f(iy) = -\frac{\pi}{2}\tanh\frac{\pi y}{2}\coth\frac{\pi k y}{2m}, \qquad f(m+iy) = -\frac{\pi}{2}\coth\frac{\pi y}{2}\tanh\frac{\pi k y}{2m}.$$

As $|\operatorname{Im} z| \to \infty$ we have $\tan(\pi z/2) \to \pm i$ and $\cot(\pi k z/2m) \to \mp i$, uniformly in $\operatorname{Re} z$ and with exponentially small error, so $f(z) \to -\pi/2$. Integrate f over the boundary of the rectangle with vertices $-iT$, $m-iT$, $m+iT$, iT. The contributions of the two horizontal sides tend to $-\frac{\pi m}{2}$ and $+\frac{\pi m}{2}$ as $T\to\infty$, and the vertical sides give $i\int_{-T}^{T}(f(m+iy)-f(iy))\,dy$. The integrand decays exponentially, and we obtain

$$C_m(k) - \frac{m}{k}F_k(m) = \frac{1}{2\pi i}\cdot i\cdot\frac{\pi}{2}\int_{\mathbb{R}}\Big(\tanh\frac{\pi y}{2}\coth\frac{\pi k y}{2m} - \coth\frac{\pi y}{2}\tanh\frac{\pi k y}{2m}\Big)\,dy.$$

With $\lambda = k/m$ and the identity

$$\tanh u\coth\lambda u - \coth u\tanh\lambda u = \frac{2(\cosh 2u - \cosh 2\lambda u)}{\sinh 2u\,\sinh 2\lambda u}, \tag{4.1}$$

the substitution $v = \pi y$ turns the right-hand side into $I(\lambda)$. □

Lemma 4.2. *For $\lambda > 0$ write $\pi\lambda I(\lambda) = \log\frac{1}{\lambda} + K_0 - \rho(\lambda)$. Then $0 \le \rho(\lambda) \le \frac{7\pi^2}{72}\lambda^2$. Moreover $I(1/\lambda) = -\lambda I(\lambda)$.*

Proof. By (4.1) and the substitution $v = 2u$, $\pi I(\lambda) = \int_0^\infty (\tanh u \coth \lambda u - \coth u \tanh \lambda u)\, du$. Write $\coth x = \frac{1}{x} + \kappa(x)$, where $0 \le \kappa(x) \le x/3$ for $x > 0$ (from $\kappa(x) = \sum_{n\ge1} \frac{2x}{x^2+\pi^2n^2}$). Using $\coth x - \tanh x = 2/\sinh 2x$ one checks that

$$\tanh u \coth \lambda u - \coth u \tanh \lambda u = \frac{\phi(2\lambda u) - \psi(2u)}{\lambda u} - (1 - \tanh u)\,\kappa(\lambda u) - (\coth u - 1)\tanh \lambda u,$$

where $\phi(x) = x/\sinh x$ and $\psi(x) = 2/(e^x+1) = 1 - \tanh(x/2)$. All three terms are integrable on $(0,\infty)$, and the last two are nonnegative. By Frullani's theorem (ϕ is continuous, $\phi(0) = 1$, $\phi(\infty) = 0$),

$$\int_0^\infty \big(\phi(2\lambda u) - \psi(2u)\big)\frac{du}{u} = \log\frac{1}{\lambda} + \int_0^\infty \big(\phi(x) - \psi(x)\big)\frac{dx}{x}.$$

For $\operatorname{Re} s > 0$ we have the Mellin transforms $\int_0^\infty \phi(x)x^{s-1}dx = 2\Gamma(s+1)(1-2^{-s-1})\zeta(s+1) = \frac{1}{s} + \log 2 + O(s)$ and $\int_0^\infty \psi(x)x^{s-1}dx = 2\Gamma(s)(1-2^{1-s})\zeta(s) = \frac{1}{s} + \log\frac{\pi}{2} - \gamma + O(s)$ (expand the integrands in powers of e^{-x}; for the second expansion use $\zeta(0) = -\frac12$ and $\zeta'(0) = -\frac12\log 2\pi$). Since $\phi - \psi = O(x)$ near 0, letting $s \to 0^+$ gives $\int_0^\infty (\phi - \psi)\, dx/x = \log 2 - \log\frac{\pi}{2} + \gamma = K_0$. Hence $\pi\lambda I(\lambda) = \log\frac{1}{\lambda} + K_0 - \lambda(J_1 + J_2)$, where

$$0 \le J_1 = \int_0^\infty (1 - \tanh u)\kappa(\lambda u)\, du \le \frac{\lambda}{3}\int_0^\infty u(1-\tanh u)\, du = \frac{\pi^2\lambda}{72},$$

$$0 \le J_2 = \int_0^\infty (\coth u - 1)\tanh(\lambda u)\, du \le \lambda\int_0^\infty u(\coth u - 1)\, du = \frac{\pi^2\lambda}{12};$$

here we used $\tanh x \le x$, $1 - \tanh u = 2\sum_{n\ge1}(-1)^{n-1}e^{-2nu}$ and $\coth u - 1 = 2\sum_{n\ge1} e^{-2nu}$. Thus $0 \le \rho(\lambda) \le (\frac{\pi^2}{72} + \frac{\pi^2}{12})\lambda^2 = \frac{7\pi^2}{72}\lambda^2$. The relation $I(1/\lambda) = -\lambda I(\lambda)$ follows from the substitution $v \mapsto \lambda v$ in (1.5). □

The constant $\frac{7\pi^2}{72}$ is optimal as $\lambda \to 0$: by dominated convergence, J_1/λ and J_2/λ tend to the two bounds $\frac{\pi^2}{72}$ and $\frac{\pi^2}{12}$.

Corollary 4.3. (a) *For odd $m \ge 1$ we have $F_m(1) = C_m(1) = I(1/m)$ and*

$$\frac{m}{\pi}(\log m + K_0) - \frac{7\pi}{72m} \le C_m(1) \le \frac{m}{\pi}(\log m + K_0).$$

(b) *For odd $k \ge 3$ and every integer r, $|F_k(r)| \le F_k(1) \le \frac{k}{\pi}(\log k + K_0)$; moreover $F_k(-1) = -F_k(1)$, and the constant K_0 is optimal: $\pi F_k(1)/k - \log k \to K_0$.*

(c) *For $0 < \lambda \le 1$, $|I(\lambda)| \le \frac{1}{\pi\lambda}\big(\log\frac{1}{\lambda} + 2\big)$.*

Proof. (a) The substitution $a = m - 2t$ gives

$$F_m(1) = \sum_{t<m/2} \tan\frac{\pi t}{m} = \sum_{\substack{a<m\\ a\ \text{odd}}} \cot\frac{\pi a}{2m} = C_m(1),$$

and Lemma 4.1 with $k = 1$ gives $C_m(1) = mF_1(m) + I(1/m) = I(1/m)$. Now apply Lemma 4.2 with $\lambda = 1/m$.

(b) If $(r,k) = 1$, then as t runs over $1,\dots,(k-1)/2$ the residues $\pm rt$ run over $1,\dots,k-1$, and $\tan(\pi(k-x)/k) = -\tan(\pi x/k)$; hence $F_k(r) = \sum_{x=1}^{(k-1)/2} \pm\tan(\pi x/k)$ and $|F_k(r)| \le F_k(1)$. If $g = (r,k) > 1$, put $k' = k/g$ and $r' = r/g$. The function $t \mapsto \tan(\pi r't/k')$ has period k' and is odd, so its sum over a complete period vanishes. Since $(k-1)/2 = k'\frac{g-1}{2} + \frac{k'-1}{2}$, the sum over $1 \le t \le (k-1)/2$ consists of $\frac{g-1}{2}$ complete periods and of the sum over $1 \le t \le (k'-1)/2$; thus

$F_k(r) = F_{k'}(r')$, and $|F_k(r)| \le F_{k'}(1) \le F_k(1)$ by the previous case and by (a) (note $k \ge 3k'$). The bound for $F_k(1)$ and the optimality of K_0 follow from (a).

(c) By Lemma 4.2, $|\pi\lambda I(\lambda)| \le \log\frac{1}{\lambda} + K_0 + \frac{7\pi^2}{72} < \log\frac{1}{\lambda} + 2$. □

4.1. **An even continued fraction expansion of F_k.** Iterating Lemma 4.1 expresses $F_k(n)/k$ exactly as a sum over the steps of a Euclidean-type algorithm. This section explains the size and the singularities of $F_k(n)$; it is not needed for the proof of Theorem C, which uses the cruder decomposition of Section 6.

Lemma 4.4 (Odd reciprocity)**.** *Let k and m be coprime odd positive integers. Then $C_m(k) = F_m(k)$ and*

$$\frac{F_k(m)}{k} = \frac{F_m(k)}{m} + \frac{1}{k} I\Big(\frac{m}{k}\Big).$$

Proof. Write the odd $a \in (0,m)$ as $a = m-2t$ with $1 \le t \le \frac{m-1}{2}$. Since k is odd and cot has period π, $\cot\frac{\pi k(m-2t)}{2m} = \cot(\frac{\pi}{2} - \frac{\pi kt}{m}) = \tan\frac{\pi kt}{m}$; summing over t gives $C_m(k) = F_m(k)$. By Lemma 4.1, $F_m(k) = \frac{m}{k}F_k(m) + I(\frac{k}{m})$; divide by m and use $I(\frac{k}{m}) = -\frac{m}{k}I(\frac{m}{k})$ (Lemma 4.2). □

Proposition 4.5 (Expansion)**.** *Let $k \ge 3$ be odd and $\gcd(n,k) = 1$. Put $n_{-1} = n$, $n_0 = k$, and while $n_i > 1$ let $\rho_i \in [1, n_i - 1]$ be the residue of n_{i-1} modulo n_i and*

$$(n_{i+1}, s_{i+1}, b_i) = \begin{cases} (\rho_i,\ +1,\ \lfloor n_{i-1}/n_i \rfloor) & \text{if } \rho_i \text{ is odd,} \\ (n_i - \rho_i,\ -1,\ \lfloor n_{i-1}/n_i \rfloor + 1) & \text{if } \rho_i \text{ is even.} \end{cases}$$

Then $n_{i-1} = b_i n_i + s_{i+1} n_{i+1}$, all n_i $(i \ge 0)$ are odd, $\gcd(n_i, n_{i+1}) = 1$, $k = n_0 > n_1 > \cdots > n_J = 1$, and with $\varepsilon_i = s_1 s_2 \cdots s_{i+1}$

$$\frac{F_k(n)}{k} = \sum_{i=0}^{J-1} \varepsilon_i J_i, \qquad J_i = \frac{1}{n_i} I\Big(\frac{n_{i+1}}{n_i}\Big) > 0.$$

Proof. Since $\gcd(n_i, \rho_i) = \gcd(n_i, n_{i-1}) = 1$, we have $\rho_i \ne 0$ while $n_i > 1$; n_{i+1} is odd because n_i is, and $n_{i+1} < n_i$. As F_{n_i} is odd and has period n_i, $F_{n_i}(n_{i-1}) = F_{n_i}(\rho_i) = s_{i+1} F_{n_i}(n_{i+1})$, and Lemma 4.4 gives $F_{n_i}(n_{i+1})/n_i = F_{n_{i+1}}(n_i)/n_{i+1} + J_i$. Starting from $F_{n_0}(n_{-1})/n_0 = F_k(n)/k$ and iterating until $F_{n_J} = F_1 = 0$ gives the formula. That $J_i > 0$ follows from Lemma 4.6(a) below. □

For $i \ge 1$ the partial quotients b_i are even (Proposition 4.8(b)), so this is an *even continued fraction*, the kind that Lägeler [18] related to the sums $S(h,k)$.

Lemma 4.6. *For $0 < \lambda < 1$: (a) $0 < I(\lambda) \le \frac{\pi}{8}(1-\lambda)(1+\lambda^{-2})$; (b) $I(\lambda) \le \frac{1}{\pi\lambda}(\log\frac{1}{\lambda} + K_0)$. Consequently, in Proposition 4.5, $J_i \le \frac{\log(n_i/n_{i+1}) + K_0}{\pi n_{i+1}}$ for every step, and $J_i \le \frac{5\pi}{8}\big(\frac{1}{n_{i+1}} - \frac{1}{n_i}\big)$ if $n_{i+1} \ge \frac{1}{2}n_i$.*

Proof. (a) Since $\cosh v - \cosh\lambda v = 2\sinh\frac{(1+\lambda)v}{2}\sinh\frac{(1-\lambda)v}{2}$, the integrand in (1.5) is positive. Using $\sinh x \le x\cosh x$ for $x \ge 0$ and $2\sinh\frac{(1+\lambda)v}{2}\cosh\frac{(1-\lambda)v}{2} = \sinh v + \sinh\lambda v$,

$$I(\lambda) \le \frac{1-\lambda}{2\pi}\int_0^\infty v\Big(\frac{1}{\sinh\lambda v} + \frac{1}{\sinh v}\Big)dv = \frac{1-\lambda}{2\pi}\cdot\frac{\pi^2}{4}(\lambda^{-2}+1).$$

(b) is Lemma 4.2 with $\rho \ge 0$. For the consequences put $\lambda = n_{i+1}/n_i$; if $\lambda \ge \frac{1}{2}$ then $1 + \lambda^{-2} \le 5$ and $\frac{1-\lambda}{n_i} = \frac{n_i - n_{i+1}}{n_i^2} \le \frac{1}{n_{i+1}} - \frac{1}{n_i}$. □

Call a step i (V,T)-*singular* if $n_{i+1} \le V$ and $n_i \ge T n_{i+1}$.

Proposition 4.7 (Bounded part)**.** *Let $k \geq 3$ be odd, $\gcd(n,k)=1$, $T \geq 2$ and $V \geq 1$. If $\mathcal{I}$ is a set of steps none of which is (V,T)-singular, then*

$$\sum_{i \in \mathcal{I}} J_i \leq \frac{2}{\pi}\log T + 2.49 + \frac{2}{\pi}\cdot\frac{\log k + K_0}{V}.$$

In particular $|F_k(n)/k|$ is bounded by the right-hand side if no step is (V,T)-singular.

Proof. By Lemma 4.6, the steps with $n_{i+1} \geq \frac{1}{2}n_i$ contribute at most $\frac{5\pi}{8}\sum_i(\frac{1}{n_{i+1}}-\frac{1}{n_i}) < \frac{5\pi}{8}$, however long the runs of such steps are. If $i<j$ are steps with $n_{i+1} < \frac{1}{2}n_i$ and $n_{j+1} < \frac{1}{2}n_j$, then $n_{j+1} < \frac{1}{2}n_j \leq \frac{1}{2}n_{i+1}$; so the values $w = n_{i+1}$ over these steps decrease by a factor larger than 2 from one to the next, and $\sum_{w>W}\frac{1}{w} < \frac{2}{W}$ for every $W>0$. The steps with $n_{i+1} > V$ therefore contribute at most $\frac{2(\log k+K_0)}{\pi V}$, and those with $n_{i+1} \leq V$, for which $n_i < Tn_{i+1}$, at most $\frac{2}{\pi}(\log T + K_0)$. Finally $\frac{5\pi}{8} + \frac{2K_0}{\pi} < 2.49$. □

To identify the singular steps define integers c_i, d_i $(i \geq -1)$ by $c_{-1}=1$, $c_0=0$, $d_{-1}=0$, $d_0=-1$ and $c_{i+1} = s_{i+1}(c_{i-1}-b_ic_i)$, $d_{i+1} = s_{i+1}(d_{i-1}-b_id_i)$. Since $n_{i+1} = s_{i+1}(n_{i-1}-b_in_i)$, induction gives

$$n_i = c_i n - d_i k \qquad (i \geq -1), \tag{4.2}$$

and $c_id_{i+1} - c_{i+1}d_i = \pm 1$, because it equals -1 for $i=-1$ and is multiplied by $-s_{i+1}$ at each step.

Proposition 4.8 (Steps and linear forms)**.** *For every step i $(0 \leq i < J)$:*
(a) *$k \mid c_{i+1}n - n_{i+1}$, and $\gcd\big(c_{i+1}, (c_{i+1}n-n_{i+1})/k\big) = 1$;*
(b) *if $i \geq 1$ then b_i is even and $b_i \geq 2$; moreover $|c_1| = 1 < |c_2| < |c_3| < \cdots$ and $c_i \equiv i \pmod 2$;*
(c) *$k = n_i|c_{i+1}| + s_{i+1}n_{i+1}|c_i|$, hence $\frac{k}{n_i+n_{i+1}} \leq |c_{i+1}| \leq \frac{k}{n_i-n_{i+1}}$;*
(d) (sign law) *$\varepsilon_i = -(-1)^{c_{i+1}}\operatorname{sgn}(c_{i+1})$.*

Proof. (a) is (4.2) together with $\gcd(c_{i+1},d_{i+1})=1$. (b) For $i \geq 1$ both n_{i-1} and n_i are odd, so $\rho_i \equiv 1 - \lfloor n_{i-1}/n_i \rfloor \pmod 2$ and in both cases b_i is even; and $b_i \geq 1$ because $n_{i-1} > n_i$. Next $c_1 = s_1$, and if $|c_i| > |c_{i-1}|$ then $|c_{i+1}| \geq b_i|c_i| - |c_{i-1}| > |c_i|$. The parity follows from $c_{i+1} \equiv c_{i-1} \pmod 2$. (c) $\Delta_i = n_ic_{i+1} - n_{i+1}c_i$ satisfies $\Delta_{-1} = -k$ and $\Delta_i = -s_{i+1}\Delta_{i-1}$, so $|\Delta_i| = k$. For $i \geq 1$ we have $b_i|c_i| > |c_{i-1}|$, hence $\operatorname{sgn} c_{i+1} = -s_{i+1}\operatorname{sgn} c_i$; so c_i and c_{i+1} have opposite signs if $s_{i+1}=1$ and equal signs if $s_{i+1}=-1$, and in the latter case $n_i|c_{i+1}| > n_{i+1}|c_i|$. This gives the identity for $i \geq 1$, and for $i=0$ it reads $k = n_0$. The bounds follow from $0 \leq |c_i| < |c_{i+1}|$. (d) We have $\operatorname{sgn} c_1 = s_1 = \varepsilon_0$ and $\operatorname{sgn} c_{i+1} = -s_{i+1}\operatorname{sgn} c_i$ for $i \geq 1$, so $\operatorname{sgn} c_{i+1} = (-1)^i\varepsilon_i$; and $(-1)^{c_{i+1}} = (-1)^{i+1}$ by (b). □

Proposition 4.9 (Converse)**.** *Let $m \geq 1$ be odd and $r \neq 0$, and suppose that $k \mid rn-m$, $\gcd(r,(rn-m)/k)=1$ and $|r|m < k/2$. Then there is exactly one step i with $(n_{i+1},c_{i+1}) = (m,r)$.*

Proof. Let $\rho'_{-1}=n$, $\rho'_0=k$, $\rho'_{j-1} = q_j\rho'_j + \rho'_{j+1}$ $(0 \leq \rho'_{j+1} < \rho'_j)$ be the ordinary Euclidean algorithm, ending with $\rho'_t = 1$, and define $C_{-1}=1$, $C_0=0$, $C_{j+1} = C_{j-1} - q_jC_j$, so that $\rho'_j \equiv C_jn \pmod k$ and $k = \rho'_{j-1}|C_j| + \rho'_j|C_{j-1}|$ (the argument of Proposition 4.8(c)); in particular $|C_j| \leq k/2$ for $j \geq 1$.

We first show that the algorithm of Proposition 4.5 produces every odd ρ'_j, $j \geq 1$, as some n_{i+1}. Say that it is *at state j* if $(n_{i-1},n_i) = (\rho'_{j-1} + e\rho'_j, \rho'_j)$ for some i and some $e \in \{0,1\}$; it starts at state 0. At state j with $\rho'_j > 1$ the residue of n_{i-1} modulo n_i is ρ'_{j+1}. If ρ'_{j+1} is odd, the next pair is (ρ'_j, ρ'_{j+1}), i.e. state $j+1$. If ρ'_{j+1} is even, the next pairs are $(\rho'_j - (u-1)\rho'_{j+1}, \rho'_j - u\rho'_{j+1})$ for $u = 1, \ldots, q_{j+1}$: indeed for $u < q_{j+1}$ we have $\rho'_j - u\rho'_{j+1} \geq \rho'_{j+1} + \rho'_{j+2} > \rho'_{j+1}$, so the residue is again the even number ρ'_{j+1}. The last of these pairs is $(\rho'_{j+1} + \rho'_{j+2}, \rho'_{j+2})$, i.e. state $j+2$,

and $\rho'_{j+2} \equiv \rho'_j$ is odd. Since two consecutive ρ'_j are never both even, every $j \geq 1$ with ρ'_j odd is reached, and then $n_{i'+1} = \rho'_j$ for a step i'. Comparing (4.2) with $\rho'_j \equiv C_j n$ gives $c_{i'+1} \equiv C_j \pmod k$; as $|c_{i'+1}| \leq k/(n_{i'} - n_{i'+1}) \leq k/2$ (odd numbers differ by at least 2), $|C_j| \leq k/2$ and k is odd, $c_{i'+1} = C_j$.

Now write $rn - m = dk$. Then $\gcd(r, d) = 1$ and $|\frac{n}{k} - \frac{d}{r}| = \frac{m}{|r|k} < \frac{1}{2r^2}$, so by Legendre's theorem d/r is a convergent P_l/Q_l of n/k (the additional convergent of the second finite expansion of n/k has denominator at least $k/2$, which is excluded by $|r| < k/(2m)$). Hence $m = |Q_l n - P_l k|$ is one of the remainders ρ'_j, $j \geq 1$; it is odd, so it equals n_{i+1} for a step i, and $c_{i+1} = C_j \equiv r \pmod k$. Both have absolute value less than $k/2$, so $c_{i+1} = r$. Uniqueness holds because the n_i are distinct. □

Theorem 4.10 (Explicit formula)**.** *Let $k \geq 3$ be odd, $\gcd(n, k) = 1$, $V \geq 1$ and $T \geq 4$, and let $\mathcal{W}$ be the set of pairs (m, r) with $1 \leq m \leq V$ odd, $r \neq 0$, $k \mid rn - m$, $\gcd(r, (rn - m)/k) = 1$ and $|r|m \leq k/T$. Then*

$$\frac{F_k(n)}{k} = -\frac{1}{\pi} \sum_{(m,r)\in\mathcal{W}} (-1)^r \operatorname{sgn}(r) \frac{\log \frac{k}{|r|m} + K_0}{m} \; + \; E,$$

where

$$|E| \leq \frac{2}{\pi} \log(T + 1) + 2.49 + \frac{2}{T} + \frac{2}{\pi} \cdot \frac{\log k + K_0}{V}.$$

Proof. By Proposition 4.9 each $(m, r) \in \mathcal{W}$ equals (n_{i+1}, c_{i+1}) for exactly one step i, and its sign is $\varepsilon_i = -(-1)^r \operatorname{sgn}(r)$ by Proposition 4.8(d). A step with $n_{i+1} \leq V$ that does not come from $\mathcal{W}$ violates only $|c_{i+1}|n_{i+1} \leq k/T$ (Proposition 4.8(a)), so $k/T < |c_{i+1}|n_{i+1} \leq k n_{i+1}/(n_i - n_{i+1})$ and $n_i < (T + 1)n_{i+1}$; by Proposition 4.7 with $T + 1$ these steps, together with those with $n_{i+1} > V$, contribute at most $\frac{2}{\pi} \log(T + 1) + 2.49 + \frac{2}{\pi} \frac{\log k + K_0}{V}$. For $(m, r) \in \mathcal{W}$ coming from step i, put $x = |r|m/k \leq 1/T \leq \frac{1}{4}$. Proposition 4.8(c) gives $n_i|r|/k \in [1 - x, 1 + x]$, so $|\log \frac{n_i|r|}{k}| \leq 2x$, and $m/n_i \leq \frac{4}{3}x$, so that $\rho(m/n_i) \leq \frac{7\pi^2}{72}(\frac{4}{3}x)^2 \leq 1.71x^2$. By Lemma 4.2, $J_i = \frac{\log(n_i/m) + K_0 - \rho(m/n_i)}{\pi m} = \frac{\log \frac{k}{|r|m} + K_0}{\pi m} + \vartheta \frac{2x + 1.71x^2}{\pi m}$ with $|\vartheta| \leq 1$. Finally the steps from $\mathcal{W}$ have $n_i \geq 3n_{i+1}$, so their values of m decrease geometrically and $\sum_{\mathcal{W}} \frac{1}{m} \leq \frac{3}{2}$; the total error is at most $\frac{3}{2\pi}(\frac{2}{T} + \frac{1.71}{T^2}) < \frac{2}{T}$. □

Corollary 4.11 (Singularities of $F_k(p)$)**.** *Let $p > k$ be prime, let $u_k \in [1, k)$ satisfy $2pu_k \equiv 1 \pmod k$, and put $\theta_k = u_k/k = ((2p)^{-1} \bmod k)/k$. Then $k \mid pr - m$ if and only if $2m\theta_k \equiv r/k \pmod 1$. If $k \mid ap + c$ with $a \geq 1$, c odd, $\gcd(a, c) = 1$ and $a|c| \leq k/T$, then the sum in Theorem 4.10 (with $n = p$, $V \geq |c|$) contains the term*

$$\operatorname{sgn}\big((-1)^a c\big) \frac{\log \frac{k}{a|c|} + K_0}{\pi|c|},$$

which is negative if and only if $(-1)^a c < 0$.

Proof. Since $p^{-1} \equiv 2u_k \pmod k$, $pr \equiv m$ is equivalent to $r \equiv 2mu_k \pmod k$. If $k \mid ap + c$, put $(m, r) = (|c|, -a \operatorname{sgn} c)$; then $k \mid rp - m$, $\gcd(r, (rp - m)/k) = \gcd(a, (ap + c)/k) = 1$ (as $\gcd(a, c) = 1$), and $-(-1)^r \operatorname{sgn}(r) = (-1)^a \operatorname{sgn} c = \operatorname{sgn}((-1)^a c)$. □

For example, if $k \mid p + 1$ then $F_k(p) = F_k(-1) = -F_k(1)$, the most negative value possible (Corollary 4.3); if $k \mid 2p - 1$ and $k \geq 9$, then for $T = 4$ and $V = k$ the conditions $r \equiv 2m \pmod k$ and $|r| \leq k/(4m)$ force $r = 2m$, and $\gcd(2m, m(2p - 1)/k) \geq m$ forces $m = 1$; so $\mathcal{W} = \{(1, 2)\}$ and $F_k(p)/k = -\frac{1}{\pi} \log k + O(1)$ as well. In the same way $k \mid 2p + 1$ gives $\mathcal{W} = \{(1, -2)\}$ and $F_k(p)/k = \frac{1}{\pi} \log k + O(1)$, and $k \mid p - 1$ gives $F_k(p) = F_k(1)$. Numerically, for the primes $10^6 < p \leq 4 \cdot 10^6$ with the most negative values of $S(p)/p - \frac{1}{2} \log p$, the deficit is carried mainly

by moduli dividing $p+1$ and $2p-1$, and by the absence of the positive contributions of moduli dividing $p-1$ and $2p+1$ (Subsection 10.5).

5. The exact decomposition and the proof of Theorem B

We now insert the reciprocity law into the dual formula (3.1). This gives the exact identity $S(p)=\Psi(p)+\Phi(p)$ of Theorem B(i), and the evaluation of the smooth part $\Psi(p)$ produces the constant c_0. For Theorem B(ii) we show that the tangent sums with moduli $k\ge\sqrt{p}(\log p)^4$ contribute only $O(p/\log p)$. A single $F_k(p)$ can be as large as $\frac{k}{\pi}(\log k+K_0)$ in absolute value, so we must use cancellation between different k: summed over k in an interval, the sums $C_p(k)$ become incomplete Kloosterman sums, and Weil's bound applies (Lemmas 5.2 and 5.3).

Proof of Theorem B(i). By Lemma 4.1 with $m=p$ we have $C_p(k)=\frac{p}{k}F_k(p)+I(k/p)$ for odd $k<p$. Inserting this into (3.1) gives $S(p)=\Psi(p)+\Phi(p)$; the term $k=1$ does not contribute to $\Phi(p)$ because $F_1=0$.

Since $\cot\frac{\pi k}{2p}=\frac{2p}{\pi k}h(\frac{k}{p})$, Lemma 4.2 gives

$$\Psi(p)=\frac{4p}{\pi^2}\sum_{\substack{0<k<p\\ k\text{ odd}}}\frac{h(k/p)}{k^2}\Big(\log\frac{p}{k}+K_0-\rho\Big(\frac{k}{p}\Big)\Big).$$

Since $0\le 1-h(x)\le x^2$ and $0\le\rho(x)\le\frac{7\pi^2}{72}x^2$, replacing $h(k/p)(\log\frac{p}{k}+K_0-\rho(\frac{k}{p}))$ by $\log\frac{p}{k}+K_0$ changes the k-th term by at most $(k/p)^2(\log\frac{p}{k}+2)$, so the total change is at most $\frac{4}{\pi^2 p}\sum_{k<p}(\log\frac{p}{k}+2)\le\frac{12}{\pi^2}$, because $\sum_{k\le p}\log\frac{p}{k}=p\log p-\log p!\le p$. Extending the sum to all odd $k\ge1$ costs at most $\frac{4p}{\pi^2}\sum_{k\ge p}(\log\frac{k}{p}+K_0)k^{-2}=O(1)$. Hence

$$\Psi(p)=\frac{4p}{\pi^2}\Big(\frac{\pi^2}{8}(\log p+K_0)-\mathcal{L}\Big)+O(1),\qquad \mathcal{L}=\sum_{k\text{ odd}}\frac{\log k}{k^2},$$

where we used $\sum_{k\text{ odd}}k^{-2}=\pi^2/8$. Differentiating $\sum_{k\text{ odd}}k^{-s}=(1-2^{-s})\zeta(s)$ at $s=2$ gives $\mathcal{L}=-\frac{\log 2}{4}\zeta(2)-\frac{3}{4}\zeta'(2)$, and therefore $\Psi(p)=\frac{p}{2}\log p+c_0p+O(1)$ with c_0 as in (1.6). □

Remark 5.1 (The complete expansion of $\Psi(p)$)**.** Put $g(\lambda)=\cot\frac{\pi\lambda}{2}\,I(\lambda)$, so that $\Psi(p)=\frac{2}{p}\sum_{k\text{ odd},\,0<k<p}g(k/p)$. Near $\lambda=0$ one has $\pi\lambda I(\lambda)=\log\frac{1}{\lambda}+K_0-\rho(\lambda)$, where ρ has the (divergent) asymptotic expansion $\rho(\lambda)\sim\sum_{j\ge1}\frac{B_{2j}}{j}\zeta(2j)\,(2^{2j}-2^{1-2j})\lambda^{2j}$ without logarithmic terms (so the coefficient of λ^2 is $\frac{7\pi^2}{72}$, as in Lemma 4.2); near $\lambda=1$, the relation $I(1/\lambda)=-\lambda I(\lambda)$ gives $I(1)=0$ and $g(1)=g'(1)=0$. The nodes k/p, k odd, form a midpoint grid at 0 and a trapezoidal grid at 1, and the generalized Euler–Maclaurin formula for integrands with algebraic–logarithmic endpoint singularities (Navot [21, 22]) gives the asymptotic expansion

$$\Psi(p)-\frac{p}{2}\log p-c_0p\sim c_1+\frac{\log 2}{6p}-\frac{\zeta(3)}{240p^3}+\frac{\pi^2}{60p^4}+\frac{\zeta(5)}{672p^5}-\frac{4\pi^2}{63p^6}-\frac{3\zeta(7)}{2560p^7}+\cdots,$$

in which the coefficient of p^{1-2j} $(j\ge2)$ is $B_{2j}(1-4^{1-j})\zeta(2j-1)/(j(2j-1))$, the coefficient of p^{-2m} $(m\ge2)$ is $2^{2m}B_{2m}g^{(2m-1)}(1)/(2m)!$, and there are no terms p^{-2} or $(\log p)/p$. The constant term is the Hadamard finite part

$$\begin{aligned}c_1=\operatorname{FP}\int_0^1 g(\lambda)\,d\lambda&=\frac{2}{\pi^2}(1-K_0)+\int_0^1\Big(g(\lambda)-\frac{2(\log\frac{1}{\lambda}+K_0)}{\pi^2\lambda^2}\Big)d\lambda\\&=-0.40762\,59135\,14130\,51147\ldots;\end{aligned}$$

we computed it by two independent quadratures, and a blind fit of exact values of $\Psi(p)$ for $101\le p\le16001$ reproduces it to $4\cdot10^{-25}$. We have verified the expansion numerically through

the term p^{-10} (the residual is $5 \cdot 10^{-34}$ at $p = 1009$), and it reproduces Table 1; we state it here without a detailed proof. An integer-relation search at 108 digits found no expression of c_1 in terms of γ, $\log 2$, $\log \pi$, $\zeta(3)/\pi^2$, $\zeta'(2)/\pi^2$, Catalan's constant$/\pi$, $\log \Gamma(\frac{1}{4})$ and the Stieltjes constant γ_1.

p	101	211	401	1009	10007	20011	100003
$\Psi(p) - \frac{p}{2}\log p - c_0 p$	-0.40648	-0.40708	-0.40734	-0.40751	-0.40761	-0.40762	-0.407625

TABLE 1. The second-order term of $\Psi(p)$ (see Remark 5.1). Since $\Psi(p)$ is computed through the exact identity of Theorem B(i), these values also confirm c_0.

Lemma 5.2 (Weil)**.** *Let c be an integer with $p \nmid c$ and let $J \subset [1,p)$ be an interval. Then*

$$\Big| \sum_{\substack{k\in J\\ k \text{ odd}}} e\Big(\frac{c\overline{k}}{2p}\Big)\Big| \le 2\sqrt{p}\,(\log p + 2).$$

Proof. Let $\overline{2} = (p+1)/2$. Since $\frac{1}{2p} = \frac{\overline{2}}{p} - \frac{1}{2}$, we have $e(cx/2p) = (-1)^c e(c\overline{2}x/p)$ for odd x. Applied to the odd residue $x = \overline{k}$, which is $\equiv k^* \pmod p$, this gives $e(c\overline{k}/2p) = (-1)^c e(c\overline{2}k^*/p)$. For $k = 2j+1$ we have $k \equiv 2(j+\overline{2}) \pmod p$, so $k^* \equiv \overline{2}\,(j+\overline{2})^*$. Hence the sum in question equals $\pm\sum_{y\in J'} e(by^*/p)$ with $b = c\,\overline{2}^{\,2} \not\equiv 0 \pmod p$ and $J' = \{j+\overline{2} :\ 2j+1 \in J\}$, a set of at most $(p-1)/2$ consecutive integers none of which is divisible by p. Completing the sum,

$$\sum_{y\in J'} e\Big(\frac{by^*}{p}\Big) = \frac{1}{p}\sum_{\ell \bmod p}\Big(\sum_{y\in J'} e\Big(\frac{\ell y}{p}\Big)\Big)\mathcal{K}(b,-\ell;p), \qquad \mathcal{K}(b,\ell;p) = \sum_{z=1}^{p-1} e\Big(\frac{bz^*+\ell z}{p}\Big).$$

Here $\mathcal{K}(b,0;p) = -1$, and $|\mathcal{K}(b,\ell;p)| \le 2\sqrt{p}$ for $p \nmid \ell$ by Weil's bound [29] (see also [15]). Since $|\sum_{y\in J'} e(\ell y/p)| \le 1/(2\|\ell/p\|)$, the sum is at most

$$\frac{|J'|}{p} + \frac{1}{\sqrt{p}}\sum_{\ell=1}^{p-1}\frac{1}{\|\ell/p\|} \le 1 + 2\sqrt{p}\,(1+\log p) \le 2\sqrt{p}\,(\log p + 2). \qquad \square$$

Lemma 5.3. *For every interval $J \subset [1,p)$,*

$$\Big| \sum_{\substack{k\in J\\ k \text{ odd}}} C_p(k)\Big| \le \frac{4}{\pi^2}\,p^{3/2}(\log p + 2)^3.$$

Proof. Let $\tilde\sigma$ be the function on $\mathbb{Z}/2p\mathbb{Z}$ that equals σ outside $\{0,p\}$ and vanishes at 0 and p. Then $\tilde\sigma = f - \delta_0 + \delta_p$ with f as in the proof of Lemma 3.1, so by (3.2) its Fourier coefficients vanish for even ℓ and equal $\hat f(\ell) - \frac{1}{p} = -\frac{i}{p}\cot\frac{\pi\ell}{2p}$ for odd ℓ; in particular the coefficient at $\ell = p$ vanishes, and the sum of the absolute values of all coefficients is $\frac{2}{p}C_p(1)$. By (3.3),

$$\sum_{\substack{k\in J\\ k \text{ odd}}} C_p(k) = \sum_{\substack{0<n<p\\ n \text{ odd}}} \cot\frac{\pi n}{2p} \sum_{\substack{0<\ell<2p,\ \ell\neq p\\ \ell \text{ odd}}} \hat{\tilde\sigma}(\ell) \sum_{\substack{k\in J\\ k \text{ odd}}} e\Big(\frac{\ell n\overline{k}}{2p}\Big).$$

Since $p \nmid \ell n$, Lemma 5.2 bounds the inner sum by $2\sqrt{p}(\log p + 2)$. Therefore the left-hand side is at most $C_p(1)\cdot\frac{2}{p}C_p(1)\cdot 2\sqrt{p}(\log p+2)$, and Lemma 3.3 gives the claim. $\square$

Proof of Theorem B(ii). Let $\sqrt{p}(\log p)^4 \le K \le p^{9/10}$ and split $\Phi(p) = \Phi_{<K}(p)+\Phi_{\ge K}(p)$ according as $k < K$ or $k \ge K$.

Large k. By Lemma 4.1, $2\cot\frac{\pi k}{2p}\frac{F_k(p)}{k} = \frac{2}{p}\cot\frac{\pi k}{2p}\big(C_p(k) - I(\frac{k}{p})\big)$. By Corollary 4.3(c) and $\cot\frac{\pi k}{2p} \le \frac{2p}{\pi k}$, the terms $I(k/p)$ contribute at most

$$\frac{2}{p}\sum_{k\ge K}\frac{2p}{\pi k}\cdot\frac{p}{\pi k}\Big(\log\frac{p}{k}+2\Big) \le \frac{4p}{\pi^2}(\log p+2)\sum_{k\ge K}\frac{1}{k^2} \ll \frac{p\log p}{K} \ll \frac{\sqrt{p}}{(\log p)^3}.$$

For the terms $C_p(k)$ we cover $[K,p)$ by the intervals $B_j = [2^jK, 2^{j+1}K)\cap[K,p)$, $j \ge 0$. On B_j the weight $\cot(\pi k/2p)$ is positive and decreasing and at most $2p/(\pi 2^jK)$, so Abel summation and Lemma 5.3 give

$$\Big|\sum_{k\in B_j}\cot\frac{\pi k}{2p}\,C_p(k)\Big| \le \frac{2p}{\pi 2^jK}\cdot\frac{4}{\pi^2}\,p^{3/2}(\log p+2)^3.$$

Summing over j and multiplying by $2/p$ we obtain at most $\frac{32}{\pi^3}p^{3/2}(\log p+2)^3/K \ll p/\log p$. Hence $\Phi_{\ge K}(p) \ll p/\log p$.

Small k. For $3 \le k < K$ we have $2\cot\frac{\pi k}{2p}\frac{1}{k} = \frac{4p}{\pi k^2}h(\frac{k}{p})$ and $0 \le 1-h(\frac{k}{p}) \le \frac{k^2}{p^2}$. By Corollary 4.3(b),

$$\Big|\Phi_{<K}(p) - \frac{4p}{\pi}R_K(p)\Big| \le \frac{4}{\pi p}\sum_{\substack{k<K\\ k\text{ odd}}}|F_k(p)| \le \frac{4}{\pi^2 p}\sum_{\substack{k<K\\ k\text{ odd}}}k(\log k+K_0) \ll \frac{K^2\log K}{p} \ll p^{4/5}\log p.$$

Combining these estimates with part (i) proves (1.7).

For the sum over all $k < p$ we may assume that p is large, take $K = \sqrt{p}(\log p)^4$ and show that $\frac{4p}{\pi}\sum_{K\le k<p}F_k(p)/k^2 \ll p/\log p$. By Lemma 4.1 this sum equals $\sum_{K\le k<p}\frac{4}{\pi k}\big(C_p(k)-I(\frac{k}{p})\big)$, and the argument for $\Phi_{\ge K}(p)$ applies verbatim with the decreasing weight $\frac{4}{\pi k}$ in place of $\frac{2}{p}\cot\frac{\pi k}{2p}$. □

6. The tangent sums as functions of θ_k

This section provides the tools for the terms of (1.7) with $k < \sqrt{p}(\log p)^4$: a formula for $F_k(p)/k$ in terms of θ_k alone, bounds for its head on average over k, and bounds for its tail for each k separately. Section 8 combines them.

Lemma 6.1. *Let k be odd with $3 \le k < p$, let $u_k \in [1,k)$ satisfy $2pu_k \equiv 1 \pmod k$, and put $\theta_k = u_k/k$.*

(a) $F_k(p) = \sum_{n\ odd,\,0<n<k}\cot\frac{\pi n}{2k}\,\sigma(n\overline{k})$.

(b) *For odd n with $0 < n < k$ we have $\sigma(n\overline{k}) = -\operatorname{sgn}(\frac12 - \{n\theta_k\})$; hence*

$$F_k(p) = -\sum_{\substack{0<n<k\\ n\ odd}}\cot\frac{\pi n}{2k}\,\operatorname{sgn}\Big(\frac12-\{n\theta_k\}\Big)$$

depends only on θ_k.

(c) *If $\overline{k}\in[1,2p)$, then $\theta_k \equiv \frac{1}{2pk} - \frac{\overline{k}}{2p} \pmod 1$.*

Proof. (a) As t runs over $1,\dots,k-1$, $x = pt \bmod k$ runs over $1,\dots,k-1$; write t_x for the corresponding t, and let $\epsilon(t) = 1$ for $t \le (k-1)/2$ and $\epsilon(t) = -1$ otherwise. Since $\sum_{x=1}^{k-1}\tan(\pi x/k) = 0$, we get $F_k(p) = \sum_x \tan(\pi x/k)\frac{1+\epsilon(t_x)}{2} = \frac12\sum_x\tan(\pi x/k)\epsilon(t_x)$. As $t_{k-x} = k - t_x$, the terms x and $k-x$ coincide, and writing $x = (k-n)/2$ with n odd, $0 < n < k$, we obtain $F_k(p) = \sum_n \cot\frac{\pi n}{2k}\,\epsilon(t_{(k-n)/2})$, where $t_{(k-n)/2} \equiv -nu_k \pmod k$. On the other hand, let $w \in (0,2p)$ with $w \equiv n\overline{k} \pmod{2p}$. Then $kw = n + 2pj$ for an integer j, and $0 < kw < 2pk$, $0 < n < k < p$ give $0 \le j \le k-1$; moreover $j \equiv -nu_k \pmod k$ and $j \ne 0$ since $k \nmid n$. Thus $j = t_{(k-n)/2}$. Finally

$\sigma(n\overline{k}) = 1$ if and only if $w < p$, i.e. $n + 2pj < pk$, i.e. $j < \frac{k}{2} - \frac{n}{2p}$; as $0 < \frac{n}{2p} < \frac{1}{2}$ this means $j \le \frac{k-1}{2}$. Hence $\sigma(n\overline{k}) = \epsilon(j)$, proving (a).

(b) With j as above, $j = k - (nu_k \bmod k) = k(1 - \{n\theta_k\})$, so $j \le \frac{k-1}{2}$ if and only if $\{n\theta_k\} > \frac{1}{2}$.

(c) Since $2pu_k + k\overline{k} \equiv 1$ modulo k and modulo $2p$, we have $2pu_k + k\overline{k} = 1 + 2pkm$ with $m \in \mathbb{Z}$; divide by $2pk$. □

Throughout the rest of the paper we write, for odd k with $3 \le k < p$,

$$g(y) = \operatorname{sgn}\Big(\frac{1}{2} - \{y\}\Big) \ \ (y \notin \tfrac{1}{2}\mathbb{Z}), \qquad c_n(k) = \frac{1}{k}\cot\frac{\pi n}{2k} = \frac{2}{\pi n}\,h\Big(\frac{n}{k}\Big) \ \ (0 < n < k), \qquad \Phi_k = \frac{F_k(p)}{k}.$$

For $p \nmid m$ we have $\sigma(m) = g(m/2p)$, because $\{m/2p\} < \frac{1}{2}$ exactly when $m \bmod 2p \in (0, p)$. By Lemma 6.1,

$$\Phi_k = \sum_{\substack{0<n<k \\ n \text{ odd}}} c_n(k)\,\sigma(n\overline{k}), \qquad \sigma(n\overline{k}) = g\Big(\frac{n\overline{k}}{2p}\Big) = -g(n\theta_k), \tag{6.1}$$

and none of the arguments of g lies in $\frac{1}{2}\mathbb{Z}$ (if $n\theta_k \in \frac{1}{2}\mathbb{Z}$ then $k \mid 2nu_k$, so $k \mid n$). For $N \ge 1$ we split $\Phi_k = \Phi_k^{\le N} + \Phi_k^{>N}$, where $\Phi_k^{\le N}$ is the part of the sum in (6.1) with $n \le N$; thus $\Phi_k^{>N} = 0$ if $k \le N + 1$. Since $0 \le h \le 1$, we have $0 < c_n(k) \le \frac{2}{\pi n}$, and since $\sum_{n \text{ odd}, 3 \le n \le X} \frac{1}{n} \le \frac{1}{2}\log X$,

$$|\Phi_k^{\le N}| \le \frac{2}{\pi}\Big(1 + \frac{1}{2}\log N\Big). \tag{6.2}$$

Lemma 6.2 (Vaaler [27]; see [13, Theorem A.6])**.** *For every integer $H \ge 1$ there are complex numbers a_ℓ, $1 \le |\ell| \le H$, with $|a_\ell| \le \frac{1}{2\pi|\ell|}$, such that for all real x*

$$\Big|((x)) - \sum_{1 \le |\ell| \le H} a_\ell e(\ell x)\Big| \le \Delta_H(x) := \frac{1}{2H+2}\sum_{|\ell| \le H}\Big(1 - \frac{|\ell|}{H+1}\Big)e(\ell x).$$

Note that $\Delta_H \ge 0$ (it is a multiple of the Fejér kernel).

Lemma 6.3 (The square wave)**.** *Let $H \ge 1$. There are b_ℓ $(1 \le |\ell| \le H)$ with $b_\ell = 0$ for even ℓ and $|b_\ell| \le \frac{2}{\pi|\ell|}$ such that, with*

$$T_H(y) = \sum_{1 \le |\ell| \le H} b_\ell e(\ell y), \qquad E_H(y) = \frac{2}{H+1}\sum_{\substack{|\ell| \le H \\ \ell \text{ even}}}\Big(1 - \frac{|\ell|}{H+1}\Big)e(\ell y),$$

we have $E_H \ge 0$ and $|g(y) - T_H(y)| \le E_H(y)$ for all $y \notin \frac{1}{2}\mathbb{Z}$.

Proof. By (2.1), $g(y) = 2((y + \frac{1}{2})) - 2((y))$ for $y \notin \frac{1}{2}\mathbb{Z}$. Applying Lemma 6.2 at y and at $y + \frac{1}{2}$ gives the claim with $b_\ell = 2a_\ell((-1)^\ell - 1)$ and error $2\Delta_H(y) + 2\Delta_H(y + \frac{1}{2}) = E_H(y)$, which is ≥ 0. □

For $K \ge 6$ let

$$\mathcal{M}_K = \max\Big\{\Big|\sum_{\substack{k \in I \\ k \text{ odd}}} e\Big(\frac{c\overline{k}}{2p}\Big)\Big| : \ p \nmid c, \ I \subset [K, 2K) \text{ an interval}\Big\}.$$

Lemma 6.4 (Reduction to initial sums)**.** *Let $6 \le K \le p/2$. Then*

$$\mathcal{M}_K \le 4\max\Big\{\Big|\sum_{1 \le n \le N'} e\Big(\frac{bn^*}{p}\Big)\Big| : \ p \nmid b, \ \frac{K}{2} - 1 \le N' \le 2K\Big\}.$$

Proof. Let $\overline{2} = (p+1)/2$. Since $\frac{1}{2p} = \frac{\overline{2}}{p} - \frac{1}{2}$, we have $e(cx/2p) = (-1)^c e(c\overline{2}x/p)$ for odd x; applied to the odd residue $x = \overline{k} \equiv k^* \pmod p$ this gives $e(c\overline{k}/2p) = (-1)^c e(ak^*/p)$ with $a = c\overline{2}$. For an interval $I = [u,v] \cap \mathbb{Z} \subset [K, 2K)$ we have $\sum_{k\in I,\, k \text{ odd}} e(ak^*/p) = \sum_{k\in I} e(ak^*/p) - \sum_{u/2\le j\le v/2} e(a\overline{2}j^*/p)$, because $(2j)^* = \overline{2}j^*$. Each of the two sums on the right is the difference of two sums $\sum_{n\le N'} e(bn^*/p)$ with $b \in \{a, a\overline{2}\}$ and $N' \in \{v, u-1, \lfloor v/2\rfloor, \lceil u/2\rceil - 1\} \subset [\frac{K}{2}-1, 2K]$. □

Lemma 6.5 (Bourgain–Garaev [8, Theorem 16])**.** *There are absolute constants C_{BG} and A_{BG} such that for every sufficiently large prime p, every $b \not\equiv 0 \pmod p$ and every N with*

$$\exp\bigl(A_{BG}(\log p)^{2/3}(\log\log p)^{8/3}\bigr) \le N < p$$

we have

$$\Bigl|\sum_{1\le n\le N} e\Bigl(\frac{bn^*}{p}\Bigr)\Bigr| \le C_{BG}\,\frac{(\log\log p)^3 \log p}{(\log N)^{3/2}}\,N.$$

Consequently,

$$\frac{\mathcal{M}_K}{K} \le 16\sqrt{2}\,C_{BG}\,\frac{(\log\log p)^3\log p}{(\log K)^{3/2}} \qquad \text{for} \quad \exp\bigl((A_{BG}+1)(\log p)^{2/3}(\log\log p)^{8/3}\bigr) \le K \le \frac{p}{2}.$$

Proof. The first statement is [8, Theorem 16] (the theorem and section numbers refer to the first arXiv version, where the theorem is stated in §2.2 and proved in §12.2). There it reads $\max_{(a,p)=1} |\sum_{n\le N} e_p(an^*)| \ll (\log N)^{-3/2}(\log p)(\log\log p)^3 N$ with an absolute implied constant, p being a large prime and $N < p$; no range of N is stated. The proof puts $\varepsilon = \log N/\log p$ and first disposes of $\varepsilon \ge \frac{4}{7}$ by Weil's bound. For smaller N it works with a parameter β that must satisfy $\beta < (\log\log p)^{-3}$ (p. 55 there), and at the end it chooses $\beta = C\log\log p/(\varepsilon^{3/2}(\log p)^{1/2})$ with a sufficiently large absolute constant C (p. 60 there). This choice satisfies $\beta < (\log\log p)^{-3}$ if and only if

$$\log N > C^{2/3}(\log p)^{2/3}(\log\log p)^{8/3}.$$

So the proof gives the bound in the range stated above, with any constant $A_{BG} > C^{2/3}$. (Below this range the stated bound would still be nontrivial, so it cannot be extended to smaller N by comparison with the trivial bound N; we do not claim that it fails there.) For the second statement, let N' be one of the lengths in Lemma 6.4. In its proof $N' \le v < 2K \le p$, so $N' < p$. As $K \ge 6$ we have $N' \ge \frac{K}{2} - 1 \ge \frac{K}{3}$, so $\log N' \ge \log K - \log 3 \ge A_{BG}(\log p)^{2/3}(\log\log p)^{8/3}$ for large p, and the first statement applies to N'. As $K \ge 9$ we also have $N' \ge \sqrt{K}$, hence $\log N' \ge \frac{1}{2}\log K$; the second statement follows from Lemma 6.4. □

We do not use Korolev's bound [17, Theorem 2], which is claimed for a slightly wider range of N; Garaev and Shparlinski note a gap in the proof of Lemma 2 of [17] [12, Section 1.1].

Lemma 6.6 (Heads)**.** *Let $9 \le K \le p/2$, let $B \subset [K, 2K)$ be an interval, and let H, N be integers with $1 \le H < p$ and $2 \le N < K$. Then*

$$\Bigl|\sum_{\substack{k\in B\\ k \text{ odd}}} \frac{\Phi_k^{\le N}}{k}\Bigr| \le (1+\log N)\Bigl(\frac{2}{H} + 10\,(1+\log H)\,\frac{\mathcal{M}_K}{K}\Bigr).$$

Proof. Put $y_{kn} = n\overline{k}/2p$, $\varphi_n(x) = h(n/x)/x$ and $w_{kn} = c_n(k)/k = \frac{2}{\pi n}\varphi_n(k) > 0$ for odd $k \in B$ and odd $n \le N$ (note $n < K \le k$). By (6.1) the sum to be estimated is $\sum_{k,n} w_{kn} g(y_{kn})$. Let $Z(\ell) = \sum_{k,n} w_{kn} e(\ell y_{kn})$. By Lemma 6.3,

$$\Bigl|\sum_{k,n} w_{kn} g(y_{kn}) - \sum_{\ell} b_\ell Z(\ell)\Bigr| \le \sum_{k,n} w_{kn} E_H(y_{kn}) \le \frac{2}{H+1}\Bigl(Z(0) + \sum_{\substack{0<|\ell|\le H\\ \ell \text{ even}}} |Z(\ell)|\Bigr).$$

Since $\sum_{k\in B}\frac{1}{k}\le 1$, we have $Z(0)\le\frac{2}{\pi}(1+\frac{1}{2}\log N)\le 1+\log N$. For $\ell\neq 0$ we have $p\nmid \ell n$. On $[K,2K]$, $0<\varphi_n\le 1/K$ and $|\varphi_n'(x)|=|\frac{n}{x^3}h'(\frac{n}{x})+\frac{h(n/x)}{x^2}|\le\frac{1+\pi^2/4}{x^2}$ (because $|h'|\le\pi^2/4$ on $[0,1]$), so Abel summation over the odd $k\in B$ gives $|\sum_k\varphi_n(k)e(\ell n\overline{k}/2p)|\le\mathcal{M}_K(\frac{1}{K}+\frac{1+\pi^2/4}{2K})\le 3\mathcal{M}_K/K$, and therefore $|Z(\ell)|\le 3(1+\log N)\mathcal{M}_K/K$. Since $\sum_{\ell\text{ odd},\,|\ell|\le H}|b_\ell|\le\frac{4}{\pi}(1+\frac{1}{2}\log H)$, we obtain the bound $(1+\log N)\big[\frac{2}{H+1}+\big(6+\frac{12}{\pi}+\frac{6}{\pi}\log H\big)\frac{\mathcal{M}_K}{K}\big]$, which implies the lemma. □

Lemma 6.7 (Tails)**.** *Let k be odd with $3\le k<p$ and let $2\le N<k$. For $N\le t<k$ put $G_k(t)=\sum_{n\ odd,\,N<n\le t}g(n\theta_k)$ and $D_k=\sup_{N\le t<k}|G_k(t)|/t$. Then* (a) $|\Phi_k^{>N}|\le\frac{\pi}{2}D_k\log\frac{k}{N}$, *and* (b) $|\Phi_k^{>N}|\le\frac{1}{\pi}\log k$.

Proof. (a) By (6.1), $\Phi_k^{>N}=-\sum_{N<n<k}c(n)g(n\theta_k)$, where $c(t)=\frac{1}{k}\cot\frac{\pi t}{2k}$ is decreasing on $[N,k]$ with $c(k)=0$. Partial summation, $G_k(N)=0$ and $c(k)=0$ give $\Phi_k^{>N}=\int_N^k G_k(t)c'(t)\,dt$. Since $\sin u\ge\frac{2u}{\pi}$ on $[0,\frac{\pi}{2}]$, $|c'(t)|=\frac{\pi}{2k^2}\csc^2\frac{\pi t}{2k}\le\frac{\pi}{2t^2}$, and (a) follows. (b) follows from $c_n(k)\le\frac{2}{\pi n}$ and $\sum_{n\text{ odd},\,3\le n<k}\frac{1}{n}\le\frac{1}{2}\log k$. □

Lemma 6.8 (Generic θ)**.** *Let $M\ge 1$ and $N\ge 2$ be integers and $0<\eta\le\frac{1}{2}$, and let $\theta\in\mathbb{R}$ satisfy $\|2m\theta\|\ge\eta$ for $1\le m\le M$. Let $\mathcal{S}$ be a set of consecutive odd integers in $(N,t]$, $t\ge N$, with $n\theta\notin\frac{1}{2}\mathbb{Z}$ for $n\in\mathcal{S}$. Then*

$$\frac{1}{t}\Big|\sum_{n\in\mathcal{S}}g(n\theta)\Big|\le\frac{1}{M}+\frac{2+\log M}{\eta N}.$$

In particular, $D_k\le\frac{1}{M}+\frac{2+\log M}{\eta N}$ if $\|2m\theta_k\|\ge\eta$ for $1\le m\le M$.

Proof. Write $\mathcal{S}=\{n_0+2j:0\le j<|\mathcal{S}|\}$. For $1\le|\ell|\le M$, $|\sum_{n\in\mathcal{S}}e(\ell n\theta)|=|\sum_j e(2\ell j\theta)|\le 1/(2\|2\ell\theta\|)\le 1/(2\eta)$. Lemma 6.3 with $H=M$ gives

$$\Big|\sum_{n\in\mathcal{S}}g(n\theta)\Big|\le\sum_{\ell\text{ odd}}\frac{2}{\pi|\ell|}\cdot\frac{1}{2\eta}+\frac{2}{M+1}\Big(|\mathcal{S}|+\frac{M}{2\eta}\Big)\le\frac{2|\mathcal{S}|}{M+1}+\frac{1}{\eta}\Big(1+\frac{2}{\pi}+\frac{1}{\pi}\log M\Big).$$

Now use $|\mathcal{S}|\le\frac{t-N}{2}+1\le\frac{t}{2}$ and $t\ge N$. For the last statement take $\mathcal{S}=\{n\text{ odd}:N<n\le t\}$ with $t<k$; by the remark after (6.1), $n\theta_k\notin\frac{1}{2}\mathbb{Z}$. □

Lemma 6.9 (Witnesses)**.** *Let k be odd with $3\le k<p$, let $1\le m<k$ and $0<\eta\le\frac{1}{2}$. If $\|2m\theta_k\|<\eta$, then there is an integer r with $0<|r|<\eta k$ and $k\mid pr-m$.*

Proof. Choose $b\in\mathbb{Z}$ with $|2mu_k/k-b|<\eta$ and put $r=2mu_k-bk$; then $|r|<\eta k$ and $pr\equiv 2pu_k\,m\equiv m\pmod k$. If $r=0$ then $k\mid 2mu_k$, hence $k\mid m$, which is impossible. □

7. Large divisors of the terms of an arithmetic progression: proof of Theorems D and D*

This section proves Theorems D and D*. It is self-contained and does not involve $S(p)$. In Section 8 the moduli k for which θ_k is close to a rational number with a small denominator turn out to be divisors larger than $T|r|$ of the numbers $pr-m$, and Theorems D and D* bound their $1/k$-weighted mass. The idea is to write a large divisor d of a term as $d=ef$, where e is the part of d composed of primes up to some y. If e is at most the length R of the progression, then the terms divisible by e are regularly spaced, and the rest f, composed of large primes, can be chosen in few ways. If $e>R$, then e has a divisor s slightly below R, and the terms divisible by s form a much shorter progression (Lemma 7.2). After a few such steps the progressions are so short that Rankin's trick (Lemma 7.1) finishes the proof.

Throughout this section $q \ge 3$ is an integer (not necessarily prime), $L = \log q$ and $X = q^2$. An *admissible progression* is a set $A = \{a + qt : t \in I\}$, where I is a nonempty set of consecutive integers, $\gcd(a, q) = 1$, and $0 < |a + qt| \le X$ for $t \in I$; its *length* is $|I|$. For $Y > 0$ put

$$\Delta(A; Y) = \sum_{t \in I} \sum_{\substack{d \mid a+qt \\ d > Y}} \frac{1}{d}.$$

We use the following simple facts.

(F1) Every divisor of an element of A is coprime to q. If $e \ge 1$ is coprime to q, the $t \in I$ with $e \mid a + qt$ form one residue class modulo e, so their number is at most $|I|/e + 1$.

(F2) If $Y \ge X$ then $\Delta(A; Y) = 0$.

We also use Mertens' bound in the form $M(y) := \prod_{\ell \le y}(1 - 1/\ell)^{-1} = \sum_{P^+(e) \le y} 1/e \le e^{\gamma}(1 + \log y)$ for $y \ge 2$, which follows from [23, (3.30)] for $y \ge 3$ and is checked directly for $2 \le y < 3$.

Lemma 7.1 (Rankin)**.** *For $Y > 0$ and $0 < \alpha < 1$, $\Delta(A; Y) \le |I|\, Y^{-\alpha} P(\alpha)$, where $P(\alpha) = \max_{1 \le n \le X} \sigma_{\alpha-1}(n)$. If $0 < \alpha \le \frac{1}{2}$ and $L \ge 3$, then $\log P(\alpha) \le 4(4L)^{\alpha}(\log\log(4L) + 1)$.*

Proof. For $n \ne 0$, $\sum_{d \mid n,\, d > Y} \frac{1}{d} \le \sum_{d \mid n} \frac{1}{d}(\frac{d}{Y})^{\alpha} = Y^{-\alpha}\sigma_{\alpha-1}(|n|)$. Next, $\sigma_{\alpha-1}(n) \le \prod_{\ell \mid n}(1 - \ell^{\alpha-1})^{-1}$. If $\ell_1 < \cdots < \ell_k$ are the prime factors of $n \le X$ and q_i denotes the i-th prime, then $\ell_i \ge q_i$ and the product is at most $\prod_{i \le k}(1 - q_i^{\alpha-1})^{-1}$; moreover $\theta(q_k) \le \log(q_1 \cdots q_k) \le \log X = 2L$, where θ is Chebyshev's function, and since $\theta(\ell) \ge \ell/2$ for primes $\ell \ge 3$ we get $q_k \le 4L$. For $\alpha \le \frac{1}{2}$ we have $\ell^{\alpha-1} \le 2^{-1/2}$ and $-\log(1 - x) \le x/(1 - 2^{-1/2}) \le 4x$ for $0 \le x \le 2^{-1/2}$, so $\log P(\alpha) \le 4\sum_{\ell \le 4L} \ell^{\alpha-1} \le 4(4L)^{\alpha} \sum_{\ell \le 4L} \frac{1}{\ell} \le 4(4L)^{\alpha}(\log\log(4L) + 1)$. □

For $2 \le y$ put $N_y = \lfloor \log X / \log y \rfloor$ and, for $Z \ge 1$,

$$G_y(Z) = \sum_{0 \le i \le \log Z / \log y} \binom{N_y}{i}.$$

Lemma 7.2 (One step)**.** *Let A be an admissible progression of length at most R, let $T' \ge 1$ and $2 \le y \le R$, and let $\mathcal{S}$ be the set of integers s with $R/y < s \le R$ and $P^+(s) \le y$. For $s \in \mathcal{S}$ let $A_s = \{n/s : n \in A,\ s \mid n\}$. Then each nonempty A_s is an admissible progression of length at most $R/s + 1 \le 2R/s$, and*

$$\Delta(A; T'R) \le \frac{2M(y)}{T'} \sum_{j \ge 0} 2^{-j} G_y(2^{j+1}T'R) + \sum_{s \in \mathcal{S}} \frac{1}{s}\, \Delta\Big(A_s; \frac{T'R}{s}\Big).$$

Proof. Let $s \in \mathcal{S}$. By (F1), $\gcd(s, q) = 1$ and the $t \in I$ with $s \mid a + qt$ are $t = t_s + sv$, with v running over a set of at most $|I|/s + 1$ consecutive integers; then $(a + qt)/s = a' + qv$ with $a' = (a + qt_s)/s$, which is coprime to q. The elements of A_s are nonzero and at most $X/s \le X$ in absolute value. This proves the first assertion.

Now consider the pairs (n, d) with $n \in A$, $d \mid n$ and $d > T'R$, and write $d = ef$, where e is the largest y-smooth divisor of d and f is y-rough.

Case $e \le R$. Fix $j \ge 0$ and consider the pairs with $2^j T'R < d \le 2^{j+1}T'R$. The map $(n, d) \mapsto (e, n, f)$ is injective. Given e (which is coprime to q), there are at most $R/e + 1 \le 2R/e$ elements n with $e \mid n$, by (F1). Given n, the prime factors of f, counted with multiplicity, form a sub-multiset of size at most $\log(2^{j+1}T'R)/\log y$ of the multiset of prime factors of n that exceed y; that multiset has at most N_y elements because $y^{\Omega} \le |n| \le X$, and a multiset with at most N_y elements has at most $\binom{N_y}{i}$ sub-multisets of size i. So there are at most $G_y(2^{j+1}T'R)$ choices for f. Since $1/d < 2^{-j}/(T'R)$ and $\sum_{P^+(e) \le y} 1/e = M(y)$, these pairs contribute at most $\sum_j \frac{2R\,M(y)}{T'R} 2^{-j} G_y(2^{j+1}T'R)$.

Case $e > R$. List the prime factors of e, with multiplicity, in non-decreasing order, and let s be the first partial product that exceeds R/y. It exists because $e > R \ge R/y$, and $s \le (R/y)\cdot y = R$ because every prime factor is at most y; so $s \in \mathcal{S}$ and $s \mid d \mid n$. The map $(n,d) \mapsto (s, n/s, d/s)$ is injective, $d/s \mid n/s$, $d/s > T'R/s$, and $1/d = \frac{1}{s}\cdot\frac{1}{d/s}$; so these pairs contribute at most $\sum_{s\in\mathcal{S}} \frac{1}{s}\Delta(A_s; T'R/s)$. □

Lemma 7.3. *Let $y \ge 2$ and $Z \ge 1$. Then $G_y(2^jZ) \le (1+N_y)^{1+j\log 2/\log y} G_y(Z)$ for $j \ge 0$. If $1+N_y \le y^{1/3}$, then $\sum_{j\ge 0} 2^{-j} G_y(2^{j+1}Z) \le 2.7\,(1+N_y)\,G_y(2Z)$. Moreover $\sum_{0\le i\le m}\binom{N}{i} \le (eN/m)^m$ for $1 \le m \le N$.*

Proof. Since $\binom{N}{i+1} \le N\binom{N}{i}$, we have $\sum_{i\le m+1}\binom{N}{i} \le (1+N)\sum_{i\le m}\binom{N}{i}$. Passing from Z to 2^jZ raises the upper limit $\lfloor \log Z/\log y\rfloor$ by at most $\lceil j\log 2/\log y\rceil \le 1 + j\log 2/\log y$, which gives the first claim. If $1+N_y \le y^{1/3}$, then $(1+N_y)^{j\log 2/\log y} \le 2^{j/3}$, so the sum is at most $G_y(2Z)\big(1 + (1+N_y)\sum_{j\ge 1} 2^{-2j/3}\big) \le 2.7(1+N_y)G_y(2Z)$. The last claim is the standard bound $\sum_{i\le m}\binom{N}{i} \le \sum_{i\le m}\frac{N^i}{i!} \le (\frac{N}{m})^m \sum_i \frac{m^i}{i!}$. □

Proof of Theorem D. Fix $\gamma_0 \in (0,1)$, let $L = \log q$ be large in terms of γ_0, and put $\kappa = \log T = L^{\gamma_0}$ and

$$\alpha = \frac{\gamma_0}{2}, \qquad \beta = \frac{\gamma_0}{8}, \qquad u = \Big\lfloor \frac{\kappa}{3\log L}\Big\rfloor, \qquad R_b = T^{\beta}, \qquad T_j = 2^{-j}T.$$

Define $x_0 = 2L$, $x_{j+1} = \log 2 + (\kappa + x_j)/u$, and $J = \min\{j : x_j \le \beta\kappa\}$. Unrolling the recursion, $x_j \le (\log 2 + \frac{\kappa}{u})\frac{u}{u-1} + 2Lu^{-j} \le 4\log L + 2Lu^{-j}$, and $u \ge L^{\gamma_0}/(4\log L)$; hence $J \le \lceil 1/\gamma_0\rceil + 1$ once L is large. We also assume that L is so large that $u \ge 1 + 8/\gamma_0$, $J\log 2 \le \beta\kappa/2$ and $u\log 2 < \kappa/2$.

For $j \le J$ and $R \ge 1$ let $B_j(R)$ be the supremum of $\Delta(A; T_jR)$ over all admissible progressions A of length at most R, and let $b_j = \sup\{B_j(R) : 1 \le R \le e^{x_j}\}$.

Base case. If $R \le R_b$, Lemma 7.1 gives

$$B_j(R) \le R^{1-\alpha}T_j^{-\alpha}P(\alpha) \le 2^{\alpha J}R_b^{1-\alpha}T^{-\alpha}P(\alpha) = \exp\big(-(\tfrac{3}{8}\gamma_0 + \tfrac{1}{16}\gamma_0^2)\kappa + \alpha J\log 2 + \log P(\alpha)\big) =: E_b,$$

because $\beta(1-\alpha) - \alpha = -\frac{3}{8}\gamma_0 - \frac{1}{16}\gamma_0^2$. By Lemma 7.1, $\log P(\alpha) \ll L^{\gamma_0/2}\log\log L = o(\kappa)$, so $E_b = \exp(-(\frac{3}{8}\gamma_0 + \frac{1}{16}\gamma_0^2)\kappa + o(\kappa))$.

Recursive step. Let $j < J$ and $R_b < R \le e^{x_j}$, and put $y = (T_jR)^{1/u}$. Since $\log R > \beta\kappa$ and $(u-1)\beta \ge 1$, we have $\log T_j \le \kappa \le (u-1)\log R$, i.e. $y \le R$. Also $\log y = (\log T_j + \log R)/u \ge (\kappa - J\log 2 + \beta\kappa)/u \ge (1+\frac{\beta}{2})\frac{\kappa}{u} \ge (3+\beta)\log L$, so $y \ge L^{3+\beta}$; and $N_y \le 2L/\log y \le L$, so $1 + N_y \le y^{1/3}$ for large L. The upper limit in $G_y(2T_jR)$ is $\lfloor u(1 + \log 2/\log(T_jR))\rfloor = u$, because $\log(T_jR) \ge \kappa/2 > u\log 2$. If $u \le N_y$, Lemma 7.3 and $N_y/u \le 2L/\log(T_jR) \le 4L^{1-\gamma_0}$ give

$$G_y(2T_jR) \le \Big(\frac{eN_y}{u}\Big)^u \le \exp\Big(\frac{\kappa}{3\log L}\big((1-\gamma_0)\log L + \log(4e)\big)\Big) = \exp\Big(\frac{1-\gamma_0}{3}\kappa + o(\kappa)\Big),$$

and if $u > N_y$ then $G_y(2T_jR) \le 2^{N_y} \le 2^u$, which is smaller. Since $\log y \le (\kappa + x_j)/u \le 2L$, we have $M(y) \le e^{\gamma}(1+2L) \le 6L$ and $1 + N_y \le L$. Lemmas 7.2 and 7.3 (with $T' = T_j$) therefore give, for every admissible A of length at most R,

$$\Delta(A; T_jR) \le \frac{2\cdot 6L}{T_j}\cdot 2.7L \cdot e^{\frac{1-\gamma_0}{3}\kappa + o(\kappa)} + \sum_{s\in\mathcal{S}}\frac{1}{s}\Delta\Big(A_s; \frac{T_jR}{s}\Big).$$

The first term is at most $E_1 := \exp(-\frac{2+\gamma_0}{3}\kappa + o(\kappa))$, since $T_j \ge 2^{-J}T$. In the second, A_s has length at most $R' := 2R/s$, and $T_jR/s = T_{j+1}R'$; moreover $2 \le R' < 2y$, so $\log R' < \log 2 + (\kappa + x_j)/u = x_{j+1}$. Hence $\Delta(A_s; T_jR/s) \le b_{j+1}$, and since $\sum_{s\in\mathcal{S}} 1/s \le M(y) \le 6L$,

$$B_j(R) \le E_1 + 6L\, b_{j+1} \qquad (R_b < R \le e^{x_j}).$$

Conclusion. Combining the two cases, $b_j \le E_b + E_1 + 6L\, b_{j+1}$ for $j < J$, and $b_J \le E_b$ because $e^{x_J} \le R_b$. Hence

$$b_0 \le (J+1)(6L)^J (E_b + E_1) = \exp\Big(-\big(\tfrac{3}{8}\gamma_0 + \tfrac{1}{16}\gamma_0^2\big)\kappa + o(\kappa)\Big) \le e^{-\gamma_0\kappa/4} = T^{-\gamma_0/4}$$

for $L \ge L_0(\gamma_0)$, because J is bounded in terms of γ_0 and $\log(6L) = o(\kappa)$. Finally let A be admissible of length at most R. If $R \ge X$ then $\Delta(A; TR) = 0$ by (F2); otherwise $\log R < 2L = x_0$ and $\Delta(A; TR) \le B_0(R) \le b_0$. □

Remark 7.4. (a) The same argument with $\alpha = \gamma_0 - \varepsilon/3$ and $\beta = \varepsilon/3$ gives the bound $T^{-\gamma_0+\varepsilon}$ for $L \ge L_0(\gamma_0, \varepsilon)$ (if $\alpha > \frac12$ one uses the bound $\log P(\alpha) \ll_\alpha L^\alpha$, which follows in the same way as Lemma 7.1). This is essentially optimal: if $R = 1$ and $a + qt$ is the product of all primes up to z with $\theta(z) \le 2L$, then the products of $k = \lceil \kappa/\log(z/2)\rceil$ distinct primes from $(z/2, z]$ are at least $(\pi(z) - \pi(z/2))^k/k^k$ divisors of $a + qt$ in $(T, 2^{k-1}zT]$, and a short computation gives $\Delta \ge T^{-\gamma_0 - (\log 2 + o(1))/\log L}$. (b) The proof uses only (F1), (F2), Rankin's trick, Mertens' theorem and the inequality $\theta(\ell) \ge \ell/2$ for primes $\ell \ge 3$; in particular it does not use Shiu's theorem [24], which would require the length of the progression to be at least a fixed power of q. The large size of T is essential: the base case needs T^α to beat $P(\alpha) = \exp(O(L^\alpha \log\log L))$, which is impossible if T is a power of $\log q$. (c) The threshold $L_0(\gamma_0)$ is effective but very large: for $\gamma_0 = 0.3$ an optimized explicit version of the argument needs L of order 10^{11}, and the constants as printed here give about 10^{19}. (d) Although the asymptotic regime of Theorem D is far out of reach, its shape can be tested. For $q = 1000003$, $R = 4^i \le 2^{16}$ and $10 \le T \le 10^5$ we computed $T\,\Delta(A; TR)$ exactly (by sieving the progression) for random a and for a equal to a product of the first primes. For random a the value stays between 0.8 and 4.5, as the random model predicts; for the smooth starting terms it grows from 16 at $T = 10$ to 186 at $T = 10^5$, roughly like $T^{0.27}$, well inside the bound $T^{1-\gamma_0/4}$ and consistent with Remark (a). In the application, the $1/k$-weighted mass of the odd $k \in (10^3, 2\cdot 10^7)$ with $\|2m\theta_k\| < 1/T$ for some $m \le 50$ is (3–6) $\cdot\, 50/T$ for $10^2 \le T \le 10^5$ and a random prime $p \approx 10^{12}$, and at most $13 \cdot 50/T$ for the primes $p = 667866432132899$ and $p = 83483304016613$, for which $p+1$, respectively $2p-1$, is divisible by every odd number below 40.

The proof of Theorem D works for much smaller T if α and β are taken of size $1/\log\log q$ instead of size γ_0. This gives Theorem D*.

Proof of Theorem D.* We keep the notation of the proof of Theorem D; only the parameters change. Put $\kappa = \log T$. If $\kappa > 2L$, then $TR \ge T > X$ and the sum vanishes by (F2). So we may assume $(\log L)^3 \le \kappa \le 2L$. Put

$$\alpha = \frac{1}{\log L}, \qquad \beta = \frac{1}{2\log L}, \qquad u = \Big\lfloor \frac{\kappa}{3\log L} \Big\rfloor, \qquad R_b = T^\beta, \qquad T_j = 2^{-j}T,$$

and define $x_0 = 2L$, $x_{j+1} = \log 2 + (\kappa + x_j)/u$ and $J = \min\{j :\ x_j \le \beta\kappa\}$. Define $B_j(R)$ and b_j as in the proof of Theorem D. In this proof "for large L" means $L \ge L_1$ for an absolute constant L_1; we take $q_1 = e^{L_1}$.

Parameters. Since $\kappa \ge (\log L)^3$, we have $u \ge \frac{\kappa}{3\log L} - 1 \ge \frac14(\log L)^2$, hence $(u-1)\beta \ge 1$, and $\kappa/u \le 3.1\log L$. Unrolling the recursion as before, $x_j \le (\log 2 + \frac{\kappa}{u})\frac{u}{u-1} + 2Lu^{-j} \le 4\log L + 2Lu^{-j}$. As $\beta\kappa \ge \frac12(\log L)^2 \ge 8\log L$, we get $x_j \le \beta\kappa$ as soon as $u^j \ge 8L/(\log L)^2$, so

$$J \le 1 + \frac{\log 8L}{\log u} \le \frac{\log L}{\log\log L},$$

because $\log u \ge 2\log\log L - \log 4$. In particular $2^J \le L$ and $J\log 2 \le \log L \le \beta\kappa/2$. Also $u\log 2 < \kappa/3$, and $x_j \le 2L$ for all j (by induction, since $\kappa \le 2L$ and $u \ge 3$).

Base case. Let $j \le J$ and $R \le R_b$. Lemma 7.1 gives

$$B_j(R) \le R^{1-\alpha} T_j^{-\alpha} P(\alpha) \le \exp\big((1-\alpha)\beta\kappa - \alpha\kappa + \alpha J \log 2 + \log P(\alpha)\big) \le \exp\big(-\beta\kappa + 1 + \log P(\alpha)\big),$$

because $(1-\alpha)\beta - \alpha = -\beta - \alpha\beta$ for $\alpha = 2\beta$, and $\alpha J \log 2 \le 1$. By Lemma 7.1 and $(4L)^{1/\log L} = e^{1+\log 4/\log L}$, we have $\log P(\alpha) \le 4(4L)^{1/\log L}(\log\log 4L + 1) \le 14 \log\log L$. Hence $B_j(R) \le E_b := \exp(-\beta\kappa + 15 \log\log L)$.

Recursive step. Let $j < J$ and $R_b < R \le e^{x_j}$, and put $y = (T_j R)^{1/u}$.

- $\log T_j \le \kappa \le (u-1)\beta\kappa < (u-1)\log R$, so $y \le R$. Also $\log T_j \ge \kappa - J\log 2 > 0$, so $T_j \ge 1$.
- $\log(T_j R) \ge \kappa - J\log 2 + \beta\kappa \ge (1 + \frac{\beta}{2})\kappa$, so $\log y \ge (1+\frac{\beta}{2})\frac{\kappa}{u} \ge (3 + \frac{3\beta}{2}) \log L = 3\log L + \frac{3}{4}$. Hence $y \ge 2L^3$, so $N_y \le 2L/\log y \le L - 1$ and $1 + N_y \le y^{1/3}$.
- Since $\log(T_j R) \ge \kappa > u \log 2$, the upper limit in $G_y(2T_jR)$ is exactly u, as in the proof of Theorem D.
- If $u \le N_y$, then $N_y/u \le 2L/\log(T_jR) \le 2L/\kappa$, and Lemma 7.3 gives

$$G_y(2T_jR) \le \Big(\frac{eN_y}{u}\Big)^u \le \exp\Big(\frac{\kappa}{3\log L}\big(1 + \log 2 + \log L - \log\kappa\big)\Big) \le e^{\kappa/3},$$

  where we used $2eL/\kappa \ge e$ and $\log\kappa \ge 1 + \log 2$. If $u > N_y$, then $G_y(2T_jR) \le 2^{N_y} \le 2^u < e^{\kappa/3}$.
- $\log y \le (\kappa + x_j)/u \le 2L$, so $M(y) \le e^\gamma(1 + 2L) \le 6L$.

Lemmas 7.2 and 7.3 with $T' = T_j$ now give, for every admissible A of length at most R,

$$\Delta(A; T_jR) \le \frac{2 \cdot 6L}{T_j} \cdot 2.7L \cdot e^{\kappa/3} + \sum_{s \in \mathcal{S}} \frac{1}{s} \Delta\Big(A_s; \frac{T_jR}{s}\Big) \le E_1 + 6L\, b_{j+1},$$

where $E_1 := \exp(-\frac{2}{3}\kappa + 4\log L)$. For the first term we used $T_j \ge T/2^J \ge T/L$ and $32.4 \le L$. The second term is treated exactly as in the proof of Theorem D: A_s has length at most $R' = 2R/s$, $T_jR/s = T_{j+1}R'$, and $\log R' < \log 2 + \log y \le x_{j+1}$.

Conclusion. We have $b_j \le E_b + E_1 + 6L\, b_{j+1}$ for $j < J$, and $b_J \le E_b$ because $e^{x_J} \le R_b$. Since $\kappa \ge (\log L)^3$, we have $E_1 \le E_b$, and therefore

$$\log b_0 \le \log\big(2(J+1)\big) + J\log(6L) - \frac{\kappa}{2\log L} + 15\log\log L \le -\frac{\kappa}{4\log L}.$$

For the last inequality note that $\frac{\kappa}{4\log L} \ge \frac{1}{4}(\log L)^2$, while $J\log(6L) \le (1+o(1))(\log L)^2/\log\log L$ and the other terms are $O(\log\log L)$. Finally, if $R \ge X$ the sum in Theorem D* vanishes by (F2); otherwise $\log R < 2L = x_0$, and the sum is $\Delta(A; T_0 R) \le b_0$. □

8. Proofs of Theorems C*, C and C♭

By Theorem B(ii) it suffices to bound the sum Σ of the $F_k(p)/k^2$ over the odd $k < \sqrt{p}(\log p)^4$. We follow the outline in Subsection 1.4. We first prove Theorem C, which needs only Theorem D, and then explain the changes that give Theorems C* and C♭.

The structure of the proof is that of Subsection 1.4: heads below K_*, heads between K_* and K_W, and tails; the constants are tracked only to show that the exponent $2/3$ is reached.

Let $\gamma_0 \in (0, \frac{2}{3})$ and let p be a prime, large in terms of γ_0. Put $L = \log p$ (this is the $L = \log q$ of Section 7 with $q = p$) and

$$T = e^{L^{\gamma_0}}, \quad \eta = \frac{1}{T}, \quad M = H = \lceil L^2 \rceil, \quad N = \big\lceil TM(2 + \log M)\big\rceil,$$

$$K_W = \sqrt{p}L^4, \qquad K_* = \exp\big(A\, L^{2/3}(\log L)^{8/3}\big), \qquad A = A_{BG} + 1,$$

where A_{BG} is the constant of Lemma 6.5. Then $\log N = L^{\gamma_0} + O(\log L)$, and for large p we have $M < N < K_* < K_W < p/4$ (here $\gamma_0 < \frac{2}{3}$ is used), and Theorem D applies with $q = p$ and this T. By Theorem B(ii) with $K = K_W$,

$$S(p) = \frac{p}{2}\log p + c_0 p + \frac{4p}{\pi}\,\Sigma + O\Big(\frac{p}{L}\Big), \qquad \Sigma = R_{K_W}(p) = \sum_{\substack{3\le k<K_W\\ k\text{ odd}}} \frac{\Phi_k}{k}, \tag{8.1}$$

and we write $\Sigma = \Sigma_{\mathrm{head}} + \Sigma_{\mathrm{tail}}$ with $\Sigma_{\mathrm{head}} = \sum_k \Phi_k^{\le N}/k$ and $\Sigma_{\mathrm{tail}} = \sum_k \Phi_k^{>N}/k$, the sums being over odd k with $3 \le k < K_W$. Recall that $\Phi_k^{>N} = 0$ for $k \le N+1$.

The heads with $k < K_*$. By (6.2) and $\sum_{k\text{ odd},\,3\le k<K_*} \frac{1}{k} \le \frac{1}{2}\log K_*$,

$$\Big|\sum_{k<K_*} \frac{\Phi_k^{\le N}}{k}\Big| \le \frac{1}{2\pi}(2+\log N)\log K_* = \Big(\frac{A}{2\pi} + o(1)\Big) L^{2/3+\gamma_0}(\log L)^{8/3}.$$

The heads with $K_* \le k < K_W$. Cover $[K_*, K_W)$ by the intervals $B_j = [2^j K_*, 2^{j+1}K_*) \cap [K_*, K_W)$, $0 \le j \le L$. Only the nonempty B_j contribute, and for them $2^j K_* < K_W$. Apply Lemma 6.6 to each nonempty B_j with $K = K_j = 2^j K_*$ (note $N < K_* \le K_j < K_W < p/2$) together with Lemma 6.5, which applies because $K_j \ge K_* = \exp((A_{BG}+1)L^{2/3}(\log L)^{8/3})$:

$$\Big|\sum_{K_*\le k<K_W} \frac{\Phi_k^{\le N}}{k}\Big| \le (1+\log N)\Big(\frac{2(L+1)}{H} + 160\sqrt{2}\,C_{BG}(1+\log H)(\log L)^3 L \sum_{j\ge 0}(\log K_j)^{-3/2}\Big).$$

Since $\log K_j = \log K_* + j\log 2$, we have $\sum_j (\log K_j)^{-3/2} \le (\log K_*)^{-3/2} + \int_0^\infty \frac{dt}{(\log K_* + t\log 2)^{3/2}} \le 4(\log K_*)^{-1/2}$. With $1 + \log H \le 3\log L$, $H \ge L^2$ and $(\log K_*)^{1/2} = A^{1/2}L^{1/3}(\log L)^{4/3}$, the right-hand side is at most $\big(1920\sqrt{2}\,C_{BG}A^{-1/2} + o(1)\big)\,L^{2/3+\gamma_0}(\log L)^{8/3}$.

Generic tails. Call an odd $k > N+1$ *generic* if $\|2m\theta_k\| \ge \eta$ for $1 \le m \le M$, and let k be generic. Since $\eta N \ge M(2+\log M)$, Lemma 6.8 gives $D_k \le \frac{1}{M} + \frac{1}{M} = \frac{2}{M}$, and Lemma 6.7(a) gives $|\Phi_k^{>N}| \le \frac{\pi}{M}\log k$. Since $\sum_{k\text{ odd},\,3\le k<X} \frac{\log k}{k} \le 1 + \frac{1}{4}(\log X)^2$ and $\log K_W = \frac{L}{2} + 4\log L$, these k contribute at most $\frac{\pi}{M}\big(1 + \frac{1}{4}(\frac{L}{2} + 4\log L)^2\big) = \frac{\pi}{16} + o(1)$ to Σ_{tail}.

Non-generic tails. Let $\mathcal{N}$ be the set of odd k with $N+1 < k < K_W$ for which $\|2m\theta_k\| < \eta$ for some $1 \le m \le M$. For $k \in \mathcal{N}$, Lemma 6.7(b) gives $|\Phi_k^{>N}| \le \frac{1}{\pi}\log k \le \frac{L}{\pi}$, and since $M < k$, Lemma 6.9 provides a *witness* (m, r) with $1 \le m \le M$, $0 < |r| < k/T$ and $k \mid pr - m$; in particular $k > T|r|$ and $|r| < K_W/T$. For $1 \le m \le M$ and $R = 2^i$ with $1 \le R < K_W/T$ consider the progressions

$$A^+_{m,R} = \{pt - m :\ R \le t < 2R\}, \qquad A^-_{m,R} = \{pt + m :\ R \le t < 2R\}.$$

They are admissible in the sense of Section 7 (with $q = p$): their differences are p, their first terms $\mp m$ are coprime to p, and their elements lie in $(0, 2pK_W] \subset (0, p^2]$. If $k \in \mathcal{N}$ has a witness (m, r) with $R \le |r| < 2R$, then k is a divisor larger than $T|r| \ge TR$ of $|pr - m|$, which is an element of $A^+_{m,R}$ if $r > 0$ and of $A^-_{m,R}$ if $r < 0$. Hence, by Theorem D,

$$\sum_{k\in\mathcal{N}} \frac{1}{k} \le \sum_{m\le M} \sum_{\substack{R=2^i\\ 1\le R<K_W/T}} \big(\Delta(A^+_{m,R}; TR) + \Delta(A^-_{m,R}; TR)\big) \le M\,(L+1)\,2T^{-\gamma_0/4},$$

and the non-generic tails contribute at most $\frac{L}{\pi}\cdot 2M(L+1)T^{-\gamma_0/4} = o(1)$ to Σ_{tail}.

Conclusion. Every odd k with $N+1 < k < K_W$ is either generic or in $\mathcal{N}$, so

$$|\Sigma| \le \Big(\frac{A}{2\pi} + \frac{1920\sqrt{2}\,C_{BG}}{\sqrt{A}} + o(1)\Big) L^{2/3+\gamma_0}(\log L)^{8/3} + \frac{\pi}{16} + o(1),$$

and (8.1) gives Theorem C with any constant $C > \frac{4}{\pi}\big(\frac{A}{2\pi} + 1920\sqrt{2}\,C_{BG}A^{-1/2}\big)$. □

Proof of Theorem C.* We repeat the proof of Theorem C, with K_W, K_* and A as there, but with

$$T = \exp\big((\log L)^3\big), \quad \eta = \frac{1}{T}, \quad M = H = \lceil L^2 \rceil, \quad N = \big\lceil TM(2+\log M)\big\rceil.$$

Then $\log N = (\log L)^3 + O(\log L)$, so $M < N < K_* < K_W < p/4$ for large p, and Theorem D* applies with $q = p$ and this T. Equation (8.1) and the splitting $\Sigma = \Sigma_{\text{head}} + \Sigma_{\text{tail}}$ are unchanged. We bound the four ranges as before.

- *The heads with $k < K_*$* contribute at most $\frac{1}{2\pi}(2+\log N)\log K_* = (\frac{A}{2\pi} + o(1))L^{2/3}(\log L)^{17/3}$.
- *The heads with $K_* \le k < K_W$.* The bound displayed in the proof of Theorem C holds as it stands. Now $1+\log N = (1+o(1))(\log L)^3$, $1+\log H \le 3\log L$, $H \ge L^2$ and $(\log K_*)^{1/2} = A^{1/2}L^{1/3}(\log L)^{4/3}$, so these heads contribute at most $(1920\sqrt{2}\,C_{BG}A^{-1/2} + o(1))L^{2/3}(\log L)^{17/3}$.
- *Generic tails.* This part uses only $\eta N \ge M(2+\log M)$, so it contributes at most $\frac{\pi}{16} + o(1)$, as before.
- *Non-generic tails.* As before, every $k \in \mathcal{N}$ is a divisor larger than TR of an element of $A^+_{m,R}$ or $A^-_{m,R}$, for some $m \le M$ and some $R = 2^i < K_W/T$; and each of these progressions consists of R consecutive terms. Theorem D*, with $\log T = (\log L)^3$, gives

$$\sum_{k\in\mathcal{N}} \frac{1}{k} \le 2M(L+1)\exp\Big(-\frac{(\log L)^2}{4}\Big),$$

  so the non-generic tails contribute at most $\frac{L}{\pi}\cdot 2M(L+1)e^{-(\log L)^2/4} \le \exp(5\log L - \frac{1}{4}(\log L)^2) = o(1)$.

Adding up,

$$|\Sigma| \le \Big(\frac{A}{2\pi} + \frac{1920\sqrt{2}\,C_{BG}}{\sqrt{A}} + o(1)\Big)L^{2/3}(\log L)^{17/3} + \frac{\pi}{16} + o(1),$$

and (8.1) gives Theorem C* with any constant $C > \frac{4}{\pi}\big(\frac{A}{2\pi} + 1920\sqrt{2}\,C_{BG}A^{-1/2}\big)$. □

Proof of Corollary 1.1. Take $\gamma_0 = \frac{1}{6}$ in Theorem C. Then

$$\frac{S(p)}{p} \ge \frac{1}{2}\log p + c_0 - C(\log p)^{5/6}(\log\log p)^{8/3} \longrightarrow \infty.$$

The upper half of Conjecture 3.3 is Theorem A. □

Proof of Theorem C♭. Let δ, θ and C be as in the hypothesis, and let p be large. We repeat the proof of Theorem C*, with the same T, η, M, H, N and K_W, but with

$$K_* = \exp(2L^\theta).$$

Since $0 < \theta < 1$, we have $N < K_* < K_W$ for large p. The tails do not involve K_*, so they contribute $\frac{\pi}{16} + o(1)$ as in the proof of Theorem C*. For the heads with $k < K_*$, (6.2) gives the bound $\frac{1}{2\pi}(2+\log N)\log K_* = O(L^\theta(\log L)^3)$.

For the heads with $K_* \le k < K_W$ we use the hypothesis in place of Lemma 6.5. Let $K_* \le K \le p/2$. The lengths N' in Lemma 6.4 satisfy $K/3 \le N' < p$ (see the proof of Lemma 6.5), so $\log N' \ge 2L^\theta - \log 3 \ge L^\theta$, and the hypothesis gives $|\sum_{n\le N'} e(bn^*/p)| \le CN'L^{-\delta} \le 2CKL^{-\delta}$. By Lemma 6.4, $\mathcal{M}_K/K \le 8CL^{-\delta}$. We apply Lemma 6.6 to the at most $L+1$ blocks B_j as in the proof of Theorem C. Since we no longer have a saving that grows with K, we simply add the bounds for the blocks:

$$\Big|\sum_{K_*\le k<K_W} \frac{\Phi_k^{\le N}}{k}\Big| \le (1+\log N)(L+1)\Big(\frac{2}{H} + 80\,C(1+\log H)L^{-\delta}\Big) = O\big(L^{1-\delta}(\log L)^4\big) + o(1).$$

Hence $\Sigma = O(L^{\theta}(\log L)^3 + L^{1-\delta}(\log L)^4)$, and (8.1) gives the asymptotic formula. Since $\max(\theta, 1-\delta) < 1$, it follows that $S(p)/p \geq \frac{1}{2}\log p - o(\log p) \to \infty$, and together with Theorem A this gives Conjecture 3.3. □

The hypothesis of Theorem C$^\flat$ holds with $\theta = \frac{9}{10}$ and $\delta = \frac{1}{4}$. Indeed, let $\log N \geq L^{9/10}$. For large p this is far inside the range of Lemma 6.5, and that lemma gives

$$\Big|\sum_{n\leq N} e\Big(\frac{bn^*}{p}\Big)\Big| \leq C_{BG}(\log L)^3 L\,(\log N)^{-3/2}N \leq C_{BG}(\log L)^3 L^{-7/20}N \leq C_{BG}L^{-1/4}N$$

as soon as $(\log L)^3 \leq L^{1/10}$. So Theorem C$^\flat$ gives a second proof of Corollary 1.1, with the error term $O(p(\log p)^{9/10}(\log\log p)^4)$. It uses Lemma 6.5 only for $\log N \geq (\log p)^{9/10}$, so its conclusion holds whatever the size of A_{BG}. This is the route of the Lean proof of Conjecture 3.3, which assumes the bound of Lemma 6.5 only for $\log N \geq \frac{1}{2}(\log p)^{9/10}$ (Appendix A).

Remark 8.1. (a) The argument uses three inputs of very different nature: Weil's bound (through Theorem B(ii)) for $k \geq K_W$, the bound of Bourgain and Garaev for the heads with $K_* \leq k < K_W$, and Theorem D or D* for the tails. No information on short Kloosterman sums of length below K_* is used: below K_* the heads are estimated trivially, which is affordable because $\sum_{k<K_*} 1/k \ll (\log p)^{2/3+o(1)}$ while each head costs only $O(\log N)$, which is $O((\log p)^{\gamma_0})$ for Theorem C and $O((\log\log p)^3)$ for Theorem C*, and the tails are controlled by the Diophantine condition $\|2m\theta_k\| \geq 1/T$, which involves no exponential sums at all. (b) The exponent $\frac{2}{3}$ comes from the range in which the proof of the Bourgain–Garaev bound applies (Lemma 6.5; equivalently, from the range of Karatsuba's method), and so does the factor $(\log\log p)^{8/3}$. With $K_* = \exp(AL^a(\log L)^b)$ the heads below and above K_* cost $(\log N)L^a(\log L)^b$ and $(\log N)L^{1-a/2}(\log L)^{4-b/2}$, up to constants. These balance exactly at $a = \frac{2}{3}$ and $b = \frac{8}{3}$, the lower end of the range of Lemma 6.5. So, with this bound for short Kloosterman sums, $(\log p)^{2/3}$ is the limit of the method; the power $\frac{17}{3}$ in Theorem C* is 3, from $\log N$, plus $\frac{8}{3}$, from $\log K_*$. A bound of the form $\mathcal{M}_K \leq K(\log K)^{-1-\delta}(\log p)^{o(1)}$, with a fixed $\delta > 0$, for all K with $\log K \geq (\log p)^{\vartheta}$ would allow $K_* = \exp((\log p)^{\vartheta+o(1)})$ and give the error term $O_\varepsilon(p(\log p)^{\vartheta+\varepsilon})$. We expect the truth to be $O(p(\log\log p)^2)$, cf. Theorem E. In the other direction, Theorem C$^\flat$ shows that Corollary 1.1 needs only a fixed power-of-log saving. (c) All ingredients are effective, so p_0 in Theorem C* and $p_0(\gamma_0)$ in Theorem C are effective in principle; they depend on the constants C_{BG} and A_{BG} of Lemma 6.5, which have not been computed, and even optimistic assumptions about them lead to thresholds $\log p_0$ of a size that is far beyond the range of any computation. So Conjectures 1.1, 3.1 and 3.2 are proved here only for sufficiently large primes.

9. Averages over primes: proofs of Theorems E and F

On average over the primes the problem is much easier, because $F_k(p)$ depends only on $p \bmod k$. We split F_k into tangent sums G_d over the reduced residues modulo the divisors d of k, and compute their averages and mean squares over residue classes. The large sieve then shows that the moduli above a power of $\log X$ contribute little for almost all primes in $(X, 2X]$ (Theorem E(i)); together with the Siegel–Walfisz theorem it gives the mean square bound of Theorem E(ii), and the Siegel–Walfisz theorem gives the means in residue classes (Theorem F). Besides Theorem B(ii), which rests on Weil's bound, these proofs use neither bounds for short Kloosterman sums nor Theorem D.

For odd $d \geq 3$ and $n \in \mathbb{Z}$ let

$$G_d(n) = \sum_{\substack{1\leq j<d/2\\(j,d)=1}} \tan\frac{\pi jn}{d}.$$

$G_d(n)$ depends only on $n \bmod d$, and $G_d(-n) = -G_d(n)$.

Lemma 9.1. *For odd $k \geq 3$ and $n \in \mathbb{Z}$, $F_k(n) = \sum_{d|k,\, d\geq 3} G_d(n)$. Consequently, for $3 \leq K_1 < K_2$,*

$$R_K(n) = \sum_{\substack{3\leq d<K \\ d \text{ odd}}} w_d(K)\frac{G_d(n)}{d^2}, \qquad R_{K_2}(n) - R_{K_1}(n) = \sum_{\substack{3\leq d<K_2 \\ d \text{ odd}}} w_d(K_1,K_2)\frac{G_d(n)}{d^2},$$

where $w_d(K) = \sum_{m \text{ odd},\, dm<K} m^{-2} \in [1, \pi^2/8]$ and $w_d(K_1,K_2) = \sum_{m \text{ odd},\, K_1\leq dm<K_2} m^{-2} \in [0, \min(\pi^2/8,\, 2d/K_1)]$.

Proof. Write $t/k = j/d$ in lowest terms; then $d \mid k$, $d \geq 3$, and $t < k/2$ if and only if $j < d/2$; conversely each such pair (d, j) comes from exactly one t. The formulas for R_K follow by writing $k = dm$, and the bound for $w_d(K_1, K_2)$ follows from $\sum_{m\geq M} m^{-2} \leq 2/M$ for $M > 0$. □

Lemma 9.2 (Conditional averages)**.** *Let $d \geq 3$ be odd, $g \mid d$, $m = d/g$ and $s \in \mathbb{Z}$.*

(a) $\frac{1}{m}\sum_{i=0}^{m-1} G_d(s+gi) = A_{d,g}(s) := \sum_{1\leq j<d/2,\,(j,d)=1} \tan\frac{\pi js}{g}$.

(b) $A_{d,1} \equiv 0$, *and* $|A_{d,g}(s)| \leq \frac{\tau(d)}{\pi}\, g(\log g + 2)$ *for $g \geq 3$. In particular* $|G_d(n)| \leq \frac{\tau(d)}{\pi} d(\log d + 2)$.

(c) *Let $(s, g) = 1$ and let $A^*_{d,g}(s)$ be the average of $G_d(x)$ over the reduced residues x modulo d with $x \equiv s \pmod g$. Then $A^*_{d,1} = 0$, and for $g \geq 3$*

$$|A^*_{d,g}(s)| \leq C\,\tau(d)^2\,(\log\log 3d)\,g\,(\log d + 2).$$

Proof. (a) For odd m and real x (avoiding poles) we have $\sum_{i=0}^{m-1}\tan(x + i\pi/m) = m\tan(mx)$; this follows by logarithmic differentiation of $\prod_{i=0}^{m-1}\sin(x + i\pi/m) = 2^{1-m}\sin mx$ and the substitution $x \mapsto x + \frac{\pi}{2}$. For $(j, d) = 1$, ji runs over all residues modulo m as i does, so $\sum_i \tan(\pi j(s+gi)/d) = \sum_i \tan(\pi js/d + \pi i/m) = m\tan(\pi js/g)$.

(b) If $g = 1$ every term is $\tan(\pi js) = 0$. Let $g \geq 3$. Since $ms \in \mathbb{Z}$, $\tan(\pi(d-j)s/g) = -\tan(\pi js/g)$, and $j \mapsto d - j$ maps the reduced residues in $(0, d/2)$ onto those in $(d/2, d)$. Hence

$$A_{d,g}(s) = \frac{1}{2}\sum_{\substack{c \bmod g \\ (c,g)=1}} \tan\frac{\pi cs}{g}\, N(c), \qquad N(c) = \sum_{\substack{0<j<d,\ (j,d)=1 \\ j\equiv c \pmod g}} \operatorname{sgn}\Big(\frac{1}{2} - \frac{j}{d}\Big).$$

Write $j = c + gi$ $(0 \leq i < m,\ 0 < c < g)$. Then $(j, d) = 1$ if and only if $(c + gi, m^*) = 1$, where m^* is the product of the primes dividing m but not g. By Möbius inversion, $N(c) = \sum_{e|m^*}\mu(e)\sum_{i:\, e\mid c+gi}\operatorname{sgn}(\frac{1}{2} - \frac{c+gi}{d})$. For $e \mid m^*$ the relevant $j = c+gi$ form an arithmetic progression with difference ge and m/e terms, one in each interval $[gel, ge(l+1))$, $0 \leq l < m/e$; since m/e is odd, the inner sum is ± 1. Hence $|N(c)| \leq 2^{\omega(m^*)} \leq \tau(d)$. Finally let $h = (s, g)$ and $g' = g/h$. As c runs over the reduced residues modulo g, cs/h runs over the reduced residues modulo g', each hit $\varphi(g)/\varphi(g')$ times; thus $\sum_c |\tan(\pi cs/g)| \leq h\sum_{x=1}^{g'-1}|\tan(\pi x/g')| = 2hF_{g'}(1) \leq \frac{2g}{\pi}(\log g + 2)$ by Corollary 4.3(a) and Lemma 3.3 (and the sum vanishes if $g' = 1$). This proves (b); the case $g = d$ gives the bound for G_d.

(c) If $g = 1$, $A^*_{d,1}$ is the average of the odd function G_d over all reduced residues, hence 0. Let $g \geq 3$. The reduced residues $x \equiv s \pmod g$ are $x = s + gi$, $0 \leq i < m$, with $(s + gi, m^*) = 1$; there are $m\varphi(m^*)/m^*$ of them. For $e \mid m^*$ (so $(e, g) = 1$ and $ge \mid d$), the i with $e \mid s + gi$ form one residue class modulo e, and the corresponding x are $s_e + gei'$, $0 \leq i' < m/e$, for some $s_e \equiv s \pmod g$. By Möbius inversion and (a) with g replaced by ge,

$$A^*_{d,g}(s) = \frac{m^*}{m\,\varphi(m^*)}\sum_{e|m^*}\mu(e)\,\frac{m}{e}\,A_{d,ge}(s_e).$$

By (b), $|A_{d,ge}(s_e)| \le \frac{\tau(d)}{\pi} ge(\log d + 2)$, so $|A^*_{d,g}(s)| \le \frac{m^*}{\varphi(m^*)} 2^{\omega(m^*)} \frac{\tau(d)}{\pi} g(\log d + 2)$, and $\frac{m^*}{\varphi(m^*)} \le \frac{d}{\varphi(d)} \ll \log\log 3d$. □

We shall use the following consequence of the Chinese remainder theorem. Let d, d' be odd, $g = (d, d')$, and let x be uniformly distributed on $\mathbb{Z}/\Lambda\mathbb{Z}$ (respectively on $(\mathbb{Z}/\Lambda\mathbb{Z})^\times$), where Λ is a common multiple of d and d'. Then, conditionally on $x \bmod g$, the residues $x \bmod d$ and $x \bmod d'$ are independent and uniformly distributed over the residues (respectively reduced residues) modulo d and modulo d' that lie over $x \bmod g$. Consequently

$$\mathbb{E}\,[G_d(x)G_{d'}(x)] = \mathbb{E}_s\big[A_{d,g}(s)A_{d',g}(s)\big] \quad \text{resp.} \quad \mathbb{E}_s\big[A^*_{d,g}(s)A^*_{d',g}(s)\big], \tag{9.1}$$

where s is uniform on $\mathbb{Z}/g\mathbb{Z}$ (resp. on $(\mathbb{Z}/g\mathbb{Z})^\times$). In particular these expectations vanish when $g = 1$.

Lemma 9.3 (Mean squares over a period)**.** *Let $3 \le K_1 < K_2$ and let Λ be a common multiple of the odd integers below K_2.*

(a) $\frac{1}{\varphi(\Lambda)} \sum_{x \bmod \Lambda,\,(x,\Lambda)=1} R_{K_2}(x)^2 \le C_0$ *with an absolute constant C_0.*

(b) $\frac{1}{\Lambda} \sum_{x \bmod \Lambda} \big(R_{K_2}(x) - R_{K_1}(x)\big)^2 \le C(\log K_2)^6/K_1$.

Proof. (a) By Lemma 9.1, (9.1) and Lemma 9.2(c), writing $d = ga$, $d' = ga'$ with $g = (d, d') \ge 3$,

$$\mathbb{E}\,[R^2_{K_2}] \le C \sum_{g\ge 3} \sum_{a,a'\ge 1} \frac{\tau(ga)^2\tau(ga')^2\,(\log\log 3ga)(\log\log 3ga')\,(\log ga + 2)(\log ga' + 2)}{g^2a^2a'^2}.$$

Since $\tau(n) \ll_\varepsilon n^\varepsilon$, the summand is $\ll g^{-2+\varepsilon}(aa')^{-2+\varepsilon}$, and the series converges.

(b) By Lemma 9.1, (9.1) and Lemma 9.2(b), and since $w_d w_{d'} \le \frac{\pi^2}{8} \cdot \frac{2}{K_1}\sqrt{dd'}$ (the geometric mean of the two bounds for w_d),

$$\begin{aligned}\frac{1}{\Lambda} \sum_{x \bmod \Lambda} (R_{K_2} - R_{K_1})^2 &\le \frac{\pi^2}{8} \cdot \frac{2}{K_1} \sum_{d,d'<K_2} \sqrt{dd'}\,\frac{\tau(d)\tau(d')g^2(\log g + 2)^2}{\pi^2(dd')^2} \\ &\le \frac{(\log K_2 + 2)^2}{4K_1} \sum_{g<K_2} \frac{\tau(g)^2}{g} \Big(\sum_{a\ge 1} \frac{\tau(a)}{a^{3/2}}\Big)^2,\end{aligned}$$

where again $d = ga$, $d' = ga'$, $g = (d, d')$, and we used $\tau(ga) \le \tau(g)\tau(a)$. Since $\sum_a \tau(a)a^{-3/2} = \zeta(3/2)^2$ and $\sum_{g\le x} \tau(g)^2/g \ll (\log x)^4$, the claim follows. □

Lemma 9.4 (Large sieve)**.** *Let $f : \mathbb{Z} \to \mathbb{C}$ have period Λ, and suppose that its Fourier coefficients $\hat f(\xi)$ on $\mathbb{Z}/\Lambda\mathbb{Z}$ vanish unless ξ/Λ, in lowest terms, has denominator less than Q. Then for every set J of N consecutive integers,*

$$\sum_{n\in J} |f(n)|^2 \le (N + Q^2)\,\frac{1}{\Lambda} \sum_{n \bmod \Lambda} |f(n)|^2.$$

Proof. Distinct fractions a/q, a'/q' with $q, q' < Q$ are at distance at least $1/(qq') > Q^{-2}$ modulo 1. Hence the large sieve inequality of Montgomery and Vaughan [20] (in its dual form; see also [15]) gives $\sum_{n\in J} |\sum_\xi \hat f(\xi) e(n\xi/\Lambda)|^2 \le (N+Q^2)\sum_\xi |\hat f(\xi)|^2$, and Parseval's identity gives the claim. □

By Lemma 9.1, R_K and $R_{K_2} - R_{K_1}$ have period Λ (any common multiple of the odd numbers below K, resp. K_2), and since G_d has period d, their spectra consist of fractions with denominators below K (resp. K_2). Lemma 9.4 therefore applies with $Q = K$ (resp. $Q = K_2$).

Proof of Theorem E(i). Let X be large, $K_2 = \sqrt{2X}(\log 2X)^4$ and $Y = (\log X)^{A+14}$. For $p \in (X, 2X]$ we have $\sqrt{p}(\log p)^4 \le K_2 \le p^{9/10}$, so Theorem B(ii) with $K = K_2$ (a cutoff that does not depend on p) gives

$$S(p) - \frac{p}{2}\log p = c_0 p + \frac{4p}{\pi}R_Y(p) + \frac{4p}{\pi}\big(R_{K_2}(p) - R_Y(p)\big) + O\Big(\frac{p}{\log p}\Big).$$

By Corollary 4.3(b), and since $(\log k)/k$ is decreasing for $k \ge 3$,

$$|R_Y(p)| \le \frac{1}{\pi}\sum_{\substack{3\le k<Y\\ k \text{ odd}}} \frac{\log k + K_0}{k} \le \frac{1}{\pi}\Big(\frac{1}{4}(\log Y)^2 + \frac{K_0}{2}\log Y + \frac{1}{10}\Big).$$

Let $\mathcal{E}$ be the set of integers $n \in (X, 2X]$ with $|R_{K_2}(n) - R_Y(n)| > 1$. By Lemmas 9.4 and 9.3(b),

$$\#\mathcal{E} \le \sum_{X<n\le 2X}\big(R_{K_2}(n) - R_Y(n)\big)^2 \le (X + K_2^2)\,\frac{C(\log K_2)^6}{Y} \le C'\,\frac{X(\log X)^{14}}{Y} = C'X(\log X)^{-A}.$$

For primes $p \in (X, 2X] \setminus \mathcal{E}$ we obtain $|S(p) - \frac{p}{2}\log p| \le \frac{(A+14)^2}{\pi^2}p(\log\log X)^2 + O(Ap\log\log X)$, which is at most $\frac{(A+15)^2}{\pi^2}p(\log\log p)^2$ for $X \ge X_0(A)$. □

Proof of Corollary 1.2. (a) Apply Theorem E(i) with $A = 2$ (note that $17^2/\pi^2 < 30$) on the dyadic intervals $(x/2^{i+1}, x/2^i]$ with $x/2^{i+1} \ge \max(X_0(2), \sqrt{x})$. The exceptional primes number $\ll \sum_i \frac{x}{2^i}(\log x)^{-2} \ll x(\log x)^{-2}$, and there are $O(\sqrt{x})$ primes below $\sqrt{x}$. (b) Apply Theorem E(ii) and Chebyshev's inequality on each dyadic interval $(X, 2X]$ with $\sqrt{x} \le X \le x$, with threshold $\min_{(X,2X]}\xi$. □

Proof of Theorem E(ii). Let X be large, $\mathcal{P}_X$ the set of primes in $(X, 2X]$, $K_2 = \sqrt{2X}(\log 2X)^4$ and $Y = (\log X)^{16}$. By Theorem B(ii),

$$\sum_{p\in\mathcal{P}_X}\Big(\frac{S(p)}{p} - \frac{1}{2}\log p - c_0\Big)^2 \le \frac{48}{\pi^2}\Big(\sum_{p\in\mathcal{P}_X} R_Y(p)^2 + \sum_{X<n\le 2X}\big(R_{K_2}(n) - R_Y(n)\big)^2\Big) + O\Big(\frac{X}{(\log X)^3}\Big).$$

As in the previous proof, the second sum is $\ll X(\log X)^{14}/Y = X(\log X)^{-2}$. For the first, Lemma 9.1 gives

$$\sum_{p\in\mathcal{P}_X} R_Y(p)^2 = \sum_{d,d'<Y}\frac{w_d(Y)w_{d'}(Y)}{(dd')^2}\sum_{p\in\mathcal{P}_X} G_d(p)G_{d'}(p).$$

For each pair let $q = \operatorname{lcm}(d, d') \le Y^2$; all $p \in \mathcal{P}_X$ are coprime to q. By the Siegel–Walfisz theorem (see [15]), uniformly for $q \le (\log X)^{32}$ and $(a, q) = 1$,

$$\#\{p \in \mathcal{P}_X :\ p \equiv a \pmod q\} = \frac{\#\mathcal{P}_X}{\varphi(q)} + O\big(X\exp(-c\sqrt{\log X})\big)$$

with an ineffective constant $c > 0$. Since every term of G_d has modulus at most $\cot\frac{\pi}{2d} \le \frac{2d}{\pi}$, we have $|G_d| \le d^2$, and therefore

$$\sum_{p\in\mathcal{P}_X} G_d(p)G_{d'}(p) = \#\mathcal{P}_X \cdot \mathop{\mathbb{E}}_{x\in(\mathbb{Z}/q\mathbb{Z})^\times}\big[G_d(x)G_{d'}(x)\big] + O\big(q\,d^2d'^2X\exp(-c\sqrt{\log X})\big).$$

The error terms add up to $O(Y^4X\exp(-c\sqrt{\log X})) = o(X/(\log X)^2)$, and since the reduced residues modulo a common multiple Λ of the odd numbers below Y project uniformly onto those modulo q, the main terms add up to $\#\mathcal{P}_X \cdot \frac{1}{\varphi(\Lambda)}\sum_{(x,\Lambda)=1} R_Y(x)^2 \le C_0\,\#\mathcal{P}_X$ by Lemma 9.3(a). This proves the claim for large X, and small X are trivial. □

For Theorem F we need one more observation. Let k be odd, $q \geq 1$, $g = (k, q)$ and $(r, q) = 1$, and let x be uniformly distributed over the reduced residues modulo k with $x \equiv r \pmod g$. Then for each $d \mid k$ the residue $x \bmod d$ is uniformly distributed over the reduced residues modulo d with $x \equiv r \pmod{(d, q)}$. (The condition $x \equiv r \pmod g$ defines a coset of the kernel H of $(\mathbb{Z}/k\mathbb{Z})^\times \to (\mathbb{Z}/g\mathbb{Z})^\times$; its image under reduction modulo d is a coset of the image of H, which by the Chinese remainder theorem consists of the reduced residues modulo d that are $\equiv 1$ modulo $(d, g) = (d, q)$, and the push-forward of the uniform distribution on a coset is uniform on its image.) Together with Lemma 9.1 this gives

$$\langle F_k\rangle_{(k,q),r} = \sum_{d\mid k,\ d\geq 3} A^*_{d,(d,q)}(r), \qquad \text{hence} \qquad c_0(q;r) = c_0 + \frac{\pi}{2}\sum_{\substack{d\geq 3\\ d \text{ odd}}} \frac{A^*_{d,(d,q)}(r)}{d^2}, \tag{9.2}$$

where we used $\sum_{m \text{ odd}} m^{-2} = \pi^2/8$; the rearrangement is justified by the absolute convergence established below. By Lemma 9.2(c), only d with $(d, q) > 1$ contribute, and $|A^*_{d,(d,q)}(r)| \ll_{q,\varepsilon} d^{\varepsilon}$.

Proof of Theorem F. By (9.2) and Lemma 9.2(c), $|\langle F_k\rangle_{(k,q),r}| \ll_{q,\varepsilon} k^{\varepsilon}$, so the series defining $c_0(q;r)$ converges absolutely. It suffices to prove that the mean over $\mathcal{P}_X(r) = \{p \in (X, 2X] : p \equiv r \pmod q\}$ tends to $c_0(q;r)$; the statement for $p \leq x$ follows by summing over dyadic intervals. Let K_2 be as before and $Y = (\log X)^{16}$. By Theorem B(ii),

$$\frac{1}{\#\mathcal{P}_X(r)}\sum_{p\in\mathcal{P}_X(r)}\Big(\frac{S(p)}{p} - \frac{1}{2}\log p\Big) = c_0 + \frac{4}{\pi}\,\frac{\sum_{p\in\mathcal{P}_X(r)}\big(R_Y(p) + (R_{K_2} - R_Y)(p)\big)}{\#\mathcal{P}_X(r)} + O\Big(\frac{1}{\log X}\Big).$$

By the Cauchy–Schwarz inequality and the bound $\sum_{X<n\leq 2X}(R_{K_2} - R_Y)^2 \ll X(\log X)^{-2}$ from the previous proof, the contribution of $R_{K_2} - R_Y$ is $\ll (\#\mathcal{P}_X(r))^{-1/2} X^{1/2}(\log X)^{-1} \ll_q (\log X)^{-1/2}$. For each odd $k < Y$ let $Q = \operatorname{lcm}(k, q) \leq qY$. By the Siegel–Walfisz theorem the primes of $\mathcal{P}_X(r)$ are equidistributed, up to an error $O(X\exp(-c\sqrt{\log X}))$ per class, among the reduced residues modulo Q that are $\equiv r \pmod q$, and by the observation preceding (9.2) (applied with Q in place of k) their reductions modulo k are uniformly distributed over the reduced residues x modulo k with $x \equiv r \pmod{(k, q)}$. Hence

$$\frac{1}{\#\mathcal{P}_X(r)}\sum_{p\in\mathcal{P}_X(r)} F_k(p) = \langle F_k\rangle_{(k,q),r} + O_q\big(k^2\, Y\, \log X\, \exp(-c\sqrt{\log X})\big),$$

and summing over the odd k with $3 \leq k < Y$ with weights k^{-2} shows that the mean of $R_Y(p)$ over $\mathcal{P}_X(r)$ equals $\sum_{3\leq k<Y,\, k \text{ odd}}\langle F_k\rangle_{(k,q),r}/k^2 + o(1)$, which tends to $\sum_{k\geq 3,\, k \text{ odd}}\langle F_k\rangle_{(k,q),r}/k^2$. This proves the convergence. Since F_k is odd, $\langle F_k\rangle_{(k,q),-r} = -\langle F_k\rangle_{(k,q),r}$, whence $c_0(q;r) + c_0(q;-r) = 2c_0$. If q is a power of 2, then $(k, q) = 1$ for odd k, and $\langle F_k\rangle_{1,r}$ is the average of an odd function over all reduced residues, hence 0. □

10. Computations

This section describes how we computed $S(p)$ exactly, how the values were checked, and what they show: the thresholds of Proposition 1.3, the statistics of $S(p)/p - \frac{1}{2}\log p$, a numerical confirmation of c_0, and the structure of the primes for which $S(p)$ is unusually small. The algorithm is the Euclid-type recursion of Lemma 2.5, which needs $O(p \log p)$ operations for one prime.

q	r	$c_0(q;r)$	observed mean	standard error	number of primes
3	1	0.6384	0.6384	0.0011	206265
3	2	−0.1585	−0.1588	0.0011	206416
5	1	0.5863	0.5859	0.0017	103113
5	2	0.4159	0.4163	0.0018	103160
5	3	0.0639	0.0633	0.0018	103238
5	4	−0.1064	−0.1066	0.0017	103170
7	1	0.4656	0.4642	0.0024	68757
7	6	0.0142	0.0130	0.0024	68849
9	1	0.7689	0.7681	0.0017	68729
9	8	−0.2891	−0.2887	0.0017	68858
4	1	$c_0 = 0.2399$	0.2393	0.0014	206252
4	3	$c_0 = 0.2399$	0.2400	0.0014	206429

TABLE 2. Limiting means $c_0(q;r)$ of $S(p)/p - \frac{1}{2}\log p$ in residue classes (Theorem F), compared with the means over the primes $10^3 < p \le 6\cdot 10^6$ in the class. The values $c_0(q;r)$ were computed from (9.2), summing over $d \le D$ with $D = 8001$ and extrapolating the tail, which decays like $1/D$; this was done for $r = 1$ (and for $r = 2$ when $q = 5$), and the remaining values follow from $c_0(q;r) + c_0(q;-r) = 2c_0$. The uncertainty of the computed values is below 10^{-4}.

10.1. **Method and checks.** We computed $S(p)$ from Lemma 2.5: $S(p) = \frac{(p-1)^3}{4} - 4\sum_a F(2p,a)$, where each $F(2p,a)$ is evaluated by the Euclidean-type recursion of that lemma (we also use its nearest-integer variant and tables for small denominators). The program is written in C and all its arithmetic is exact (it uses 64-bit integers, and floating-point numbers only to represent integers below 2^{53}); it needs $O(p\log p)$ operations, and for $p \approx 10^{10}$ one evaluation takes about 100 seconds on one core. The values were checked against (i) a brute-force evaluation of the definition (1.2) for all 668 odd primes $p < 5000$; (ii) an independent $O(p\log p)$ program based on Dedekind sums and their reciprocity law, for all 17983 odd primes $p < 2\cdot 10^5$; (iii) for the larger primes, the prediction of Theorem B, see Table 4. No discrepancy occurred.

10.2. **All primes up to $6\cdot 10^6$.** We computed $S(p)$ for all 412848 odd primes $p \le 6\cdot 10^6$: first for the 283145 odd primes $p \le 4\cdot 10^6$ with the program described above, then for the remaining ones with a new implementation of the same algorithm in exact 128-bit integer arithmetic. As a further check, 640 of the values, chosen at random from all ranges and among the extreme values, were recomputed by a second program that shares no code or method with the first: it writes $(-1)^x = 1 - 2x + 4\lfloor x/2\rfloor$ and evaluates each inner sum over j as a floor sum of a linear function by a Euclid-type algorithm. All 640 values agree. During the revision of this paper all 129703 values for $4\cdot 10^6 < p \le 6\cdot 10^6$ were recomputed by a separately written program based on the same formula (Lemma 2.5); no discrepancy occurred.

Proposition 1.3 summarizes the threshold behaviour; the last primes $p \le 6\cdot 10^6$ with $S(p) \le mp$ are 5, 233, 3119, 58013, 869399 and 5935409 for $m = 1,\dots,6$, and $S(p) > 0$ throughout. The minimum of $S(p)/p$ over $3119 < p \le 6\cdot 10^6$ is 3.0254, attained at $p = 4139$; over $58013 < p \le 6\cdot 10^6$ it is 4.0055 (at $p = 90089$), and over $869399 < p \le 6\cdot 10^6$ it is 5.1332 (at $p = 1021019$). For $5\cdot 10^6 < p \le 6\cdot 10^6$ we have $S(p)/p > 5.82$. Over all odd primes $p \le 6\cdot 10^6$, $S(p)/p - \frac{1}{2}\log p$ lies between -2.160 (at $p = 750749$) and 2.386 (at $p = 4158001$), and over $10^3 < p \le 6\cdot 10^6$ its mean is 0.2396 and its standard deviation 0.626. Table 3 shows that its distribution is stationary, with mean close to c_0 and standard deviation about 0.63, in accordance with Theorems E(ii) and F.

The companion paper [32] explains the stationarity: $S(p)/p - \frac{1}{2}\log p$ has a limiting distribution over the primes, with standard deviation 0.6264 [32, Remark 3.5].

10.3. **The constant c_0.** By Theorem B, $S(p)/p - \frac{1}{2}\log p - \frac{4}{\pi}R_K(p)$ should fluctuate around c_0 with a much smaller variance than $S(p)/p - \frac{1}{2}\log p$ itself. With $K = 3002$ (i.e. $k \le 3001$), the mean of this quantity over the 412418 primes $3001 < p \le 6\cdot 10^6$ is 0.23994, with standard error $3.5\cdot 10^{-5}$ (sample standard deviation 0.022), in agreement with $c_0 = 0.23993\,41036\,32069\,46768\ldots$. Table 1, which is based on the exact identity of Theorem B(i), confirms c_0 to much higher accuracy.

range of p	primes	mean	st. dev.	minimum (at p)	maximum (at p)
$(10^3,\ 2\cdot 10^5]$	17816	0.2387	0.624	−1.719 (58013)	2.219 (150151)
$(2\cdot 10^5,\ 10^6]$	60514	0.2400	0.627	−2.160 (750749)	2.279 (910981)
$(10^6,\ 2\cdot 10^6]$	70435	0.2397	0.625	−1.892 (1767149)	2.363 (1735651)
$(2\cdot 10^6,\ 3\cdot 10^6]$	67883	0.2407	0.626	−2.041 (2972969)	2.320 (2238391)
$(3\cdot 10^6,\ 4\cdot 10^6]$	66330	0.2391	0.625	−1.957 (3950099)	2.360 (3387931)
$(4\cdot 10^6,\ 5\cdot 10^6]$	65367	0.2396	0.626	−1.840 (4046543)	2.386 (4158001)
$(5\cdot 10^6,\ 6\cdot 10^6]$	64336	0.2390	0.628	−1.939 (5525519)	2.349 (5694151)

TABLE 3. Statistics of $S(p)/p - \frac{1}{2}\log p$.

10.4. **Large and extremal primes.** The fluctuations of $S(p)/p - \frac{1}{2}\log p$ are dominated by the small moduli k. If $k \mid p+1$ then $F_k(p) = F_k(-1) = -F_k(1)$ is as negative as possible (Corollary 4.3), and if $k \mid p-1$ then $F_k(p) = F_k(1)$. Primes for which $p+1$ (resp. $p-1$) is divisible by many small odd numbers therefore have unusually small (resp. large) values of $S(p)/p$. We searched for such primes and computed $S(p)$ exactly for 28 of them between $3\cdot 10^6$ and $5.4\cdot 10^{10}$, as well as for 28 further primes near 10^7, 10^8, 10^9 and 10^{10}. Table 4 lists some of them. The smallest value found is $S(p)/p - \frac{1}{2}\log p - c_0 = -3.008$ at $p = 9166207049$, where $S(p)/p = 8.70$. In all 56 cases the exact value agrees with $c_0 + \frac{4}{\pi}R_{30002}(p)$ to within 0.03.

p	structure	$S(p)$	$\frac{S(p)}{p} - \frac{\log p}{2} - c_0$	$\frac{4}{\pi}R_{30002}(p)$
$10^{10}+19$		115344445654	−0.2184	−0.2165
$10^{10}+33$		117633975024	0.0105	0.0155
$10^{10}+61$		112907100532	−0.4621	−0.4495
9166207049	$p+1 = 2\cdot 3^4\cdot 5^2\cdot 7^2\cdot 11\cdot 13\cdot 17\cdot 19$	79755583112	−3.0083	−2.9884
53542288799	$p+1 = 2^5\cdot 3^2\cdot 5^2\cdot 7\cdot 11\cdot 13\cdot 17\cdot 19\cdot 23$	514925507950	−2.9746	−2.9478
5237832599	$p+1 = 2^3\cdot 3^4\cdot 5^2\cdot 7\cdot 11\cdot 13\cdot 17\cdot 19$	44836822926	−2.8693	−2.8626
3170267099	$p+1 = 2^2\cdot 3^4\cdot 5^2\cdot 7\cdot 11\cdot 13\cdot 17\cdot 23$	26344372166	−2.8686	−2.8548
2677114441	$p-1 = 2^3\cdot 3^2\cdot 5\cdot 7\cdot 11\cdot 13\cdot 17\cdot 19\cdot 23$	36439344488	2.5175	2.5157
20078358301	$p-1 = 2^2\cdot 3^3\cdot 5^2\cdot 7\cdot 11\cdot 13\cdot 17\cdot 19\cdot 23$	299323341516	2.8064	2.7878

TABLE 4. Exact values of $S(p)$ for some large primes, compared with the prediction of Theorem B truncated at $k \le 30001$.

10.5. **Anatomy of the extremal primes.** Using the expansion of Proposition 4.5 (with a tabulated I) one can evaluate $R_K(p)$ for $K = 4\sqrt{p}$ in a few milliseconds; for the 354 primes $p \le 4\cdot 10^6$ for which we compared, $\frac{4}{\pi}R_{4\sqrt{p}}(p)$ differs from $S(p)/p - \frac{1}{2}\log p - c_0$ by at most 0.06 (standard deviation 0.016). We computed $R_{4\sqrt{p}}(p)$ for all 204648 primes $10^6 < p \le 4\cdot 10^6$; this analysis was not extended to $6\cdot 10^6$. For the 100 primes with the most negative values, the mean deficit relative

to a random sample of primes is -1.47. Writing $R_K(p) = \sum_d w_d(K)G_d(p)/d^2$ as in Lemma 9.1, this deficit is carried almost entirely by moduli d that divide one of a few linear forms in p: 53% by $d \mid p+1$, 11% by $d \mid 2p-1$, 24% by the absence of the positive contributions of $d \mid p-1$ and $d \mid 2p+1$, and most of the rest by other forms $ap+c$ with small $|a|, |c|$; the moduli d that divide no form $ap+c$ with $|a|, |c| \le 12$ contribute $+0.003$. Moreover 57% of the deficit comes from $d < 16$, and every modulus $k \ge 256$ with $F_k(p)/k \le -1.5$ among these primes has θ_k just above 0 or just below $\frac{1}{2}$, in accordance with Corollary 4.11.

11. Remaining questions

We list the questions that this paper leaves open, and say what stands in the way of each.

Explicit thresholds. Conjectures 1.1, 3.1 and 3.2, and the inequality $S(p) > p$ for $p > 5$, are now known for $p \le 6 \cdot 10^6$ (Proposition 1.3) and for $p \ge p_1$, where p_1 can in principle be computed but is very large. Via Theorem C, Corollary 1.1 needs $\gamma_0 < \frac{1}{3}$, and for $\gamma_0 = 0.3$ an optimized explicit version of Theorem D alone requires $\log p$ of order 10^{11} (Remark 7.4(c)). The threshold p_1 also depends on the constants C_{BG} and A_{BG} of Lemma 6.5, which have not been computed. Even if $C_{BG} = 1$, the bound of Lemma 6.5 beats the trivial bound for some $N \le \sqrt{p}$ only when $\log p > 8(\log\log p)^6$, that is, when $\log p > 5.2 \cdot 10^8$. For a single prime the margin is not large either: at $p \approx 6 \cdot 10^6$, Conjecture 3.2 asks for $S(p)/p - \frac{1}{2}\log p > -4.8$, while $S(p)/p - \frac{1}{2}\log p - c_0$ reaches -3.0 at $p = 9166207049$ (Table 4). So, with this method, a proof for the primes just beyond the computed range would need explicit bounds for the sum in (1.7) that are sharp up to a constant factor, and in the middle range of moduli these require explicit cancellation in short Kloosterman sums far beyond the present state of the art. Closing the gap requires an explicit form of the whole argument with drastically better constants, or a different method for the moderately large primes.

The error term. Theorem C* gives $S(p) = \frac{p}{2}\log p + O(p(\log p)^{2/3}(\log\log p)^{17/3})$, and the exponent $\frac{2}{3}$ is the limit of the method (Remark 8.1(b)). Theorem E and the computations suggest that $S(p) = \frac{p}{2}\log p + O(p(\log\log p)^2)$ for every prime. Heuristically this would be optimal: by Linnik's theorem there are primes p with $p \equiv -1$ modulo all odd $k \le c\log p$, and for these the moduli $k \le c\log p$ contribute $-(\frac{1}{\pi^2} + o(1))(\log\log p)^2$ to $S(p)/p$; we cannot, however, control the larger moduli for such special primes. The companion paper [32, Theorem 3] proves unconditionally that $S(p)/p - \frac{1}{2}\log p - c_0 = \Omega_\pm((\log\log p)^2)$, with the constant $2e^\gamma/\pi^2 = 0.3609\ldots$; so this conjecture, if true, is optimal up to the constant.

The constant c_1. We do not know whether the constant term $c_1 = -0.40762\,59135\ldots$ of $\Psi(p)$ (Remark 5.1) can be expressed in terms of classical constants.

Formalization. A formal proof of the hypothesis of Theorem C$^\flat$, for instance of the Bourgain–Garaev bound for $\log N \ge \frac{1}{2}(\log p)^{9/10}$, or a proof of Corollary 1.1 that avoids short Kloosterman sums, would make the formal proof of Conjecture 3.3 described in Appendix A unconditional. Not formalized are, among others, Theorems E and F, Subsection 4.1, Lemmas 2.3, 2.4, 2.6 and 2.7, and Theorem B(ii) in its sharp form.

Appendix A. The Lean formalization

This appendix describes what has been checked by the Lean proof assistant, and under which hypotheses. A formal proof removes the possibility of a gap in the long chain of estimates behind Corollary 1.1, except in the inputs that are assumed. We state the hypotheses exactly, list the main formal theorems, and say where the formal proofs differ from the paper.

Numbering of the conjectures. We follow the numbering and wording of the third arXiv version of [1]. In the first version Conjecture 1.1 is stated for every odd prime; as $S(3) = 2$ and $S(5) = 4$, the two forms are equivalent. The formal-conjectures file [11] follows the first arXiv

version. There Conjecture 1.1 appears as `conjecture_1_1`, and Conjectures 3.1 and 3.2 appear as `conjecture_4_2` and `conjecture_4_3`. The statement `conjecture_4_1` is the inequality $S(k) > k$ for primes $k > 5$, which is not among the four conjectures of Subsection 1.1; it is covered by Proposition 1.3 for $p \le 6 \cdot 10^6$ and by Corollary 1.1 for large p. Finally, `conjecture_4_4` asserts that for every $n \in \mathbb{N}$ one has $nk < S(k)$ for all sufficiently large primes k; this is the lower half of Conjecture 3.3.

The project. The formalization is a Lean 4 project [34] built on Mathlib [33] (Lean and Mathlib version `v4.35.0`), of about 17000 lines, at https://github.com/samsonmaximus/theta-sum. It contains no `sorry` and no `native_decide`, and for every theorem named below `#print axioms` lists only `propext`, `Classical.choice` and `Quot.sound`; the file `ThetaSum/MainTheorems.lean` prints the statements and axioms of all of them. The definitions `S'` and `S` are copied verbatim from the formal-conjectures file `FormalConjectures/Arxiv/2501.03234/ArithmeticSumS.lean` [11], so that the theorems refer to exactly the formalized conjectures:

```
def S' (h k : ℕ) : ℤ :=
  ∑ j ∈ Finset.Ico 1 k, (-1 : ℤ) ^ (j + 1 + ⌊(h * j : ℚ) / k⌋₊)
def S (k : ℕ) : ℤ := ∑ h ∈ Finset.Ico 1 k, S' h k
```

Unconditional results.

- Theorem A with K_0 replaced by 2, that is, $|S(p)| \le \frac{2}{p} C_p(1)^2 \le \frac{2}{\pi^2} p(\log p + 2)^2$ (`theorem_A`, in the namespace of the file of [11]); the bound $|S(p)| \le 5pH_p^2$ (`S_abs_le`); and the upper half of Conjecture 3.3 (`ThetaSum.conjecture_3_3_upper`). These are also in a separate, smaller project for the upper half.
- Lemmas 3.1–3.3, Theorem B(i) (`TheoremB.S_eq_Psi_Phi`) and the reciprocity law, Lemma 4.1 (`Recip.reciprocity`). The formal proof of the reciprocity law uses no contour integration; it combines an exact polynomial identity at roots of unity with the partial fraction expansion of the cotangent.
- Theorem B(ii) in a weaker form: for every $\varepsilon > 0$ and all large primes p, $|S(p) - \frac{p}{2}\log p - c_0 p - \frac{4p}{\pi} R_K(p)| \le \varepsilon p$ with $K = \lfloor p^{3/4}(\log p)^4 \rfloor$ (`ThetaSum.theorem_B_c0`), and the closed form (1.6) of c_0 (`ThetaSum.c0_closed_form`). In place of Weil's bound this uses Kloosterman's bound $|\mathcal{K}(b, \ell; p)| \le 2p^{3/4}$ [35], which is proved in Lean by the fourth-moment method (`ThetaSum.hK34`); this is why the cutoff is $p^{3/4}(\log p)^4$.
- Theorem D, exactly as stated (`ThetaSum.theorem_D`), and Theorem D* with $q_1 = \exp(e^{1000})$, that is, for $\log\log q \ge 1000$ (`ThetaSum.theorem_D_star`).
- Vaaler's lemma, Lemma 6.2 (`Vaaler.vaaler_lemma`), by a finite argument, and the other tools of Section 6 except Lemma 6.5: the sign form, the square wave, the reduction to initial sums, the heads, the tails, an elementary version of Lemma 6.8 without Vaaler's lemma, and the witnesses.
- Conjectures 1.1, 3.1 and 3.2, and $S(p) > p$ for $p > 5$, for all primes $p \le 2 \cdot 10^4$ (`ThetaSum.S_pos_upto`, `conjecture_4_1_upto` in the namespace of the file of [11], `ThetaSum.S_gt_two_mul_upto`, `ThetaSum.S_gt_three_mul_upto`), together with $S(233) = 440$ and $S(3119) = 9194$ (`S_233`, `S_3119`), which show that the thresholds 233 and 3119 are sharp. Lemma 2.5 is proved in Lean (`S_eq_lemma25`), and the values are computed from it by the Lean kernel itself (`decide +kernel`).

The hypotheses. The bound of Bourgain and Garaev is not formalized. It enters through one of the following three propositions, in which p is prime, $b \not\equiv 0 \pmod p$, n^* is the inverse of n modulo p, and "for large p" means for all primes p larger than some bound.

(HKS) There are $\delta > 0$, $\theta < 1$ and C such that for large p and all $N < p$ with $\log N \ge (\log p)^\theta$, $\left|\sum_{n \le N} e(bn^*/p)\right| \le CN(\log p)^{-\delta}$.

(HBG) There is C such that for large p and all $N < p$ with $\log N \geq \frac{1}{2}(\log p)^{9/10}$, $\left|\sum_{n\leq N} e(bn^*/p)\right| \leq C(\log\log p)^3(\log p)(\log N)^{-3/2}N$.

(HBGfull) There are C and A such that the bound of (HBG) holds for large p and all $N < p$ with $\log N \geq A(\log p)^{2/3}(\log\log p)^{8/3}$.

(HBGfull) is Lemma 6.5. (HBG) is the same bound in a smaller range, which lies inside the range of Lemma 6.5 for large p. Lean proves that (HBG) implies (HKS), with $\theta = \frac{9}{10}$ and $\delta = \frac{1}{4}$ (`ThetaSum.HKS_of_HBG`); so (HKS), a saving of any power of $\log p$ for short Kloosterman sums of length at least $\exp((\log p)^\theta)$, is the weakest of the three. It is the hypothesis of Theorem C$^\flat$ (there we ask $\theta > 0$, which is no restriction).

Conditional results.

- Conjecture 3.3 assuming (HKS) (`ThetaSum.conjecture_3_3_of_HKS`), that is, the qualitative part of Theorem C$^\flat$, and hence assuming (HBG) (`ThetaSum.conjecture_3_3_of_BG`); likewise the statement `conjecture_4_4` of [11] for every n (`conjecture_4_4_of_HKS`).
- The inequalities of Conjectures 1.1, 3.1 and 3.2 for all sufficiently large primes, assuming (HBG) (`ThetaSum.conjecture_1_1_eventually_of_BG` and its analogues for 3.1 and 3.2). The statements for every prime are not proved; see Section 11.
- Theorem C* assuming (HBGfull) (`ThetaSum.theorem_C_star`, with some constant c in place of c_0; `ThetaSum.theorem_C_star_c0`, with c_0), and Theorem C assuming (HBGfull), for every $\gamma_0 \in (0, \frac{2}{3})$ and with a constant C depending on γ_0 (`ThetaSum.theorem_C`; `ThetaSum.theorem_C_c0`). At this size of the error term the forms with c and with c_0 are equivalent.

The first of these reads, in the notation of [11],

```
theorem conjecture_3_3_of_HKS (hK : HKS) :
    ∀ C₁ C₂ ε : ℝ, 0 < C₁ → 0 < C₂ → 0 < ε →
    ∀ᶠ k : ℕ in Filter.atTop, k.Prime →
      C₁ * k < (S k : ℝ) ∧ (S k : ℝ) < C₂ * (k : ℝ) ^ (1 + ε)
```

Differences from the paper. The formal proof of the lower half of Conjecture 3.3 follows Sections 6–8, with four changes. It replaces Weil's bound by Kloosterman's bound, as described above. Where the proof of Theorem C$^\flat$ takes $K_* = \exp(2(\log p)^\theta)$ and uses Theorem D*, it takes $K_* = \exp((\log p)^{\theta'})$ with $\theta' = \frac{1+\theta}{2} < 1$, and $T = \exp((\log p)^\gamma)$ with a small $\gamma > 0$ together with Theorem D; this gives only $S(p) = \frac{p}{2}\log p + o(p\log p)$, which suffices for $S(p)/p \to \infty$. And it proves Theorems D and D* without Mertens' theorem, with a different form of Rankin's trick and different parameters; the statements are the same. Finally, Lemma 4.2 is proved in a weaker form, which suffices for the constant c_0.

The upper bound. The proof of $|S(p)| \leq 5pH_p^2$ follows the Fourier approach of Section 3 but avoids cotangents. With $\zeta = e^{\pi i/p}$, $w_l = \zeta^{2l+1}$ and $c_l = \sum_{m=1}^{p} w_l^m$, one has the exact expansion $p(-1)^{\lfloor n/p\rfloor} = -\sum_{l<p} c_l w_l^n$ for all $n \geq 0$; hence $pS(p) = -\sum_{l<p} c_l \sum_{a,b=1}^{p-1} w_l^{ab}$. Every geometric sum $\sum_{m\leq i<n} \zeta^{ri}$ with $0 < r < 2p$ has modulus at most $1/\sin(\pi r/2p) \leq p/\min(r, 2p-r)$ by Jordan's inequality; for $2l+1 \neq p$ the map $a \mapsto (2l+1)a \bmod 2p$ is injective on $[1, p)$, which gives $\sum_a |\sum_b w_l^{ab}| \leq 2pH_p$, while the term $2l+1 = p$ is bounded trivially. This yields $p|S(p)| \leq 4p^2H_p^2 + p^2$, hence $|S(p)| \leq 5pH_p^2$; finally $H_p \leq 1 + \log p$ (proved by telescoping) and $\log p \leq p^\delta/\delta$ give the asymptotic statement.

Use of AI

I used Claude Opus 5.5 (Anthropic) throughout this project: to search the literature, to develop and test the arguments, to write and run the code, to write the Lean formalization, and to draft the text. I then had every part checked by separate Claude sessions acting as referees. They re-derived

each proof, compared each cited statement with its source, and recomputed the computational claims, in part with code written from scratch. Their reports led to corrections that are included here; one of them found that an earlier version used the Bourgain–Garaev bound outside the range its proof covers. These checks were made with the same AI system and do not replace expert refereeing. Every computational claim can be reproduced from the ancillary files. The proof of Conjecture 3.3 is also checked in the Lean proof assistant, about 17,000 lines with no unproved steps, assuming only the Bourgain–Garaev bound for short Kloosterman sums, and the values of $S(p)$ up to $6 \cdot 10^6$ were computed twice by separately written programs. I have read the final manuscript and take full responsibility for its contents; any remaining errors are mine.

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Independent researcher